DEVELOPMENT OF AN ONLINE CLASS CONFERENCE PLATFORM: INTEGRATING FACIAL EXPRESSION RECOGNITION UTILIZING FACE++ WEB API FOR LEARNERS EMOTION DETECTION

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DEDICATION

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ABSTRACT

VirtuSense: An online conference platform with emotion detection via facial expression recognition was developed to address the problem of the students' academic performance during online classes. This project was developed by using Agora for video conferencing, TypeScript and React for the frontend side, Firebase for the backend, and Face++ Web API for emotion detection. The project used agile scrum methodology. Software testing, stress testing, compatibility testing, and user acceptance testing were used to assess the project's reliability and usability, achieving "Passed" and "Compatible" remarks across all tests. The project yielded an overall mean of 3.53 in the ISO 25010 survey, indicating that the system has met all the model's criteria. This meant that the system is a reliable and usable tool for an online conference platform with emotion detection. The web application successfully achieved its objectives by providing a responsive online conference platform that supports learners during online classes, integrating emotion detection into the online class platform, and meeting the evaluation criteria based on the Technology Acceptance Model (TAM). Based on the findings, it is recommended to develop a mobile application version, enhance interactivity, incorporate AI-generated insights in the record section, and add an archive for older sessions.

Keywords: *Online Conference Platform, Emotion Detection, Face++, Agile Scrum, ISO25010, Software Testing, Stress Testing, Compatibility Testing, User Acceptance Testing*

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DEFINITION OF TERMS

Action Unit (AU)	The basic component of the Facial Action Coding System (FACS), representing a specific facial muscle movement associated with certain emotions.
Agora	a real-time communication platform that provides APIs and SDKs for embedding video calls, voice calls, and interactive live streaming features into web or mobile applications.
Application Programming Interface (API)	is a set of rules and protocols that allows software applications to communicate and exchange data with each other.
Artificial Intelligence (AI)	refers to a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, decision making, and pattern recognition.
Face++	A cloud-based AI service offering facial recognition, detection, and emotion analysis through APIs for integration into applications.
Facial Expression Recognition (FER)	is a computer vision technique that automatically identifies human emotions by analyzing facial features and movements captured through images or video frames.
Firebase	A backend-as-a-service platform by Google that provides tools such as authentication, real-time databases, cloud storage, and hosting for building web and mobile applications.
Integrated Development Environment (IDE)	A software application that provides a complete environment for coding, including a code editor, debugger, and tools for building and running applications.
ISO 20510	An international standard that provides guidelines for implementing effective service and support processes, ensuring quality, consistency, and user satisfaction in system development.
Likert Scale	A psychometric scale commonly used in surveys to measure attitudes or perceptions, typically using a 5-point or 7-point response range from “strongly disagree” to “strongly agree.”

Perceiver Dependent	A concept in emotion recognition that states emotional interpretation varies depending on the individual observer's experiences, background, and perception.
Pre-trained	Refers to an AI model that has already been trained on a large dataset, allowing it to perform specific tasks without requiring training from scratch.
Software Development Kit (SDK)	a collection of tools, libraries, documentation, and code samples that developers use to build applications for a specific platform or service.
Technology Acceptance Model (TAM)	A theoretical model that explains how users come to accept and use technology, based on perceived usefulness and perceived ease of use.
Valence Arousal	A framework used in emotion research that classifies emotions based on two dimensions: valence (positive to negative) and arousal (low to high intensity).
Web-Based	Refers to applications that run online through a web browser rather than requiring installation on a device.

CHAPTER 1

INTRODUCTION

Project Context

The emergence of online classes as part of teaching methods has both positive and negative impacts on the students' academic performance and learning process. In the Philippines, undergraduate nursing students experience lower satisfaction and a decline in their academic performance due to stress from online learning (Oducado & Estoque, 2021). Another study shows that the more engaged and satisfied students are in online learning, the better their academic performance will be (Valladolid, 2022).

As distance learning becomes increasingly prevalent in both academic and training environments, there is a growing need to research various aspects of the learning experience in an online setting (Dennen et al., 2007). Limited interaction is one of the many challenges in distance learning that could hinder rapport between learners and educators (Ong & Quek, 2023). It is also proven that the students who are not as open with their teacher about their academic problems are more likely to perform worse academically (Karaivanova, 2016). Additionally, assessment is not only treated as a way of improving students' performance but also can refine teaching methods and achieve better relationships between the students and the educators (Clipa & Mihai, 2019). Feedback and evaluation are important for the students because they seek to improve their academic performance and learning process (Moreira et al., 2023). Students' academic performance can be enhanced with formative assessment techniques in which the educator offers constant and effective feedback through a mentoring system (Sirianansopa, 2024).

Emotion serves as a crucial indicator of a student's engagement or interest in online learning or even in physical class. It also plays a significant role in creating strategies for the teaching and learning process (Krithika et al, 2016). A happy and excited student can learn a specific subject more than a sad or unhappy learner, which often leads to distraction in learning (Wibawanto et al, 2017). During the pandemic, learners often experience depression, academic stress, or anxiety in online learning because it is new to them, which has a negative impact on the learning process. An individual is said to be more focused if his/her mind is at a state of peace, which can result in better understanding of a specific topic (Lek et al, 2023). Moreover, enjoyment in academic activities is beneficial to academic performance. Also, academic emotion has a strong impact on the learners' academic achievement or performance (Camacho-Morles et al, 2021). Individuals' positive emotions, such as happiness during classes, can help boost their interest in studying. However, negative emotions like sadness, fear, and anger can negatively affect the students' behavior during exams or the learning process (Rothkhrantz, 2017). Having these problems leads to the emergence of emotion detection through facial expression recognition. A study shows that the level of students' engagement can be identified through emotion detection, particularly with their facial expression (Maqableh et al., 2022).

Recent advancements in emotion detection have been made. In 2021, a university in Thailand created an online conference platform with emotion detection and achieved 88.97% satisfactory results after applying it to real online classes. The system detects boredom, neutral, happy, and confused, and attains higher accuracy results than any other convolutional neural network-based emotion recognition system (Panichkriangkrai et al, 2022).

Researchers in India developed a system that also detects facial expression during online classes using artificial intelligence, which allows educators to gain valuable feedback on the effectiveness of their online teaching methods and make data-driven decisions to improve the learning experience for their students (Mathews & Aravindhar, 2024). In addition, a tech company in China developed an artificial intelligence that can detect micro movements of facial muscles and can identify positive and negative emotions. This technology is currently used by some teachers to monitor the students' focus during classes (Agarwal, 2022)

Having a deep understanding of students' emotions during online classes can give numerous benefits for educators and institutions, such as improving teaching styles and creating new and engaging strategies that can enhance learners' interest in online classes. Through advancements in artificial intelligence technology, the development of an online conference platform integrated with emotion detection using an Application Programming Interface (API), such as Face++, has become possible. The creation of this type of software is important for refining the online learning environment not only for students' academic gain but also for supporting educators in monitoring engagement and emotional responses in real time. Furthermore, this system can assist institutions in making data-driven decisions to improve the overall effectiveness of online education delivery.

Research Objectives

The aim of this capstone project is to develop an online conference platform capable of detecting the emotions of learners in online classes through facial expression recognition. This capstone project specifically aims to:

1. Develop a responsive web application that serves as an online conference platform to support learners during online classes.
2. Integrate real-time emotion analysis features through the integration of emotion detection that allows educators to monitor learners' emotional states as they participate in online classes.
3. Evaluate the acceptability of the system using the Technology Acceptance Model (TAM) and ISO 25010.

Project Purpose

The purpose of this project is to track students' emotional identification during online classes using Facial Expression Recognition (FER) technology. The development of this system greatly helps the education sector that still uses online classes as one of its class modalities, specifically in teaching strategies during distance learning. Furthermore, this project aligns with Sustainable Development Goal (SDG) 4: Quality Education by improving students' learning environments, including online classes, where their engagement and emotions are constantly monitored and addressed. Aside from education, this project also targets SDG 9: Industry, Innovation, and Infrastructure by innovating online learning platforms, integrating automatic emotion recognition via FER technology, which can lead to more personalized, data-driven teaching strategies. Lastly, it also contributes to the SDG 3: Good Health and Well-Being goal by assessing students’ emotional and mental health during online classes, which can lead to a healthier educational experience.

Educators. The integration of emotion detection into the system may help the target audience identify and understand their students' emotions during online class sessions by automatically detecting them from facial

expressions. The data analytics module and real-time emotion monitoring feature help educators analyze emotion trends to know when students exhibit a particular emotion and adjust their teaching approach accordingly. Learners' emotion records can track emotional responses over time, potentially helping educators provide targeted support to students based on their emotional responses. Additionally, the system allows educators to monitor emotional patterns in every session, enabling them to tailor their teaching methods based on the data given, which can lead to a better learning environment.

Future Researchers. This project serves as a pillar for future researchers interested in innovating current educational technology by integrating Facial Expression Recognition (FER) and for those who want to study students' academic engagement during online classes. The report management module, which includes records of detected emotions, emotional trends, and emotion classification, might help future researchers identify and analyze emotional trends and patterns and their connection to learners ' academic engagement. These features make the system a valuable tool for future researchers who want to study the correlation between students' emotions and learners' academic performance.

Scope and Limitations

The project was developed to analyze data and draw conclusions about learners' behavior by identifying their emotions during online classes. This project used Face++ API, which offers emotion detection as part of its vision technology. The project focused on the six basic emotions, such as happy, angry, surprised, fear, sadness, and disgust, with the addition of neutral emotion.

Accounts Management Module. This module allows users, such as learners and educators, to access the conference platform after signing in using a Google Account on the website. Users are given a chance to log in with their Google Account. The creation of sessions in online conference platforms is available for both users. In addition, the users who created the conference session are considered the hosts, and they have access to the recorded emotions of all the participants.

Conference Module. This module focuses on the online video conference platform in which users can create/join meetings. This module contains these features: On/Off Camera/Microphone; this feature is available for both the host and the participants, allowing them to turn on or turn off their camera or microphone whenever they want. Virtual Whiteboard: this feature is only available for the host to trigger, but once the host starts a virtual whiteboard, the participants can now see the displayed whiteboard by the host, but they can't interact with it. Chats: allow the participants to privately chat with the host or publicly chat with all the participants. Host Settings: This function is limited only to the host and allows the host to handle the participants. However, this module has limitations in terms of participants' handling. The platform could cater to a maximum of 32 participants in each session to make the project run smoothly without having problems. Also, users are not allowed to exit the browser during an active session because it automatically turns off the camera, and the emotion detection stops.

Emotion Detection Module. This module is the core of the project in which the participants' emotions are monitored by the host. The system can detect happiness, sadness, anger, fear, disgust, surprise, and neutral emotions. While the system can identify the emotion projected, it cannot

determine the reason behind it. This feature is available throughout the entire session. However, problems with lighting, camera positioning, and occlusion affected the emotion detection setting negatively, where it might not be able to detect or return false emotions of the participants who are having these challenges. In addition, emotion detection settings are only available if the participant’s camera is open.

Report Management Module. This module is concentrated on the data analytics of the platform. This saved and displayed the emotion detected for each session. The host can view all the participants' emotions during their hosted sessions through graphs, which also display a summary of the total distributed emotion shown throughout the entire online class or session. The profiles or the names of the students stay anonymous to avoid personal issues and to protect their privacy. The record module also generates general emotion-based suggestions, regardless of the session context.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature

This chapter contains a compilation of relevant literature and studies, such as articles, journals, and conference papers, that helped to strengthen the relevance of the present study. Additionally, it presents a summary of all the literature to fully comprehend the study.

Deep Learning and Application Programming Interface (API) in Facial Expression Recognition

Deep Learning (DL) is an emerging technology; however, much more work and improvement needs to be done. It can be used in different types of applications such as speech and audio recognition, natural language modeling and processing, information retrieval, object and facial recognition, and multi-modal and multitask learning (Deng et al, 2014). DL is a category under Machine Learning, which deals with artificial neural networks that use models to understand and interpret large datasets through pattern-based recognition without having to use precise programming. It is dominantly used in terms of image recognition, natural language processing, and autonomous systems (LeCun et al, 2015).

In the field of speech and audio recognition, recent improvements have posed a threat to automatic speech recognition because it relies on large training datasets, including confidential ones, and demands substantial computational and storage resources. However, advanced deep learning techniques such as deep transfer learning (DTL), which allows high-performance models that require small datasets, federated learning (FL), which enables training with confidential data without dataset possession, and

deep reinforcement learning (DRL), which enhances decision making, solve these issues (Kheddar et al, 2024). Meanwhile, automatic facial recognition made a large leap by using different kinds of algorithms such as Deep Convolutional Neural Network (DCNN), Deep Neural Network (DNN), and Reinforcement Learning (RL), there are large training datasets that can be used for developing an automatic facial recognition, these datasets include illumination, expression, pose variations, and occlusion (Fuad et al, 2021).

Several studies show that using an Application Programming Interface (API) for emotion recognition is as good as building a model from scratch. Some APIs achieved a good rate of accuracy in detecting emotion. Sightcorp achieved 92.11% accuracy compared to other API, which achieved a decent rate of accuracy, such as the Face API, which achieved 67.75% accuracy, while SkyBiometry got 67.58% accuracy (Bhattacharjee et al, 2018). In another study, Face++ API showed almost no dropout during the testing and performance evaluation using more naturalistic facial expression, while FaceReader API demonstrated an outstanding rate of accuracy performance in detecting standardized prototypical facial expression, surpassing human performance, and Azure got the superior advantage in terms of performance among all the API tested (Küntzler et al, 2021). Another study about the performance of API in FER shows that Google's Vision API has excellent performance on detecting processed images, with the exception of Fear and Disgust emotion, while Sighthound is the next in ranking, while Face++ and other API showed nearly equivalent matching scores to Sighthound (Bryant & Howard, 2019). In using emotion detection APIs, the confidence score is one of the most important aspects you need to consider to avoid logging false emotion recognition. One of the companies that is known for computer vision,

Leverage, said that the higher the confidence score of an emotion recognition model, the more accurate its prediction will be. The confidence level threshold that is considered high is 0.70-0.80 or 70 to 80 percent (White, 2025). In AI, it is recommended to use high confidence scores because of legal requirements and consequences of incorrect detection, which is important in emotion detection for educational purposes (Wenkel, 2021).

Facial Expression Recognition is a vast research area that can be used to study human emotions and can be applied in different fields, such as biometric security, intelligent human-computer interaction, robotics, and the medical field (Jan, 2017).

However, FER still has interesting challenges or problems (Mollahosseini et al, 2016). Although significant improvements have been made, there are still existing problems in FER, such as illumination or lighting and pose variations (Dagnes, 2019). However, in using the Region Attention Network (RAN), which compiles and embeds various region features produced by a Backbone Convolutional Neural Network into a compact representation, the problem in terms of occlusion and pose variants had greatly improved (Wang et al., 2020).

Furthermore, another challenge in the field of facial expression recognition systems is the Action Unit (AU) detection or the tiniest facial movement of a facial muscle group (Valstar et al, 2011). Detecting AU has a great impact on the result of evaluation metrics in facial recognition, particularly in FER (Hinduja, 2020).

Smartphones are emerging as the most popular and practical means for connecting to Internet-enabled services. They also have many sensors that can have a powerful effect on the user's cognition. Face recognition is

among the fastest-growing mobile applications, mainly due to its many potential applications. According to Hassan and Elgazzar (2016), face recognition is a very complex process and requires computational resources, which are above the power of even modern mobile devices. As the performance of CNN is getting better, the computational complexity becomes larger. Therefore, CNNs are usually performed on high-performance devices. But for ordinary users, these devices are expensive and cumbersome, meaning that for mobile devices with limited storage, memory, and computation capabilities, traditional CNNs are not easily deployable and generalizable (Guo et al., 2020).

Moreover, issues about privacy in FER, such as data minimisation, fairness, and transparency, have also been widely discussed (Felzmann et al., 2019). The main hindrance to the employment of machine learning in life-altering ventures where sensitive and personal information has to be processed is data protection concerns. Taking data inputs and operating as services puts machine learning applications under the purview of data privacy and data processing laws, which include frameworks such as the European Union General Data Protection Regulation (EU GDPR), United Kingdom General Data Protection Regulation (UK GDPR), and newer legislations like the AI Act and the Data Governance Act. These rules impose constraints on machine learning use cases. If considering the example of the data minimisation principle under the GDPR, it seeks to ensure that the data collected is proportionate to the purpose of the project. This implies that the developers should decide on the amount of data and features needed for every model. This is technically difficult because it is not always possible to know what a model learns from data and how this information will be molded

to obtain the predictions. In addition, restricting features from a training set can result in reducing the model's generalisation, and consequently impacting the general performance of the model (Mestari et al., 2024).

Face++ in Facial Expression Recognition

Megvii Technology developed an AI open platform that is widely used for cloud-based facial recognition, offering deep learning-based image analysis recognition with APIs and SDKs, called Face++. Several studies evaluated the performance of different API (e.g., Azure Face API, Microsoft; Face++, Megvii Technology; FaceReader, Noldus Information Technology) to assess their performance in facial emotion recognition. Face++ shows that 80% of the images are in the correct category with no face detection dropouts on non-standard images. Therefore, this result supports the use of Face++ in situations wherein consistent face localization under various input circumstances is reliable (Küntzler et al., 2021). Another study shows the comparison between eight classifiers and concluded that none were perfect, but cloud APIs such as Face++ performed competitively with the human observers, depending on the emotion (Dupré et al., 2020). Moreover, Wyman and Zhang (2023) further indicate that modern emotion APIs have a high level of true positive rates and are used in psychological interventions, highlighting their increasing reliability as well as their practical significance.

Emotion and Academic Performance Relationship

The face can communicate varied personal information, such as communicative intent, and cognitive appraisal (Cohn et al., 2003). Facial behaviour is one of the most significant bases for understanding human emotion and the intentions of people. It has a growing application in the field of remote education, interactive and entertainment games, and intelligent

transportation systems (Hassanat et al., 2021). While the development of several FER tools has been conducted throughout the many years, the question of the coherence between subjective feelings and facial expressions is still a subject of debate today (Tcherkassof & Dupré, 2020). Humans tend to use their facial expressions to communicate their emotions or feelings, which has an essential value in the visual impact of their facial behaviour, so facial expression recognition algorithms should handle a vast dataset of different kinds of facial expressions (Pereira, 2022). However, posed expressions of emotion are not always the same as genuine feelings. Automatic detection of spontaneous and posed facial expressions has become increasingly important in human behavior analysis as a result of the emerging and increasingly supported theory that facial expressions do not always reflect our genuine feelings (Jia et al., 2021). In addition, a study said that facial expression or emotion alone can't determine student engagement because of so many possible reasons for a person to have that specific emotion (Whitehill et al, 2024).

The "Basic Emotion View" conceives emotions as genetically programmed in humans and is authentically shown through facial expressions. Consequently, emotion recognition is perceiver independent, meaning that the ability to recognize emotions in others does not rely on the individual perceiving those emotions. Instead, emotions can be understood by almost anyone because they are shown in universal ways, like smiling when happy or frowning when sad. On the other hand, the constructivist approach conceives emotions as socially constructed, the emotional meaning of a facial expression being inferred by the perceiver. Hence, emotion recognition is perceiver-dependent (Tcherkassof & Dupré, 2020). To add more, emotions

can be grouped as positive and negative emotions through the valence arousal. According to Paul Ekman's framework, the six basic emotions carry specific valence, like the fear emotion, which carries unpleasant or negative valence. This became the basis of grouping the six basic emotions and the neutral emotion.

In recent years, intelligent emotion recognition has received strong emphasis in computer vision research, relating to human-computer interaction (HCI). Recognizing emotions from facial expressions remains relatively unexplored. Nonetheless, the results of the experiment, engaging a Deep Neural Networks (DNN) model, show that Particle Swarm Optimization (PSO) worked very well and the method achieved a recognition success rate of 98.76% on the MUG dataset and 97.79% on the GEMEP datasets (Vaijayanthi & Arunnehru, 2024).

The ability to understand a person's emotion is a vital skill to initiate and maintain social interactions and interpersonal relationships, and one of the bases of it is the face. However, facial expression can be influenced by a person's body language, which can tell a completely different emotion, contradicting what their facial expression is telling (Day et al., 2024).

Recognizing emotion through facial expression still has a significant problem in computer vision and human-computer interaction, but there is a recent advancement that shows 92.5% accuracy of facial expression detection by using the combination of CNN, VGG16, and VGG19 architectures, which affects the effectiveness of automated emotion recognition (Mishra et al., 2024).

There are many studies that explore the relation between emotion and academic performance. One of the key factors shaping students' academic

achievement is their emotions, which affect their motivation, cognitive processes, and their overall educational experiences (Zeidner et al., 2007). Emotions such as (e.g., enjoyment, anger, anxiety, and boredom that arise during the learning process) can significantly affect students' academic performance because emotions are ubiquitous in academic settings. (Pekrun & Linnenbrink-Garcia, 2012). Emotions and sentimentality are very important because they play an important role in the process of acquiring new knowledge while studying (MacIntyre, 2017). In addition, they also have a positive role in making students attain academic commitment and be more accomplished in school. Students who experience negative emotions, such as being anxious or frustrated, will be more confused while taking an exam than students who experience positive emotions (Richards, 2020). Furthermore, positive academic emotions promoted the learning effect and concluded that positive academic emotions are better than negative academic emotions at improving academic performance (Tan et al., 2021).

Learners' emotional state in learning naturally varies with some factors, such as their cognitive appraisal, situational antecedents, individual learning, and academic achievement, as they reflect differences in learners' perceived control and task value. Additionally, If the activity doesn't offer anything interesting or challenging, students can easily lose interest, which also affects their emotions (Pekrun, 2006). According to Acosta-Gonzaga and Ruiz-Ledesma (2022), this diversity in emotional response influences the learning outcomes of learners. Negative emotion arises when learners have low values of self-efficacy, which reduces effort, while positive emotion promotes motivation and willingness of learners. Study shows that differences in learners' prior knowledge and understanding about a certain topic can

affect their engagement in class. That's why different types of teaching strategies should be explored by instructors. (Munna and Kalam, 2021).

However, according to one definition, emotion regulation is the “extrinsic and intrinsic” process that is responsible for monitoring, evaluating, and modifying emotional reactions, especially their intensive and temporal features, to accomplish one’s goals (Berking et al., 2012). Emotion regulation is highly dependent on the ability to control one’s emotions (Burić et al., 2016). In addition, looking from another perspective, emotion regulation, such as cognitive reappraisal and expressive suppression, is a significant predictor of the academic performance of the students. Moreover, cognitive reappraisal indicates a positive association with academic performance, while expressive suppression indicates a negative association with academic performance (Nadeem et al., 2023).

Synthesis and Relevance of the Study

Emotion has a significant impact on the academic performance of students, especially during online classes. Various studies show that positive emotion can boost academic performance. However, negative emotions can hinder their learning process. Thus, the integration of Facial Expression Recognition (FER) technology to automatically detect emotion in an online conference platform is crucial for monitoring and visualizing the emotions of learners throughout the entire online class session, providing a clearer overview of learners’ emotional patterns during online classes. The advancement of FER technology has greatly increased because of the development of cloud-based API for emotion detection, making it more accessible. These types of technologies can further improve all the existing computer vision technology which can be integrated in different sectors.

These types of API, like Face++, use a pre-trained deep learning model for emotion detection. Although challenges such as lighting conditions, facial angles, and occlusions can still affect detection accuracy, the continuous improvement of these APIs helps enhance the reliability and efficiency of emotion recognition systems.

Overall, integrating Facial Expression Recognition (FER) in online conference platforms can enhance student and educator engagement and improve the overall effectiveness of online education. However, continuous improvements in FER models and emotion recognition accuracy remain necessary for widespread implementation. Future developments should also focus on improving system adaptability across diverse learning environments and varying student demographics.

Table 1. Matrix of Key Findings from Reviewed Literature

Author(s)	Title	Method/Algorithm	Key Findings
Bhattacharjee et al., 2018	Evaluation of API performance in Facial Emotion Recognition	Sightcorp API, Face API, SkyBiometry	Sightcorp achieved 92.11% accuracy, outperforming Face API (67.75%) and SkyBiometry (67.58%) — proving APIs can effectively detect emotions.
Küntzler et al., 2021	Comparative evaluation of cloud-based emotion recognition APIs	Face++, FaceReader, Azure Face API	Face++ showed minimal dropout and strong consistency; FaceReader surpassed human accuracy; Azure outperformed all in overall emotion recognition performance.
Bryant & Howard, 2019	Performance of commercial APIs in FER	Google Vision API, Sighthound, Face++	Google Vision API showed high accuracy except for Fear and Disgust; Sighthound ranked next; Face++ achieved nearly equivalent results — confirming, reliability of commercial APIs.

Table 1.1 Continuation of Matrix of Key Findings from Reviewed Literature

Author(s)	Title	Method/Algorithm	Key Findings
Dupré et al., 2020	Comparative study of eight classifiers including cloud APIs	Face++, Azure, FaceReader	No classifier was perfect, but Face++ performed competitively with human observers depending on emotion type.
Wyman & Zhang, 2023	Reliability of modern emotion APIs in psychological interventions	Various APIs (Face++, Azure, FaceReader)	Emotion APIs demonstrated high true-positive rates, supporting their use in psychological and educational applications.
Fuad et al., 2021	Advancements in Automatic Facial Recognition	DCNN, DNN, Reinforcement Learning (RL)	Deep learning-based algorithms effectively address illumination, pose, and occlusion challenges in emotion recognition.
Wang et al., 2020	Overcoming occlusion and pose issues in FER	Region Attention Network (RAN) with Backbone CNN	RAN significantly improved recognition under occlusion and pose variations, enhancing facial feature robustness.
Vaijayanthi & Arunnehru, 2024	Emotion recognition using Deep Neural Networks	DNN + Particle Swarm Optimization (PSO)	PSO achieved 98.76% accuracy on MUG dataset and 97.79% on GEMEP, proving optimization techniques enhance deep emotion recognition models.
Mishra et al., 2024	Improving accuracy of facial expression detection	CNN, VGG16, VGG19	Combined architectures achieved 92.5% accuracy, increasing the effectiveness of automated emotion recognition.
Kheddar et al., 2024	Improving DL systems for small datasets	Deep Transfer Learning (DTL), Federated Learning (FL), Deep Reinforcement Learning (DRL)	These methods improve emotion recognition by enabling high performance with limited data and protecting privacy.

Conceptual Framework

Figure 1 shows the conceptual framework, which includes all the components, architecture, and methods used to create the online conference platform with emotion detection. This framework visually illustrates the interaction between various parts of the system, such as the frontend interface, the backend services, and the emotional recognition API. It describes the flow of information through the system and user interaction, and provides insight into how the API can be used to recognize, analyze, and interpret emotions over a range of real-time or recorded online sessions.

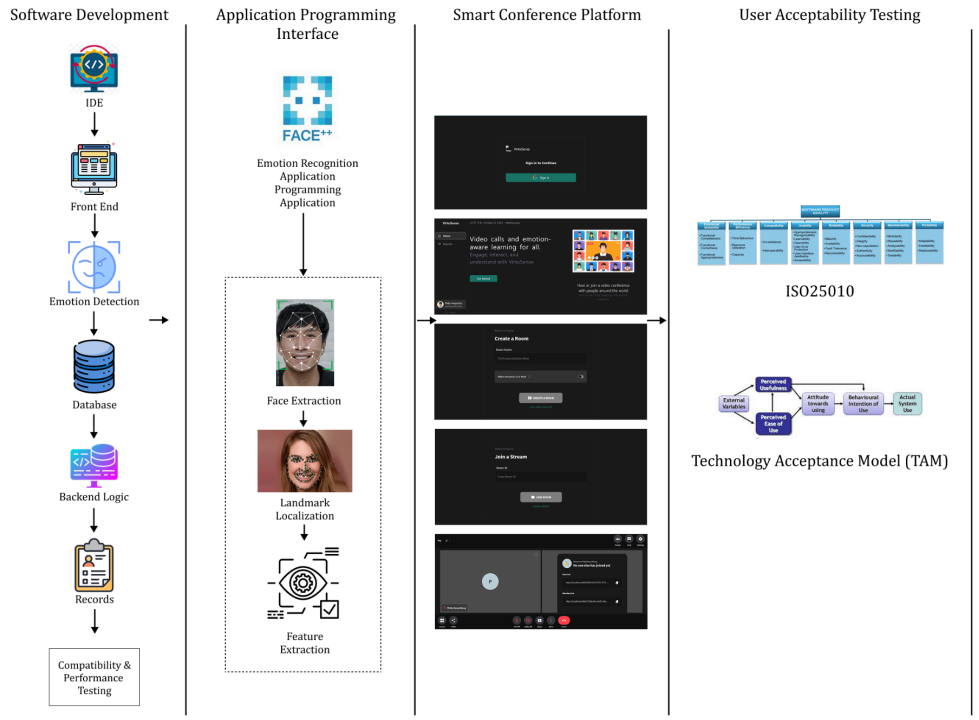

Figure 1. Visual Representation of the Project's Concept

CHAPTER 3

METHODOLOGY

This chapter showcases all the methodologies used during and after the development process of the system. Different kinds of methodologies are used as a guide to systematically develop the application. These include the project design, project development, testing, and evaluation.

Project Design

Figure 2 shows the system architecture of the online conference platform, detailing the interrelationship between each module and its roles within the system. The system is divided into four main components. The Account Module handles user login, providing secure access to the platform. The Conference Module enables essential meeting features such as creating or joining meetings, controlling mic and camera, chatting, using the virtuboard or the system's virtual whiteboard, screen sharing, and host settings. The Emotion Detection Module records facial images, processes them for emotion recognition, and utilizes an emotion recognition API to classify emotions such as happiness, sadness, anger, disgust, fear, surprise, and neutral emotion providing both labels and confidence scores for each participant's emotional state. Finally, the Record Module tracks emotional distribution with visual graphs and generates an automatic suggestion system based on detected emotions, offering insights into emotional trends and it also generated a tabulated student engagement summary which indicates whether the students' have positive, neutral, or negative emotion. Each module is linked to a central database that is responsible for secure data storing, managing, and retrieving. Modules exchange data from that database to interact with

each other. After being processed, the responses are provided to users in meeting logs.

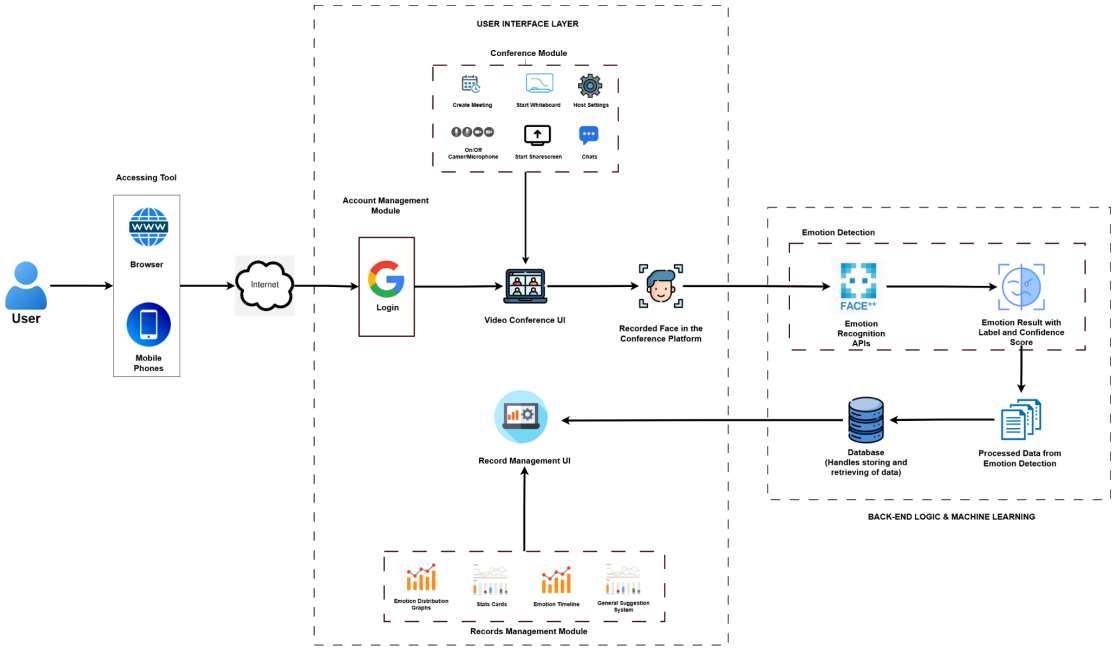

Figure 2. Architectural Flow of Online Conference Platform

Materials

This section showcases the software and hardware used in the development process of the project. It includes platforms and tools such as Integrated Development Environments (IDEs), databases, and programming languages. In addition, it covers the hardware specification required for the best performance, which can ensure that the system runs smoothly and meets all the hardware requirements.

Software

Software is one of the fundamental components in developing a system. The following are software applications that were used for developing the system application, along with their features.

Figma. is solely used to design the user interface and user experience of the project. It is used to create a prototype for the storyboard of the

application. It allows the developers to visualize the application’s interface and flow.

Visual Studio Code. serves as the primary Integrated Development Environment (IDE) for the development of the project. It is used to write, manage, and debug the project's source code efficiently. React is utilized to build the structure and functionality of each user interface component, Tailwind CSS is employed to handle the styling and visual design of the system, and TypeScript is used to enhance code reliability and maintainability by providing strong typing and improved developer tooling support throughout development.

Firebase. is used for backend services such as real-time database, authentication, and cloud storage, ensuring data is securely stored and efficiently managed. It provides a scalable infrastructure that supports dynamic user interactions within the platform. The integration of Firebase simplifies backend development and enhances overall application performance.

Node.js. is utilized to manage server-side operations, handle API requests, and serve as the runtime environment for JavaScript outside the browser. It allows asynchronous processing, which improves system responsiveness and efficiency. Its lightweight nature and wide library support make it a practical choice for building fast and scalable server-side applications.

React. is used as the main frontend library to build the user interface of the video conference platform. It enables the creation of modular, reusable components and ensures efficient rendering through its virtual DOM, resulting in a smooth, responsive, and highly interactive user experience.

TypeScript. is used to design the behaviour of the system with seamless and more organized code writing. TypeScript is used to design the behaviour of the system with seamless and more organized code writing. It also adds type safety, which helps prevent errors during development and improves code maintainability.

Tailwind CSS. is used to style the user interface of the system. It helps to improve the system's aesthetic and to make it more engaging for the target users. It also enables faster development with its utility-first classes, ensuring a consistent and responsive design across devices.

Agora. is used to create the video conferencing functionality of the system, enabling real-time video and audio communication between users. It provides reliable low-latency streaming and interactive features essential for smooth online collaboration.

Github. is used to monitor the changes between files and code during the development process, minimizing the loss of every version of the system. It also provides a secure platform for code hosting, version management, and project documentation.

Vercel. is used for web-hosting and to deploy the system on the internet to make it accessible for users regardless of their distance. It also ensures scalability, fast performance, and continuous deployment for seamless updates.

JMeter. is used to execute stress testing for the system. It pushes the system beyond its normal capacity to see how it behaves under extreme load. This helps identify the breaking point and ensures the system can recover properly after heavy usage.

Lambda. is used for the compatibility testing of the system. This software offers different Windows versions, Android versions, and Browsers.

Hardware

Table 2 shows the specifications of the hardware used in developing the online conference system. The hardware has the following specifications: 8 GB RAM to smoothly multitask during development, such as running code editors, testing environments, and browser tools simultaneously. Intel Pentium Gold for compiling code, running applications, and managing software. Windows 11 provides a user-friendly environment for software development and testing. Intel ® UHD Graphics for graphics and visual analysis. 2.00GHz to ensure responsive performance during day-to-day programming and troubleshooting.

Table 2. Hardware Specifications of Device Used in Development

Hardware	Specifications
RAM	8GB
CPU	Intel Pentium Gold
OS	Windows 11
GPU	Intel (R) UHD Graphics
CLOCKSPEED	2.00GHz

Face++ Data Requirements

Face++ web API is a public service that focuses on computer vision. However, this API doesn’t publicly document the dataset utilized in training their model. The composition, size, demographic distribution, and data sources of the training dataset are not publicly accessible. Face++ was always used as a “black-box,” or it was focused on the output or performance of the model when integrated, rather than the internal technicalities, such as

how it is trained, tuned, and its training dataset. A study in 2022 employed MEGVII Face++ Detect API to create a new dataset containing different facial expressions from *Facebook* profiles of 2,234 Italian people. The pictures contain different kinds of facial expressions, and Face++ was used to categorise each face into emotion categories such as happy, sad, fear, angry, surprise, disgust, and neutral. (Marengo et al, 2022). In an experiment comparing different emotion recognition API, Face++ achieved an overall 81% precision in the detection of standardized facial emotions. (Küntzler et al, 2021).

Table 3. Sensitivity and Precision Scores of Küntzler's Study

MEGVII Face++ Detect	**Sensitivity**	**Precision**
Neutral	0.94	0.70
Joy	0.99	0.96
Anger	0.49	0.84
Disgust	0.89	0.77
Sadness	0.81	0.75
Fear	0.40	0.95
Surprised	0.97	0.71
Average	0.79	0.81

Project Development

The development of the online conference platform follows the Agile Scrum methodology. The project starts with the listing of product backlogs, followed by sprint planning, then sprint backlogs. Daily scrum meetings are done at the end of every sprint.

In this way, the task and functions of the project are turned into a smaller process that involves the entire team for every sprint. This leads to a faster and organised project development.

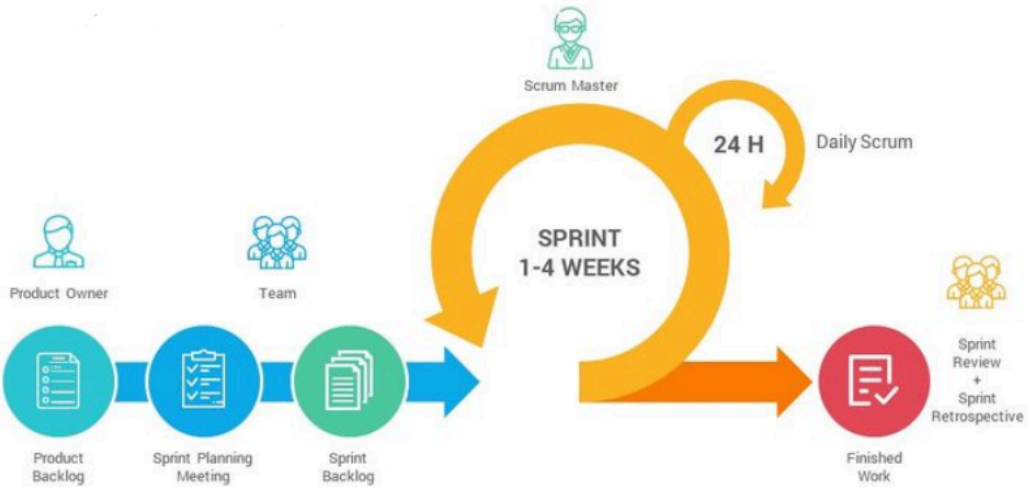

Figure 3. Scrum Sprint Cycle

Product Backlog

In this phase of Scrum, the developers determine all the functions and features of the application and rank them from highest priority to lowest priority. The developers also listed the allotted estimated hours per day for each feature and function.

Table 4 shows the product backlog with its corresponding priority rank. The developers prioritized features that serve as foundational components, those without which other functions could not be implemented. This approach ensured that high-priority features, which directly impact the feasibility of dependent functions, were addressed first in the team's development plan. The product backlog consists of the priority number of a specific task, the user story, the description of the user story, and the estimated hours that need to be allocated to making that function. This structure provides a clear overview of the scope and complexity of each task, allowing the development team to manage time and resources effectively. Moreover, it helps in setting realistic expectations regarding project timelines and deliverables. By organizing tasks based on priority, the team was able to minimize development risks and avoid

potential bottlenecks. Overall, the use of a well-defined product backlog made the development process more planned, organized, and efficient compared to performing random tasks on a daily basis.

Table 4. Product Backlog Overview for Sprint Planning

Priority	User Story	Description	Est. hr/day
1	Emotion Detection	As a user, I want the system to detect and analyze emotion during the online session (excluding the host).	2 hrs
2	Sign In	As a user, I want to sign in to access the system.	1 hr
3	Homepage	As a user, I want to create/join a video conference.	1 hr
4	Video Conference	As a user, I want to be able to participate in a video conference.	3 hrs
5	Records	As a user, I want to access emotion analysis records from past sessions.	1hr
6	On/Off-Mic/ Camera	As a user, I want to control my microphone and camera during a session.	1hr
7	Chats	As a user, I want to view chat messages during a session.	1hr
8	Share Screen	As a user, I want to share my screen during a conference.	1hr
9	Participants	As a host, I want to see a list of participants in the session.	1hr
10	Host Management/Controls	As a host, I want to manage participants (e.g., mute/unmute, remove) and control the session (e.g., allow anyone to join/ask for permission before joining)	1hr
11	Whiteboard	As a host, I want to use a virtual whiteboard during sessions for collaboration and present it to my participants.	1hr

Sprint Planning

The sprint planning is the phase where the project manager communicates with the project team to discuss and assign tasks from the product backlog to the sprint backlog. This shows how the team can focus on what needs to be done first.

System Flowchart

Figure 4 shows the flow of the system on both the host and the participant. The user has the option to either host or participate in a session. After that, the host may access the online video conference interface that includes features such as turning on/off the mic and camera, share screen, chats, host settings, end call, virtuboard that allows collaboration, and on/off camera that enables the classification of emotion. After classifying the emotion, all emotions from the participants are saved in data analytics that may be accessible for record management.

The record management displays the emotion distribution graph in which all the gathered emotion during the session is collected and displayed as a graph. This emotion distribution graph is available in the general summary and in specific session records. Statistical cards are also available in a general record and specific session records displaying total sessions, total participants, the most abundant emotion, and total emotion recorded. The emotion timeline can be seen inside the particular session record, in which it is displayed as a stacked area chart where the user can see the emotions gathered in a span of 5 minutes, and lastly, the general suggestion system, which can generate a general recommendation based on the gathered or collected data in the session, which is the emotion data. In addition, a student engagement graph is displayed in the record section.

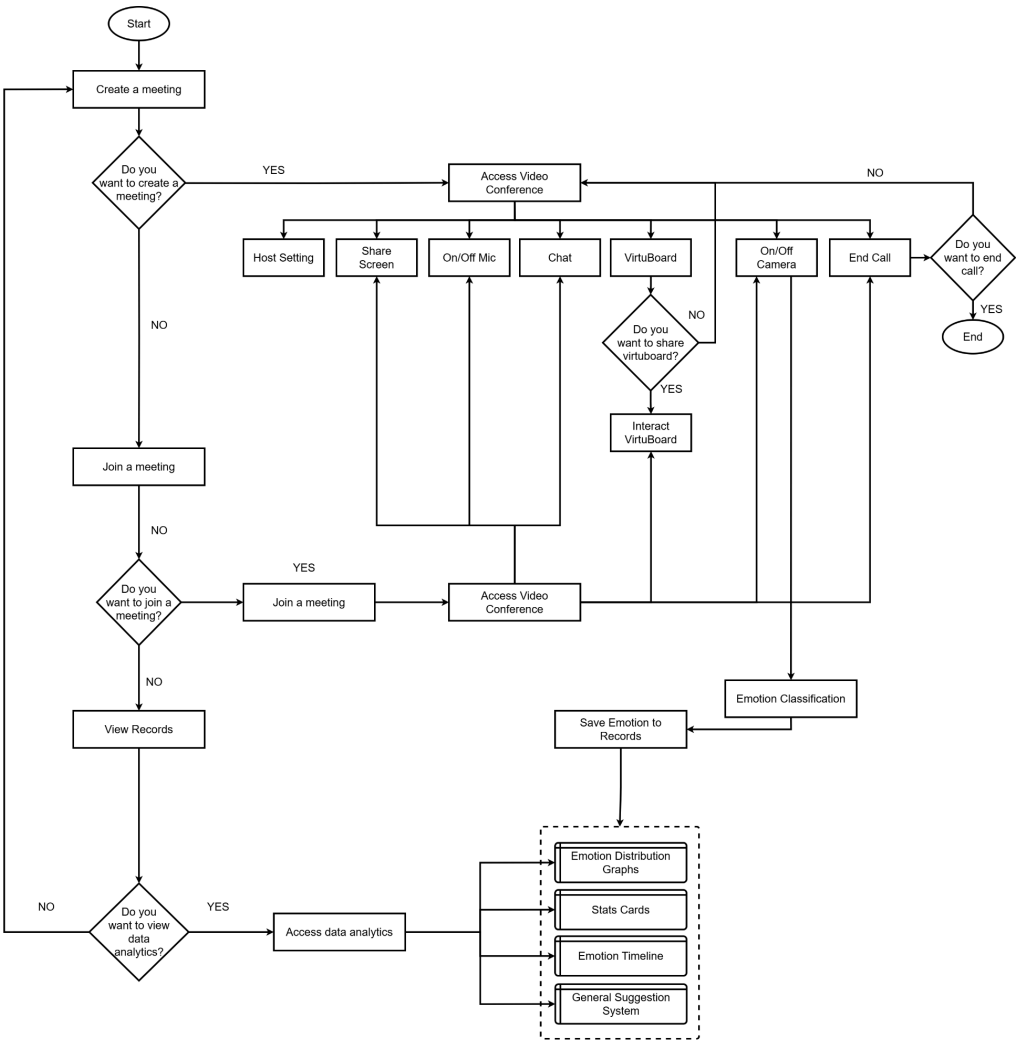

Figure 4. System Flow Diagram for Online Conference

Use Case Diagram

Figure 5 shows the features or functionalities that the project consists of. The two primary users are categorized by the participants and hosts. The tasks that are covered by the host are the Emotion Detection Result: This feature allows the host to detect the emotion of his/her participant and also see the result in the record section of the system. Virtual Whiteboard: This feature authorizes the host to trigger a whiteboard that can be used during the session. Host Settings: This functionality enables the host to manage the participants and modify his/her participants settings.

On the other hand, the Login/Registration, View Records, Video Conference, Share Screen, On/Off Camera/Mic, and Chats are both covered by the participants and the host. Login/Registration: This allows the host and the participant to create their own accounts. View Records: This allows the host to see all the results of the detected emotion of all his participants. Share Screen: This lets the host and the user share their own screen during the session. On/Off Camera/Mic: This feature allows the host and the user to turn their mic or camera on or off. Lastly, Chats: This functionality lets the host and the user interact with each other through private messages or all chats.

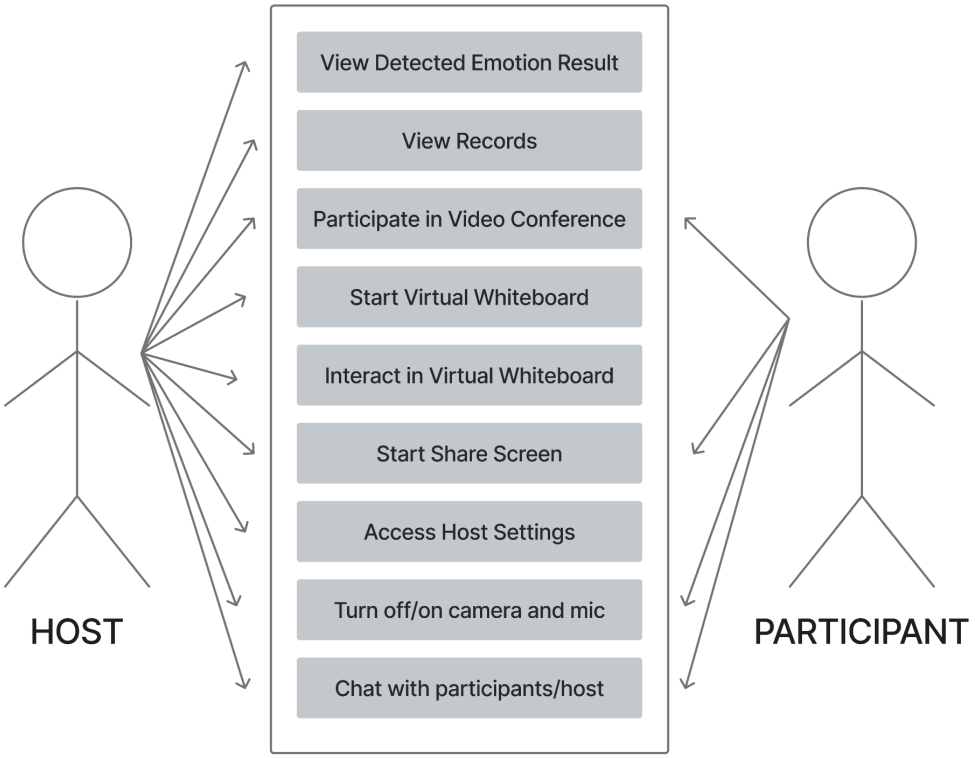

Figure 5. Use Case Representation of User Interactions

Sprint Backlog

Sprint Backlog is the set of carefully selected items taken from the product backlog during sprint planning that the development team commits to

completing within a defined sprint period. It acts as a detailed and actionable plan that breaks down high-level requirements into smaller, manageable tasks, allowing the team to clearly understand what needs to be accomplished and how the sprint goal is achieved. The sprint backlog also supports transparency and accountability by outlining specific responsibilities and timelines for each task.

As presented in Table 5 and 5.1, the sprint backlog provides a comprehensive overview of the prioritized tasks for this development phase. Key items include implementing emotion detection analysis through facial recognition technology, which is central to the system's core functionality, as well as designing a user-friendly login interface to ensure accessibility and ease of use for end users. Additionally, the development of a dashboard layout is included to enable users to view upcoming sessions, notifications, and other relevant information in an organized and intuitive manner.

The sprint consists of several tasks with estimated completion times ranging from 1 to 4 hours per day. This time allocation allows the team to maintain a balanced workload while ensuring steady progress throughout the sprint. Regular monitoring and updates of the sprint backlog help the team track progress and make necessary adjustments, ultimately contributing to the successful and timely completion of sprint objectives. In addition, breaking tasks into manageable time frames helps reduce developer fatigue and improves focus on specific deliverables. This approach also enables the team to quickly identify delays or issues during the sprint. Daily task reviews promote accountability and encourage consistent productivity among team members. Furthermore, effective time estimation supports better sprint planning and improves the accuracy of future development schedules.

Table 5. List of Tasks in the Sprint Backlog

Priority	Feature / User Story	Task Group(s) Covered	Est. Hrs/Day
1	As a user, I want the system to detect and analyze emotion during the online session (excluding the host).	Integrate Emotion Detection API Write unit test for emotion detection	2 hrs
2	As a user, I want to register and login to access the system.	Design a login and registration form Write the front-end and back-end code of the form Test login and registration form	1 hr
3	As a user, I want to create/join a video conference.	Layout dashboard/homepage and user navigation flow Develop backend logic and integrate front-end components Test conference creation/joining functions	1 hr
4	As a user, I want to be able to participate in a video conference.	Design a video conference platform interface Develop and write the code for the conference platform design Write unit testing of the conference platform page	3 hrs
5	As a user, I want to control my microphone and camera during a session.	Design UI for mic and camera toggles Develop and write the code to implement media controls Write unit testing of the media toggle features	1 hr
6	As a user, I want to view chat messages during a session.	Design a real-time chat interface for the session Develop and write the code for the chat system Write unit testing of the real-time chat module	1 hr

Table 5.1 Continuation of List of Tasks in the Sprint Backlog

Priority	Feature / User Story	Task Group(s) Covered	Est. Hrs/Day
7	As a host, I want to see a list of participants in the session.	Design the participant list interface for the host Develop and write the code for displaying participants and their statuses Write unit testing of the participants panel	1 hr
8	As a host, I want to manage participants (e.g., mute/unmute, remove) and control the session (e.g., allow anyone to join/ask for permission before joining)	Design the host control panel for managing participants and permissions Develop and write the code for participant management and session control Write unit testing of the host management features	1 hr
9	As a host, I want to use a virtual whiteboard during sessions for collaboration	Design the collaborative whiteboard interface Develop and write the code to integrate and sync the whiteboard Write unit testing of the whiteboard collaboration feature	1 hr

Face++ Process Flow

Face++ API was used in this project for processing the emotion detection module. Face++ uses its own deep learning framework, which is the MegEngine. Figure 6 shows the process of Face++, where it starts once the system sends a frame captured in the online conference platform. After receiving the frame, Face++ first extracts the face detected in that frame, and after extracting, the API starts detecting all the landmarks in the detected face for the basis of Facial Expression Recognition. Once all the landmarks have been detected, it determines the emotion shown in the face.

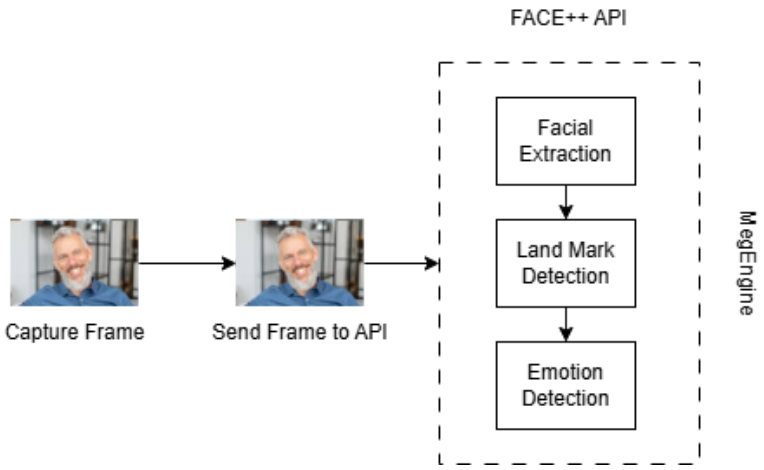

Figure 6. Face++ API Process

Agora Process Flow

Figure 7 shows the process flow of the Agora for streaming conferencing. It started by setting up the application that integrates Agora Video Conferencing. After setting up the project, the Agora SDK is initialized, and then Agora handles the server-side process in the video streaming. It includes rendering video streams in which the user's local video stream is captured, and remote user connection, in which, when the users join the same channel, their video streams are available. Lastly, the audio and video transmission manages capturing, encoding, transmitting, decoding, and rendering audio and video streams in real-time.

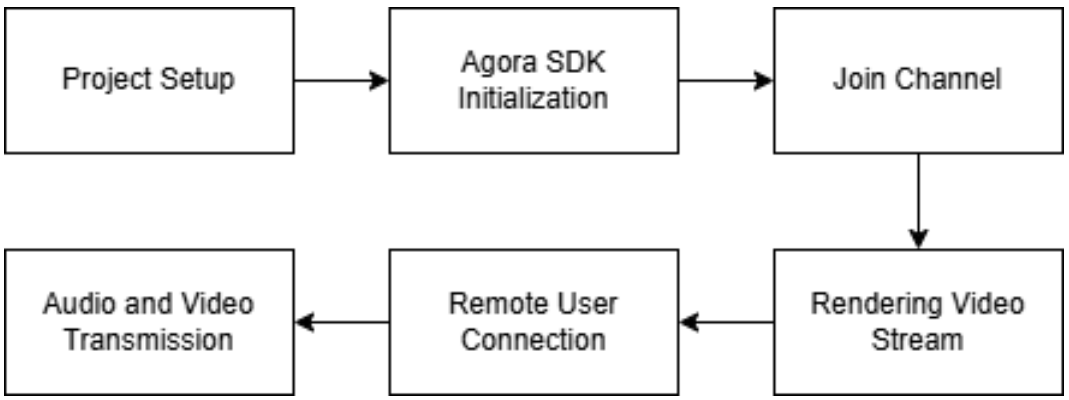

Figure 7. Agora Process Flow

Software Testing

Unit Testing

Unit Testing is conducted in every module of the system, such as the Accounts Management Module, Conference Module, Emotion Detection Module, and Records Management Module. This ensures that every function inside each module is working as expected. It also ensures that each module can work independently without any issues and does not depend on other modules. The developers use a pass or fail evaluation method for fast identification of issues inside each module.

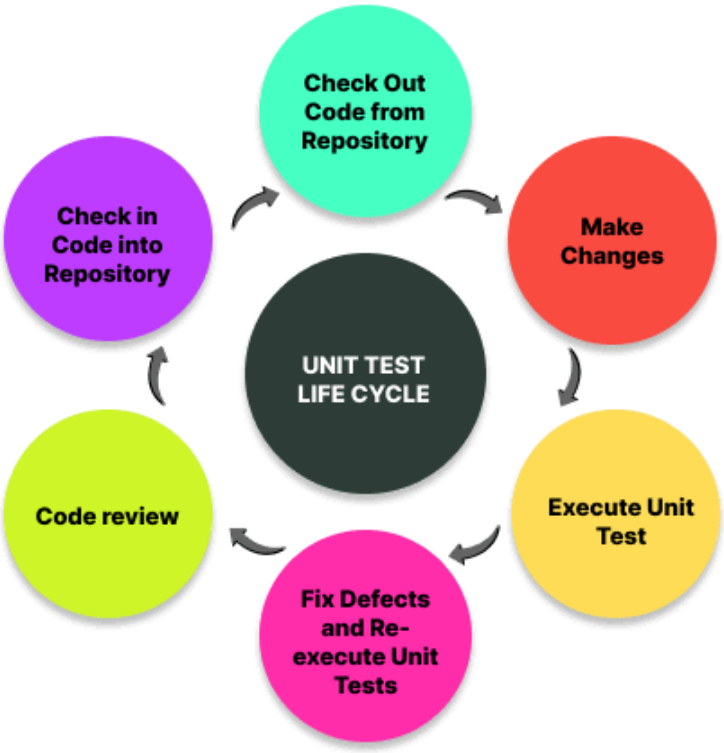

Figure 8. Unit Test Life Cycle

Integration Testing

Integration Testing was used to test if all the modules work together seamlessly. This ensures that the Account Management Module, Conference

Module, Emotion Detection Module, and Records Management Module run smoothly as a group. This testing also used a pass or fail evaluation method for a quick identification of errors, such as data discrepancies and communication between all the modules of the system. This testing phase allows the developer to see if all the modules work when combined and if the system works as intended.

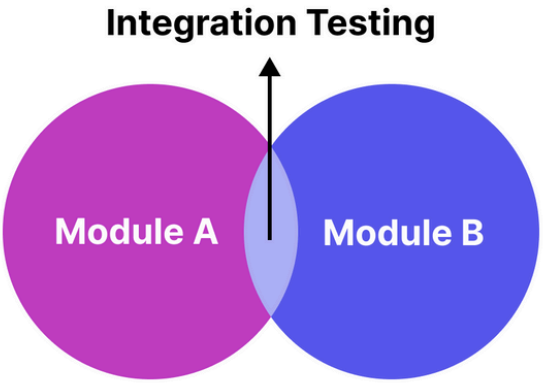

Figure 9. Process of Integration Testing

Acceptance Testing

This testing phase is the final step of the testing process in which the developer tests the performance of the system in a real-world activity. In this study, the developer used the Technology Acceptance Model (TAM) to assess the user acceptance of the system and ISO25010 to test the system's quality, usability, reliability, and functionality.

Compatibility Testing

Figure 10 shows the compatibility testing that was used to test the system's overall performance in different browser platforms, Android versions, and Windows versions. The browsers that were used in this testing are Chrome, Safari, Edge, Firefox, and Opera. These browsers are the most used browsers among different kinds of devices and users. Android versions that

were tested are Android 13, 14, 15, 16, and the Windows versions that were tested are Windows 7, 8, 10, and 11. Through this testing, the developers can identify and fix performance issues across different devices and browsers.

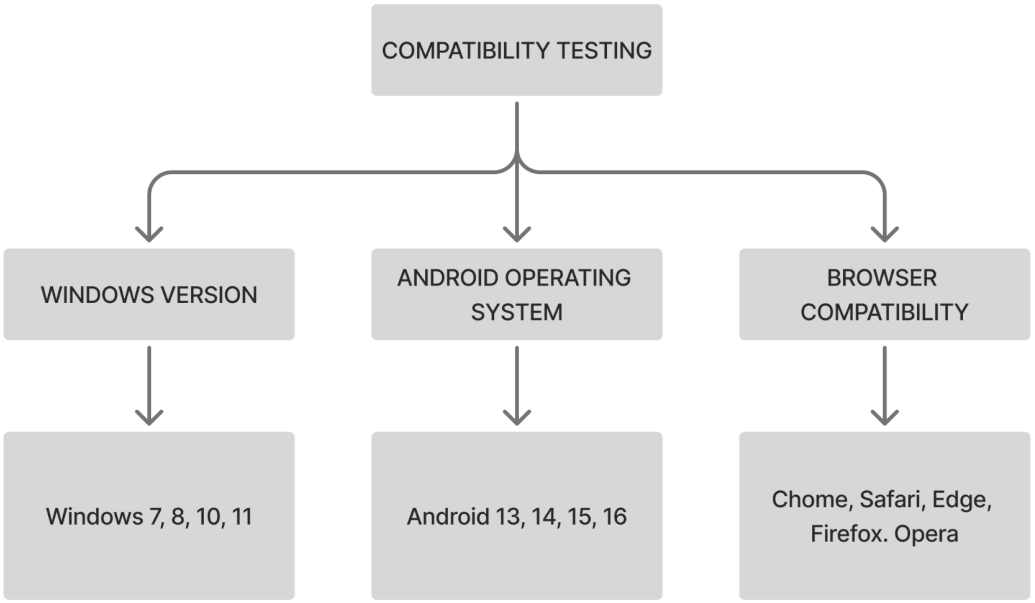

Figure 10. Classification of Compatibility Testing

Performance Testing

Figure 11 illustrates the stress testing conducted during the performance testing phase. Apache JMeter was used to evaluate the system's capabilities and limitations under varying load conditions. The testing focused on the server side of the system, where the team measured server response time while gradually increasing the number of simulated users until the system became slow and eventually inaccessible. Although the stress testing was not conducted in a real-world environment, it provided valuable insights into how the system performs under different load scenarios. The results helped identify potential bottlenecks and performance thresholds of the server. Furthermore, these findings can serve as a basis for future system optimization and scalability improvements. This ensures that the system can better accommodate increased user demand in future deployments.

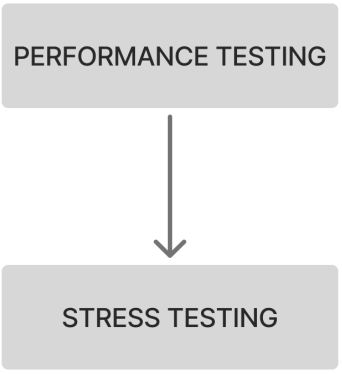

Figure 11. Performance Testing Strategy

Population of the Study

A total of 54 respondents participated to evaluate the system. This includes IT professionals and educators. IT experts focused on the system's functionalities, identifying bugs and errors, and securing the smooth flow of data and emotion detection features. Educators assess the system's usability and helpfulness in the real online class environment.

Sampling Design

The developers used purposive sampling as their sampling design. It is used when the team purposely picks respondents with unique characteristics in the community. In this case, three (3) IT professionals and fifty-one (51) Educators were picked to evaluate the application's suitability in the real world, specifically in the education sector.

Data Collection Instrument

In assessing the system's usability and to give feedback about the system, a two-set survey questionnaire was formed in accordance with the ISO25010 and TAM. The first set of surveys focuses on collecting data about the project's objectives that follow the criteria and standards of ISO25010 software evaluation standard, which aims to answer the system's functionality, reliability, usability, efficiency, maintainability, and portability. Figure 12 shows

that the software product quality was based on the ISO25010 standards. It was used as a standard for the systems evaluation procedure. It focuses on the functionality, reliability, usability, efficiency, maintainability, and portability of the Online Conference Platform. Every module and feature of the system was tested based on these standards. The second set of the survey questionnaire focused on the usefulness and fulfilment of the user requirements, and it was analyzed using TAM, which includes Perceived Usefulness, Perceived Ease of Use, Social Influence, Information Awareness, Perceived Cost, and Willingness to Adopt. These categories determined how the respondents viewed the system from their own perspectives, and the result helped the team to assess the real-world usefulness of the online conference platform.

Table 6 presents the reliability analysis of the Technology Acceptance Model (TAM) constructs using Cronbach's Alpha. Each category shows a corresponding Cronbach's Alpha value along with its verbal interpretation based on standardized thresholds. The overall Cronbach's Alpha value of 0.965 indicates an excellent level of internal consistency among the survey items. This result implies that the measurement instrument used in the study is highly reliable and produces consistent responses across different respondents. These findings strengthen the validity of the collected data and support the credibility of the evaluation results. Furthermore, the consistency of responses indicates that the respondents clearly understood the survey statements. This suggests that the developed application is user-friendly and well-aligned with user expectations. Overall, the high reliability scores demonstrate that the system is suitable for real-world application and further research use.

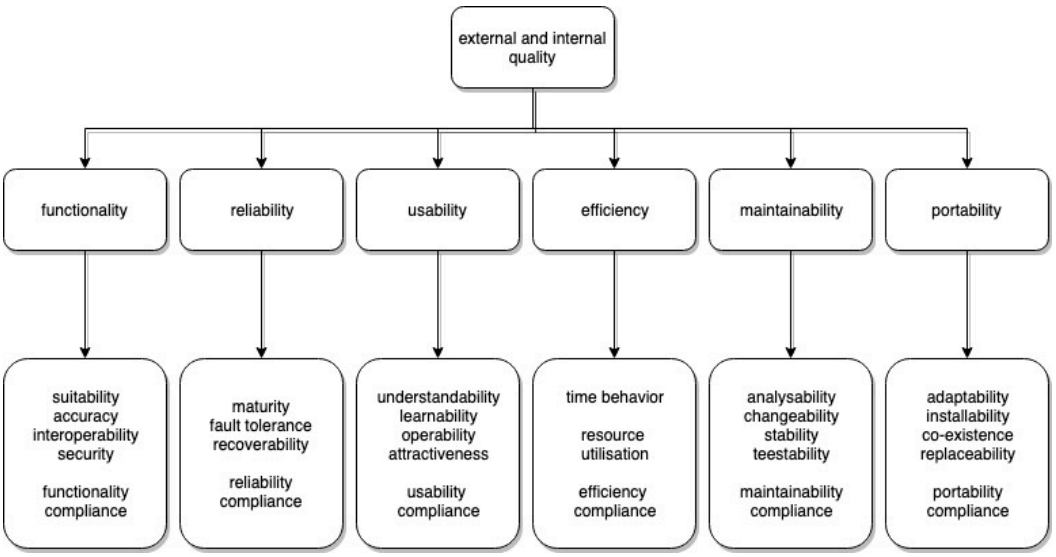

Figure 12. Quality Characteristics of Software Products

Table 6. Cronbach's Alpha Reliability Test Method

Tam Categories	Cronbach's Alpha	Verbal Interpretation
Perceived Usefulness	0.824	Good
Perceived Ease of Use	0.873	Good
Social Influence	0.869	Good
Information Awareness	0.809	Good
Perceived Cost	0.892	Good
Willingness to Adopt	0.842	Good
Overall Result	**0.965**	**Excellent**

Legend: above 0.90 = Excellent, 0.80-0.89 = Good, 0.70-0.79 = Acceptable, 0.60-0.69 = Questionable, 0.50-0.59 = Poor, below 0.50 = Unacceptable

Technology Acceptance Model

Figure 13 shows the Technology Acceptance Model (TAM). This model helped the target users understand and accept the developed software. TAM was used as a guide or standard for the participants' answers in the survey questionnaire. It focuses on the Perceived Usefulness and Perceived Ease of Use of the software's features, such as emotion detection and the online conference.

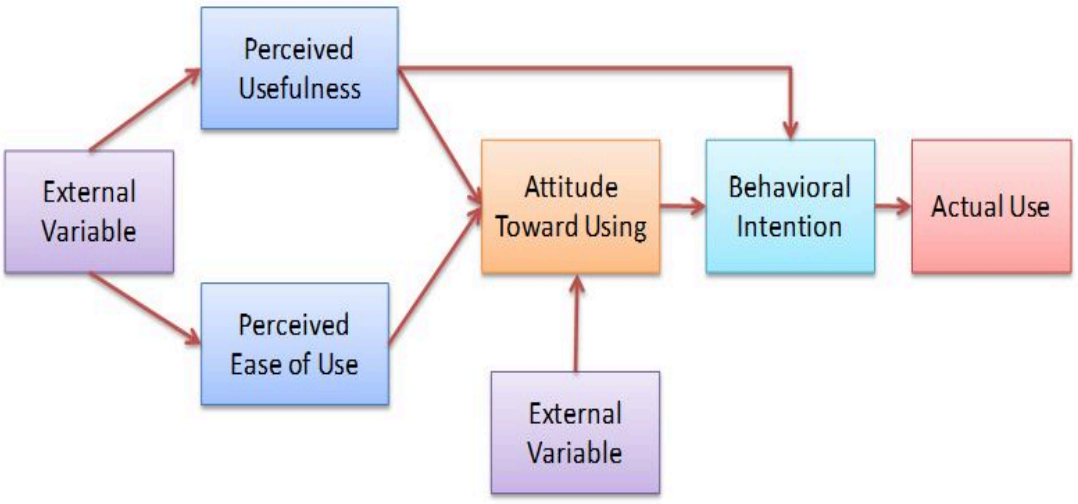

Figure 13. User Acceptance of Technology Framework

Statistical Treatment

The two sets of surveys that have been used in the data collection used a 4-point Likert Scale for both TAM and ISO25010, and it was disseminated to the participants, who were purposely selected through purposive sampling. The survey was answered as four (4) - Strongly Agree, three (3) - Agree, two (2) Disagree, and one (1) - Strongly Disagree. Table 7 demonstrates the point range and its interpretation. This type of approach eliminates neutral answers from the respondents, allowing the survey to achieve more reliable answers from the participants. As a result, the responses more accurately reflect the participants' evaluations of the system.

Table 7. 4-Point Likert Scale for TAM and ISO 25010 Survey

Point	Scale Range	Verbal Interpretation
4	3.5-4.00	Strongly Agree
3	2.5-3.49	Agree
2	1.5-2.49	Disagree
1	1.00-1.49	Strongly Disagree

Ethical Consideration

This capstone project deals with the information of its users, which is why it raises ethical considerations. This project involves the collection of student data, which was only utilised for the development of the software, specifically for testing and training of the system. All personal data from the students was treated confidentially, saved securely, and not published. Data collection was explained prior to questioning, and only if the target people gave their consent were they considered.

CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter discusses the results of the created system, which answers the methods used in the development process of this project.

Results by Objectives of the Project

Developed a Responsive Web Online Conference Application

Account Management Module

Figures 14 and 15 show the sign-in page and homepage of the online conference platform. The sign-in page offers Google Account sign-in for safer and easier registration of users. On the homepage, it displays the “Get Started” button, which leads to the creation of the online meeting session. This interface is necessary to allow users to access the main page of VirtuSense and use all the other features of the online conference platform. The users can also view and read the “Terms of Services and Privacy Policies” and agree by checking the checkbox under the sign-in button.

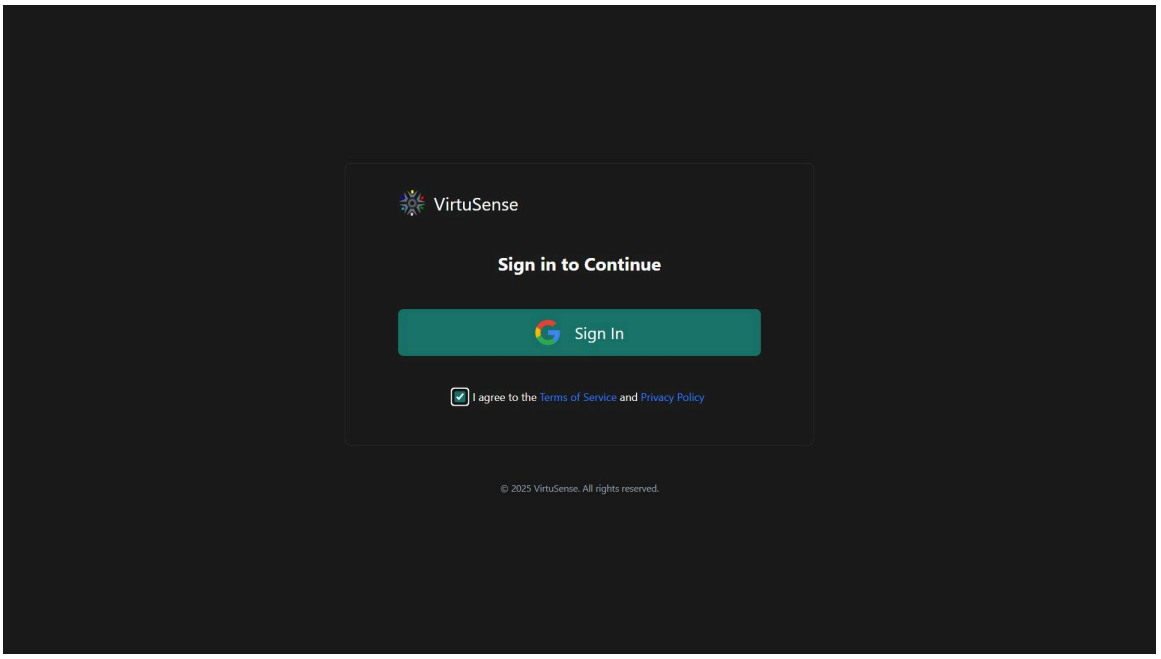

Figure 14. Sign-In Page of VirtuSense

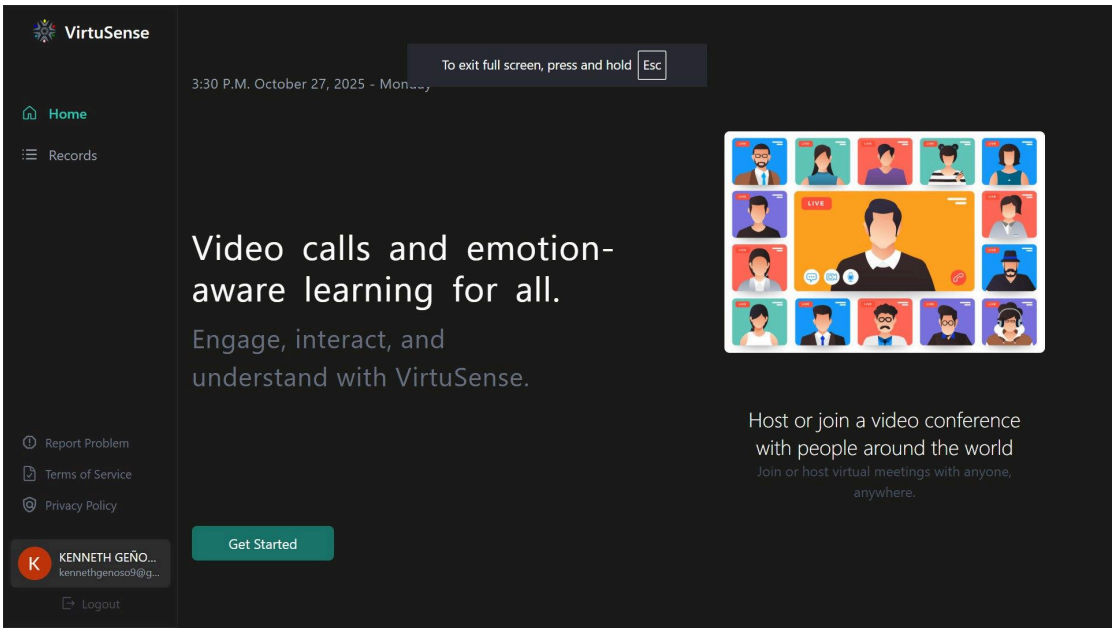

Figure 15. Home Page of VirtuSense

Conference Module

Figures 16 and 17 illustrate the creation page of online meeting sessions in which the users are required to put a name or title in the session they are making, what it is all about, or anything else they want to put in. In addition, the users were given an option to make every participant who joins a co-host. Also, inside the creation page and after the user fills in the required name of the session, the button “Create a Room” directs them to another process page in which the joining links of the online room are displayed, whether the joining link for participants or a co-host. Lastly, the “Start Room” button redirects them to the main conference platform interface.

However, for the option of joining a meeting, when the user has just joined an existing room, they just need to click the “Join with a Room ID” text, and it directs them to the participant joining process page, which requires them to enter the room ID provided by the host. Then they are able to join the conference if the host allows their admission after clicking the “Join Room” button, which is shown in Figure 18. Then, after the user joins the conference

session, they are allowed to use all the features and access available in their assigned roles. In addition, there is another way of joining an existing conference room by having the invitation link from the host of the meeting. After the invitation link was sent by the host, the participant can just click it and if they are already logged in it will redirect them to the joining page of the virtual room.

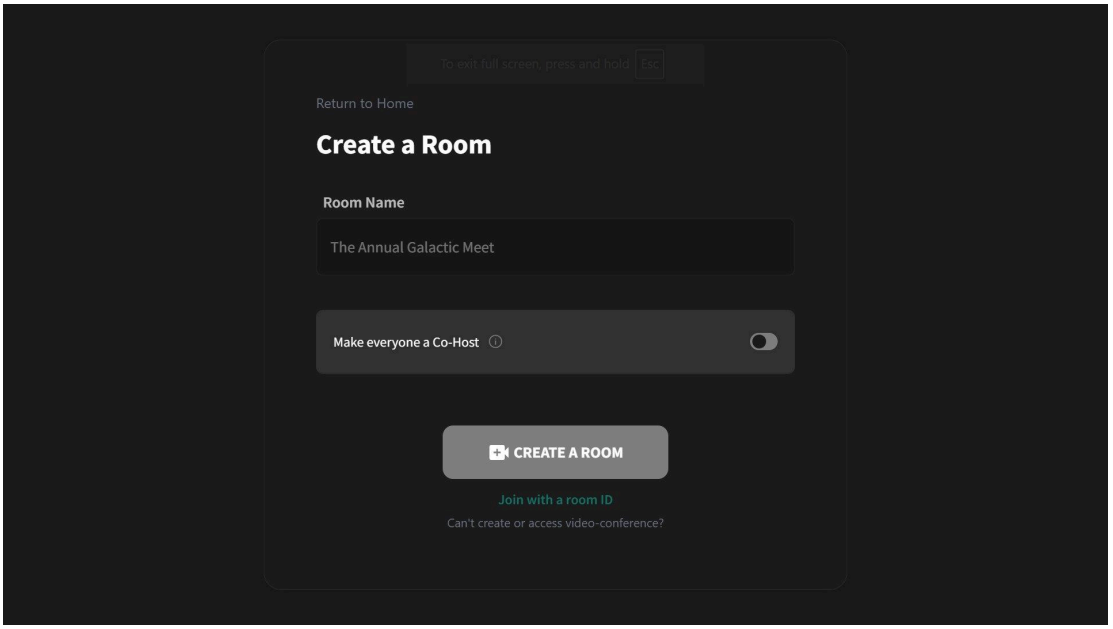

Figure 16. Online Room Creation of VirtuSense

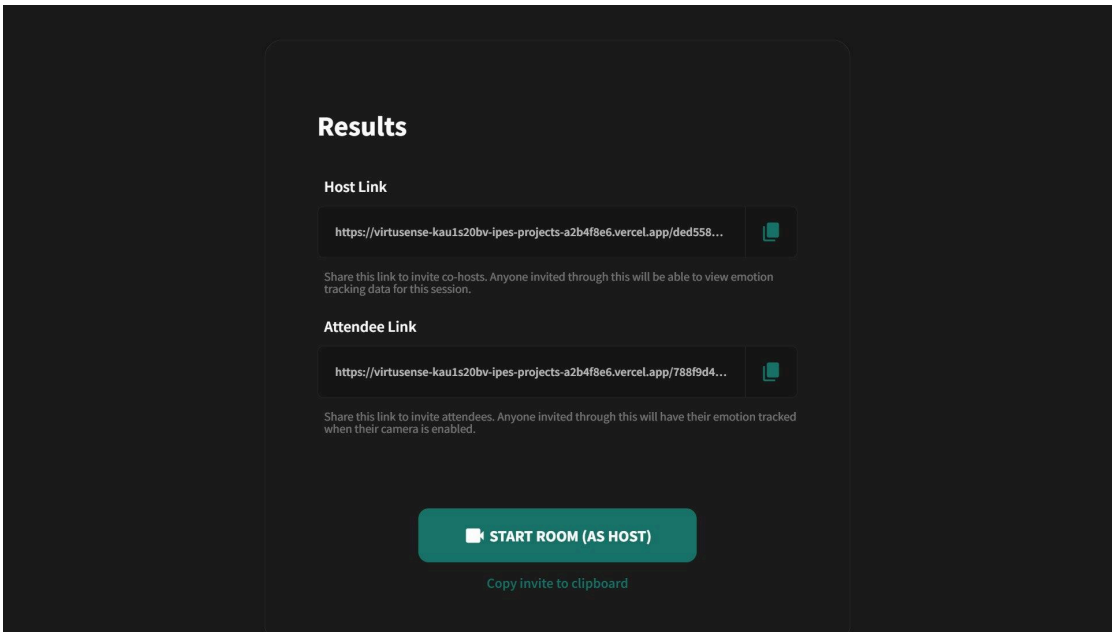

Figure 17. Online Room Creation Links of VirtuSense

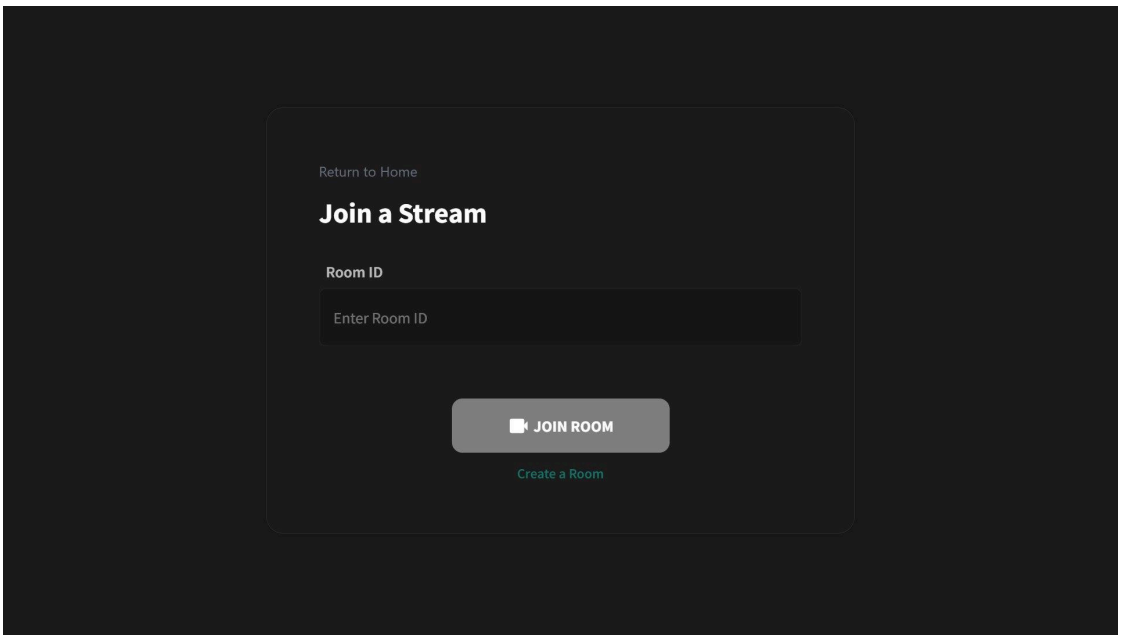

Figure 18. VirtuSense Joining a Room Page

Figure 19 shows the main interface of the online conference room. This page caters to the most important feature of the project, which is the emotion detection, which works if the participants have their cameras on. Also, on this page, the users can see all the features available, and they can use them to connect with their participants, such as the on/off camera and microphone, sharescreen, public and private chats, host settings, virtual whiteboard, page layout, and the invitation link.

Interactive features such as sharing screen allow the user to display their screen seamlessly, also if the display being shared is a video, the audio is still available. In using the virtual whiteboard, only the host can trigger and interact with it. It will just be displayed among the participants of the conference. This whiteboard aims to make the online environment more engaging. For the chats, the users can interact with other users privately, so that only a specific user can see the messages, or publicly, so that everyone in the virtual room can see the messages. The invitation link served as the ticket to the meeting room.

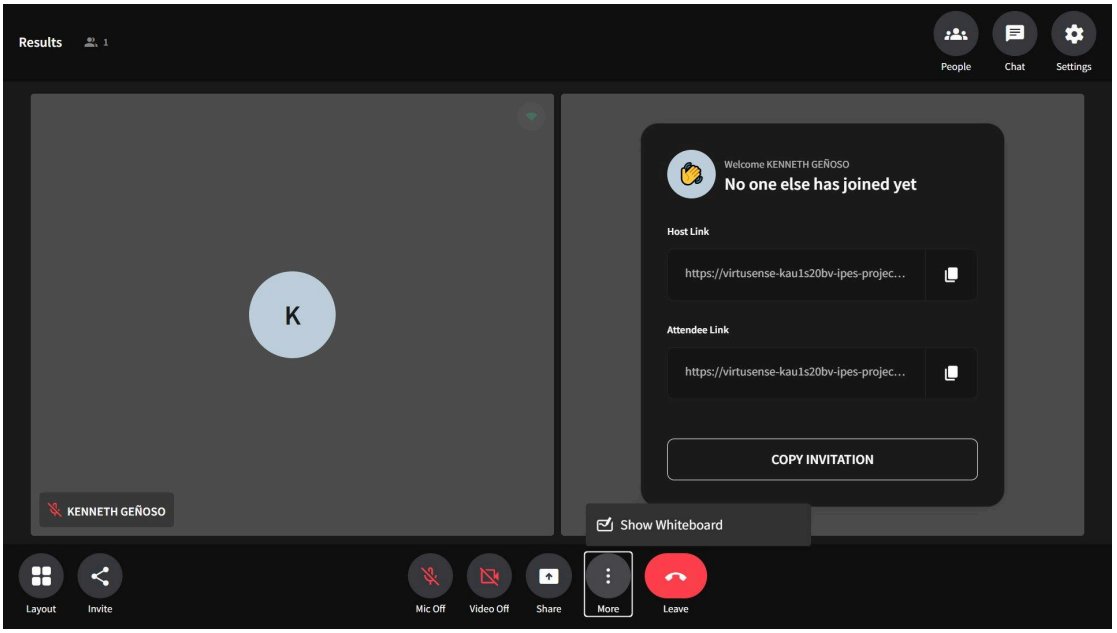

Figure 19. Online Conference Room of VirtuSense

Report Management Module

Figure 20 shows the record section of the online conference platform in which the host can see the accumulated detection among his/her participants during the online session. The system provided data analytics in this section, such as the general statistics card, which displays the total sessions, total participants, average session duration, and total emotions recorded. The general record section also shows general reminders and suggestions, an emotion distribution graph, an engagement assessment graph, and general recommendations.

The record section has another part where the host can view the data inside a particular session, and inside that page, the host can see the number of participants in that session, duration, emotion recorded, and the most frequent emotion displayed during the session. In addition, there is also an emotion timeline graph, a session engagement and emotion distribution graph, and session insights based on the gathered data. These data are important for the instructors to study the emotion patterns during each

session, and they can also be used to tailor their teaching practices based on the given data in the report management module.

These graphs shown in the record section were based on the accumulated data in each session, such as the engagement assessment table in which the six (6) emotions were categorized as positive, negative, or neutral emotions.

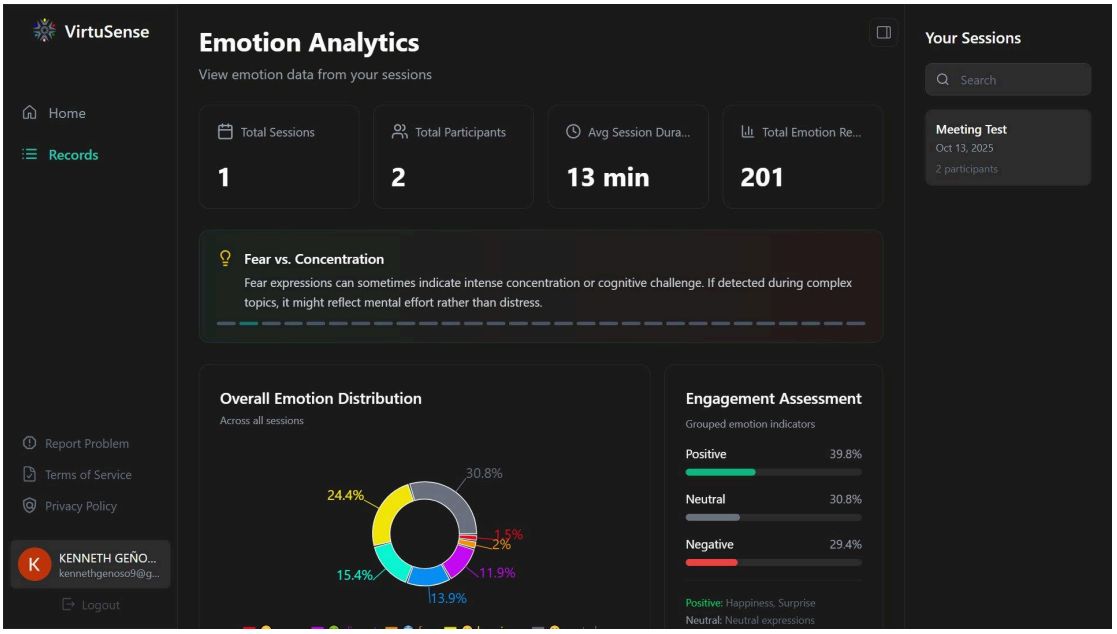

Figure 20. Record Section of VirtuSense

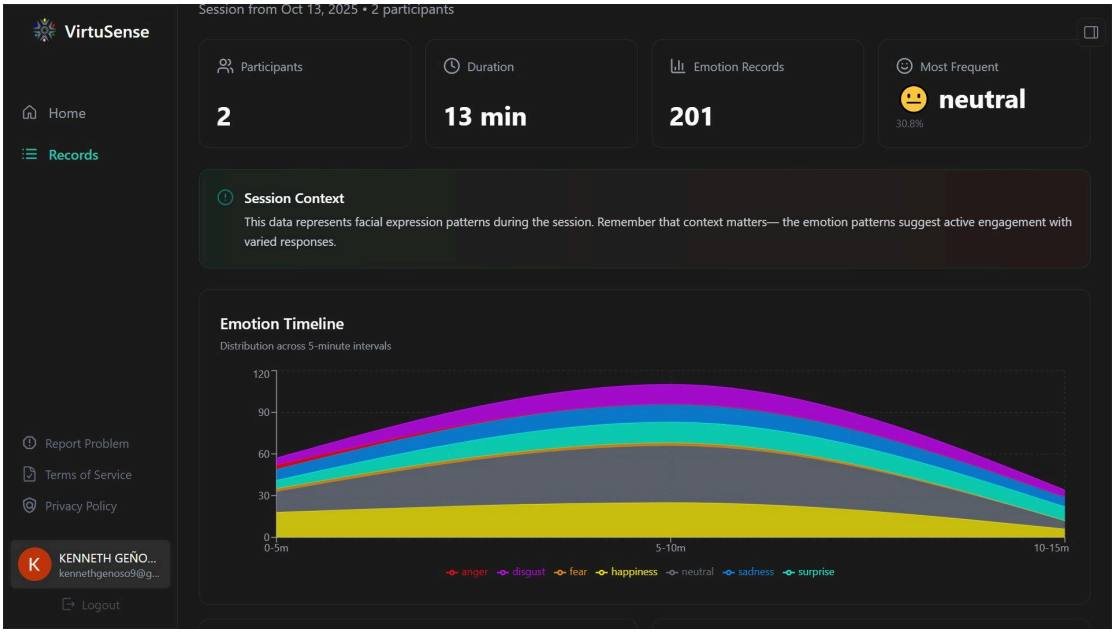

Figure 20.1. Records Section of VirtuSense

Integration of Emotion Detection Technology through Facial Expression Recognition in the Online Conference Platform

Emotion Detection Module

Figure 21 illustrates the emotion detection during online conference sessions. Although the system's emotion detection didn't really have emotion logs during the conference to keep the participants' identities anonymous for data privacy, the team made temporary console emotion logs to demonstrate that the emotion detection is working. These logs were saved or transported to the database of the system and were analyzed and displayed in the record section of the web application. The emotions were saved if the system detects a change in the facial expression of the participants, and it passes the required confidence level, which is 75%.

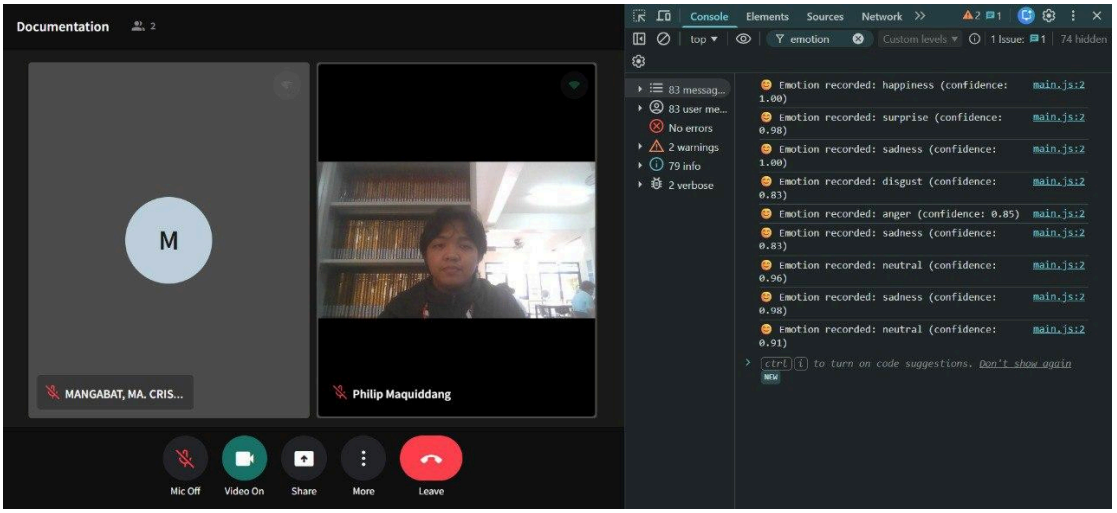

Figure 21. Emotion Detection Temporary Logs

Figure 22 shows the emotion logs being absorbed in the database; the data being transported in the backend includes the emotion detected, timestamp, and confidence level. This data is necessary for the report management module since the data analytics were based on the collected data of the system during each session.

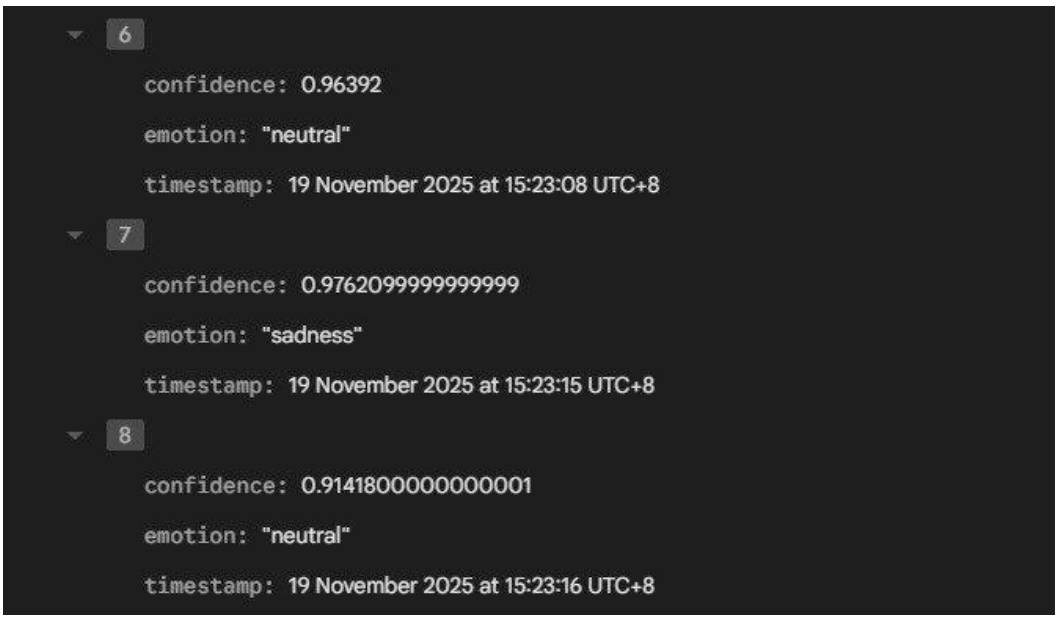

Figure 22. Emotion Detection Logs in Database

Implementation of the System in a Real Online Class Session

Figure 23 shows the interface of VirtuSense in a real online class session. This session is titled “CMSC 307” with 32 participants at its peak. The students are required to open their camera to trigger the emotion detection of the system. While some students have not opened their cameras, all the available faces in the conference were successfully detected by Face++ API and were recorded in the record section of the platform.

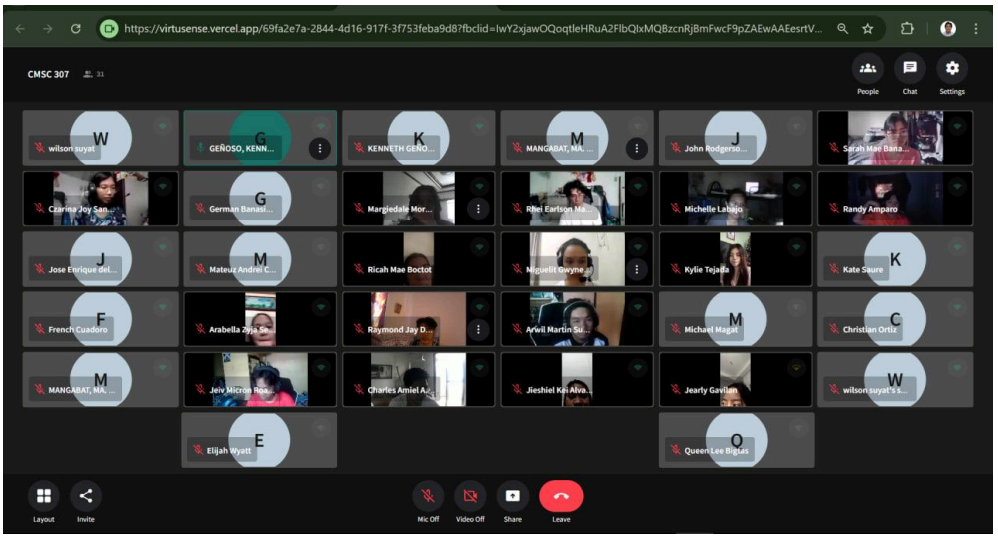

Figure 23. Real-Life Online Class Session

Figure 24 shows the interface of the online classroom while on mobile. The figure shows that all the tiles or frames are visible, and also their camera and microphone settings, with indicators whether they are on or off. This class is the same class as Figure 23.

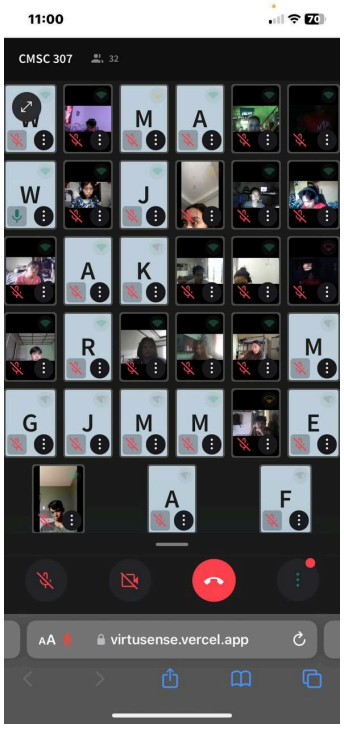

Figure 24. Mobile Interface of VirtuSense

Figure 25 demonstrates one of the features of VirtuSense, which is screen sharing. This photo was taken during the presentation of one of the students in the class “System Integration and Administration”.

Figure 25. Screen Sharing in VirtuSense

Figure 26 illustrates the statistics card and emotion timeline graph, which is based on the gathered data in the class “CMSC 307”. The statistical cards show that there are 25 detected participants’ emotions, and their duration is thirty-two (32) minutes. It also displayed the emotion records, which indicated that there were 222 emotions detected during the session. Also, the cards show the most frequent emotion detected in the class, which is “neutral”. Lastly, the emotion timeline graph shows emotion patterns during the class. In 0-10 minutes, the system detected zero emotion since the participants had their camera off. However, in 10-15 minutes, the system started to detect some emotions such as happiness, surprise, and sadness.

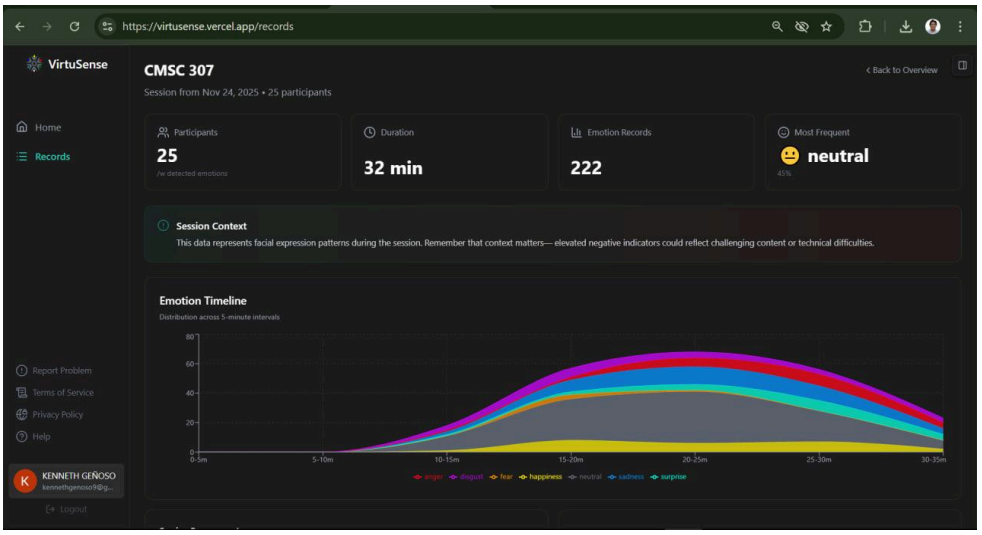

Figure 26. Statistical Card and Emotion Timeline Graph

Figure 27 displayed the session engagement, session emotion distribution, session insights graphs, and the peak moments. Session engagement categorized each emotion into positive, neutral, and negative. Happiness and surprise emotion is categorized as positive, neutral emotion is categorized as neutral, and all the remaining emotions are categorized as negative. The session engagement in the class “CMSC 307” shows neutral emotion with a 10% margin, with negative emotion. This result was used as

the basis for the session insights, which displayed low positive engagement since the session engagement displayed that there were only 19% positive emotions detected during that class. The information shown in the session insight can vary depending on the emotions shown by the students during the conference or class. These insights can help the instructors to visualize how their teaching strategies affect their student's emotion. Another graph is the session emotion distribution, which displays neutral emotion as dominant over all other emotions detected. Lastly, the peak moments information, which displays the timeline where most emotions are detected and which emotion is the most detected in that timeline.

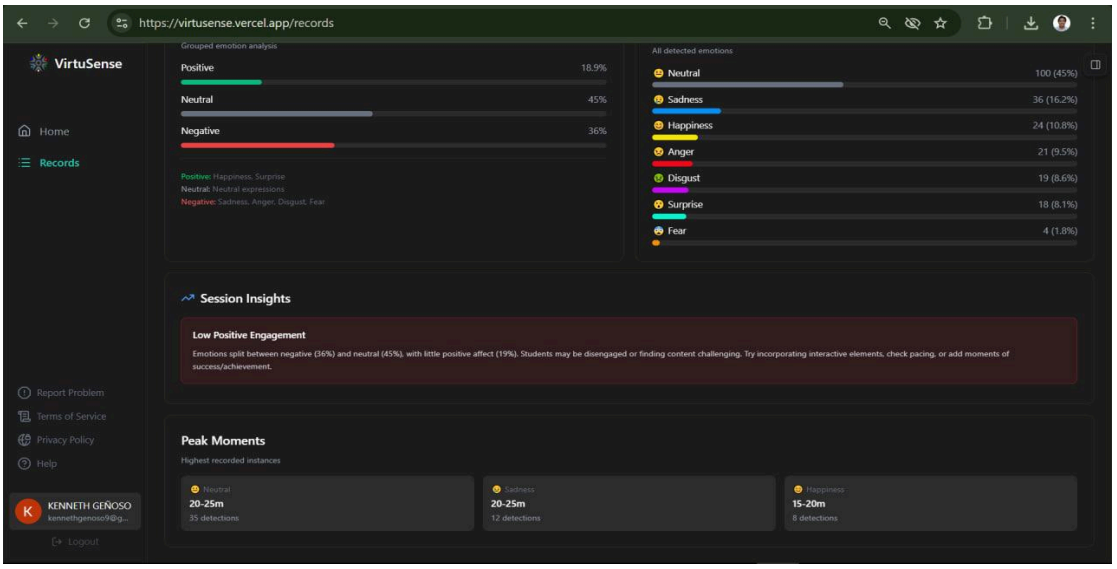

Figure 27. Session Engagement and Emotional Analysis Report

Figure 28 shows the real-time emotion summary notification. This feature helps the host or the instructor to monitor the emotional environment during the online session. This notification can only be seen by the host, and with this feature, the host can easily adjust their strategies according to the information shown by the system. It notifies the host every 5 minutes, and it determines whether the collected emotions from the participants are neutral, positive, or negative.

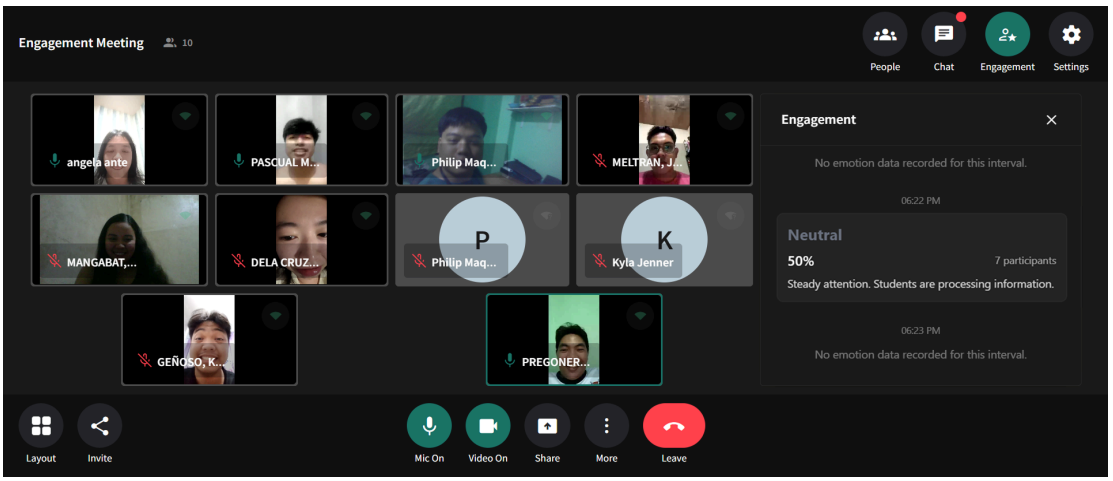

Figure 28. Real-time Emotion Summary Notification

Figure 29 shows the ratings of the students in using VirtuSense as the online class platform, which shows three to four ratings out of five. The majority of the students found VirtuSense a reliable option for conducting online classes. The host or the teacher of the class asked the students how they rated the system and gave some feedback. The students focused on the technical side of the system, which includes improving the system's features and transforming it into an app. Aside from the rated numbers by the students, they also gave their comments on how the project can be further improved.

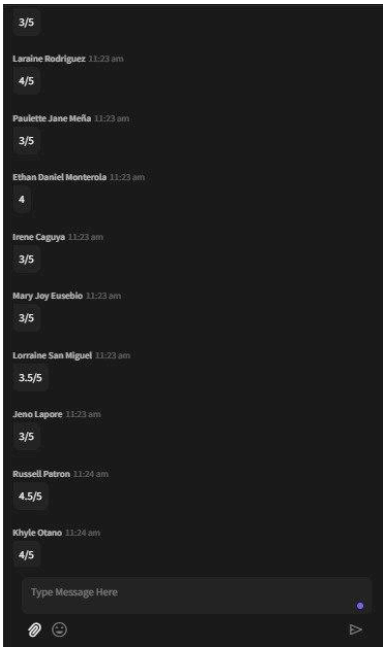

Figure 29. Ratings of Students for VirtuSense

Survey Results from Respondents Using TAM

Table 8. Evaluation Results Answered by the Respondents

Item	Mean	SD	Interpretation
Perceived Usefulness (PU)	3.67	0.00	Strongly Agree
Perceived Ease of Use (PEOU)	3.72	0.09	Strongly Agree
Social Influence (SI)	3.55	0.06	Strongly Agree
Information Awareness (IA)	3.59	0.05	Strongly Agree
Perceived Cost (PC)	3.68	0.08	Strongly Agree
Willingness to Adopt (WTA)	3.72	0.08	Strongly Agree

n= 51, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Table 9 shows the perceived usefulness of VirtuSense for the target users, who are the educators. The table shows consistent results, obtaining a mean score of 3.67 with “Strongly Agree” remarks listed in the Perceived Usefulness category and a result of standard deviation ranging from ±0.52 to ±0.59, which shows that the respondents mostly agree but with slight differences. This indicates that educators generally perceive VirtuSense as a beneficial tool in supporting their instructional activities.

Table 9. Perceived Usefulness of VirtuSense

Item	Mean	SD	Interpretation
The platform improves how I monitor students’ engagement in online classes.	3.67	0.52	Strongly Agree
The emotion detection feature helps me understand the overall emotional atmosphere of the class.	3.67	0.55	Strongly Agree
The system can contribute to a more effective online learning experience.	3.67	0.59	Strongly Agree
The platform’s emotion detection provides useful insights into the class’s general emotional state during online sessions.	3.67	0.52	Strongly Agree
Composite Mean	**3.67**	**0.00**	**Strongly Agree**

n= 51, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Table 10 indicates that the mean scores for the perceived use of VirtuSense range from 3.59 to 3.78, with “Strongly Agree” remarks appearing in the perceived ease of use category, and standard deviation values ranging from ±0.42 to ±0.57, showing that most participants provided similar ratings. This suggests that the platform is generally regarded as easy to navigate, simple to learn, and efficient to operate during online class activities. The consistently high scores and low variability indicate that users share a common positive perception of VirtuSense’s usability, highlighting its effectiveness in providing an intuitive and user-friendly environment that supports engagement and smooth operation in online learning sessions. These results also imply that the platform has an easy navigation panel that new users can tolerate.

Table 10. Perceived Ease of Use of VirtuSense

Item	Mean	SD	Interpretation
The platform is easy to navigate and use.	3.75	0.48	Strongly Agree
I learned to use the system without much difficulty.	3.59	0.57	Strongly Agree
Users can quickly access the features needed during an online class.	3.75	0.48	Strongly Agree
Interacting with the platform, such as starting sessions and viewing feedback, is straightforward.	3.78	0.42	Strongly Agree
Composite Mean	**3.72**	**0.07**	**Strongly Agree**

n= 51, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Table 11 shows that respondents strongly agree that they are willing to use the platform and willing to recommend it to their peers, and agree that their institution supports the use of emotion-monitoring technologies. This also indicates the result of standard deviation ranging from ±0.57 to ±0.61, which

shows that participants have different ratings. Overall, it indicates that the system has a positive perception of social influence.

Table 11. Social Influence of VirtuSense

Item	Mean	SD	Interpretation
I recommend the platform to my peers.	3.59	0.57	Strongly Agree
My institution supports using emotion-monitoring technologies.	3.47	0.61	Agree
I am more likely to use the system because others recommend it.	3.61	0.60	Strongly Agree
Using this system is positively viewed by my peers.	3.55	0.58	Strongly Agree
Composite Mean	**3.55**	**0.06**	**Strongly Agree**

n= 51, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Table 12 shows that respondents strongly agree that the system significantly enhances their information awareness during instruction. The average of information awareness ranges from 3.55 to 3.63. This also indicates the results of standard deviation ranging from ±0.56 to ±0.61, which shows that most participants chose similar ratings. This implies that the system effectively improves teachers' access to relevant emotional and engagement-related information.

Table 12. Information Awareness of VirtuSense

Item	Mean	SD	Interpretation
The platform provides real-time emotional feedback from students.	3.63	0.56	Strongly Agree
I am more aware of student engagement during lessons.	3.55	0.58	Strongly Agree
The system helps me become aware of the overall emotional trends among students during class.	3.63	0.60	Strongly Agree
The platform helps me make better teaching strategies.	3.55	0.61	Strongly Agree
Composite Mean	**3.59**	**0.56**	**Strongly Agree**

n= 51, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Table 13 shows that respondents strongly agree that the system is not costly, affordable to implement, justifies any cost, and saves time and resources in the long run, with the average ranging from 3.59 to 3.76. This also indicates that respondents have different ratings, with the result of standard deviation ranging from ±0.47 to ±0.64. This implies that respondents perceive the system as a financially practical solution with clear economic benefit for institutional use.

Table 13. Perceived Cost of VirtuSense

Item	Mean	SD	Interpretation
The platform does not require costly equipment.	3.71	0.50	Strongly Agree
The platform is affordable to implement.	3.76	0.47	Strongly Agree
The value of the system justifies any cost involved.	3.59	0.64	Strongly Agree
The system saves time and resources in the long run.	3.65	0.52	Strongly Agree
Composite Mean	**3.68**	**0.08**	**Strongly Agree**

n= 51, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Table 14 illustrates the respondents' willingness to adopt the VirtuSense system, as reflected by the computed mean scores and standard deviation values. The results show that the mean ratings range from 3.61 to 3.76, all of which fall within the descriptive interpretation of "Strongly Agree." These high mean values indicate that the respondents expressed a strong intention and readiness to adopt VirtuSense, suggesting that the system is perceived as beneficial, relevant, and acceptable for its intended purpose. The consistently high ratings further imply that the respondents recognize the system's potential to enhance current processes and address existing needs effectively. Overall, these findings imply a consistently very positive reception

of VirtuSense among the respondents. The strong agreement and minimal variation in responses demonstrate a collective confidence in the user's willingness to use the platform. This positive willingness to adopt suggests that VirtuSense has a high likelihood of successful implementation and sustained use, provided that it is properly deployed and supported in actual operational environments.

Table 14. Willingness to Adopt of VirtuSense

Item	Mean	SD	Interpretation
I am willing to use this platform in future classes.	3.76	0.51	Strongly Agree
I would encourage my institution to adopt this platform for online learning	3.75	0.52	Strongly Agree
I am open to using this system regularly.	3.61	0.57	Strongly Agree
I am interested in exploring similar systems.	3.76	0.47	Strongly Agree
Composite Mean	**3.72**	**0.08**	**Strongly Agree**

n= 51, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Testing and Evaluation Results of the Software Using ISO 25010

Table 15 illustrates the ISO 25010 answered by three (3) IT experts, indicating that VirtuSense achieved an acceptable evaluation in each criterion. However, the system still has a lot to improve in terms of functional accuracy and appropriateness, user interface and aesthetics, maturity, recoverability, non-repudiation, accountability, analysability, modifiability, modularity, and testability since it only got a score ranging from 3.00 - 3.33. Overall, the system's weighted mean is 3.53, which indicates that the system was a reliable tool for online class learning, which needs constant improvement to achieve the outstanding performance of the platform.

Table 15. Results of Beta Testing answered by the IT Specialist

Item	Mean	Interpretation
Functional Completeness	3.67	Strongly Agree
Functional Accuracy	3.33	Agree
Functional Appropriateness	3.33	Agree
Time Behaviour	4.00	Strongly Agree
Resource Utilization	3.67	Strongly Agree
Co-Existence	3.67	Strongly Agree
Interoperability	3.67	Strongly Agree
Learnability	4.00	Strongly Agree
Operability	3.67	Strongly Agree
User Interface Aesthetics	3.33	Agree
Maturity	3.00	Agree
Fault Tolerance	3.67	Strongly Agree
Recoverability	3.33	Agree
Confidentiality	3.67	Strongly Agree
Integrity	4.00	Strongly Agree
Non-Repudiation	3.00	Agree
Accountability	3.33	Agree
Authenticity	4.00	Strongly Agree
Analysability	3.33	Agree
Modifiability	3.00	Agree
Modularity	3.00	Agree
Testability	3.00	Agree
Adaptability	4.00	Strongly Agree
Installability	3.67	Strongly Agree
Replaceability	4.00	Strongly Agree
Overall Weighted Mean	**3.53**	**Strongly Agree**

n=3, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Software Testing

Unit Testing Results

The following tables show the result of unit testing for every module that VirtuSense consists of, such as the Accounts Management Module, Conference Module, Emotion Detection Module, and Report Management Module:

Table 16 shows the evaluation of functions in the account management module. The first function that has been evaluated is the capability of the system to allow users to sign in using their Google Account, which achieved a “Passed” remark as it functions as expected. The second test was to see if the system could assign a proper role, such as host and participant, which passed the expected output. The third function that has been evaluated is the capability of the system to display the records of every session that the user is considered as a host ,which also achieved “Passed” remarks.

Table 16. Unit Testing for Accounts Management Module

Tested Case	Expected Output	Actual Output	Response Time	Remarks
Signing Using Google Account	The system allows Google account sign in for new and existing users.	The system can use Google Sign in for new users and existing users.	2-5 secs	Passed
Assigning roles in online conference room	The user who created the session should be considered as host, and the user who joined should be considered as participant	The user created the session became host and the user who joined counted as participants	2 secs	Passed
Viewing of Records	Only the host should see all the data gathered in the created session	The host can see all the emotion data gathered in his/her created session	4-7 secs	Passed

The first evaluation conducted in the Conference Module is the capability of the system to allow users to create or join a room, which results

in a “Passed” remark as the actual output is aligned with the expected output. The second test is to see if the system can generate a joining link for the participants, which also achieved a “Passed” remark. The third up until the sixth test was focused on the features of the conference room, such as the camera and microphone settings, virtual whiteboard, screen sharing, and chats, which all achieved a “Passed” remark, as all of the expected output was seen during the testing.

Table 17. Unit Testing for Conference Module

Tested Case	Expected Output	Actual Output	Response Time	Remarks
Create/Join Meeting	The user can successfully creates or joins a meeting	The user created/joined a meeting successfully.	5 secs	Passed
Generating Joining Link	A unique and shareable meeting link is generated for participants.	The system generated a valid meeting link that can be used by others to join.	1 sec	Passed
Microphone and Camera Settings	Users can enable/disable the microphone and camera before and during the meeting.	The microphone and camera toggled properly and updated in real-time.	1 sec	Passed
Virtual Whiteboard	The whiteboard feature loads and can be used only by hosts and can be viewed by participants	The whiteboard was loaded and displayed among the participants.	1-2 secs	Passed
Share Screen	The user can share their screen without delay, and others can view it clearly.	Screen sharing worked smoothly, with minimal latency and clear display.	1-2 secs	Passed
Chats	Users can send and receive messages in real-time during the meeting.	Messages were sent and received instantly with proper timestamps.	1-2 secs	Passed

Table 18 displays the unit testing for the Emotion Detection Module. The first test is to see if the system can analyze the faces and emotions of each participant with their camera open without any visual trace, which

achieves a “Passed” remark. The second test was to analyze if all the participants’ faces and emotions could be detected, which also achieved a “Passed” remark. The third evaluation focuses on the record side, in which the system is tested to determine if the emotions that have been analyzed from the participants are saved in the database with a timestamp, which actually did what was expected. Lastly, the fourth test is to see if the system did not record an emotion or detect an emotion if there is no face detected or no participants have an open camera, which also achieved a “Passed” remark.

Table 18. Unit Testing for Emotion Module

Tested Case	Expected Output	Actual Output	Response Time	Remarks
Background Emotion Recognition	The system analyzes participants’ facial emotions in real time without displaying any visual indicators.	Emotion detection ran silently in the background; emotions successfully processed	1-2 secs	Passed
Multiple Faces Handling	The system detects and analyzes all participants’ faces whose cameras are turned on during the session.	All active participants’ faces were detected and analyzed simultaneously in the background.	1 secs	Passed
Emotion Log Storage	Detected emotions are stored in the database with timestamps for research or analysis.	Emotion data successfully saved with timestamps	1-2 secs	Passed
No Face Detected Handling	When no face is visible, the camera is off, and no host is in the session, emotion detection pauses automatically and resumes when a face reappears.	Detection paused and resumed correctly, verified via backend logs.	1 secs	Passed

Table 19 illustrates the unit testing for the Report Management Module. The first test was to see if the system could retrieve emotion data from the database and display it in the record section, which aligns with the actual output of the first test. The second test was to see if the system could

generate the necessary graphs for data visualization, which achieved a “Passed” remark as it aligned with the expected output of the second test. The third test is focused on the emotion timeline display, which requires the system to display an emotion timeline that shows all the emotions gathered every five minutes, which also achieved a “Passed” remark. The last test was to see if the system could generate a recommendation based on the analyzed emotions displayed in graphs, which aligns with its actual output during the testing.

Table 19. Unit Testing for Report Management Module

Tested Case	Expected Output	Actual Output	Response Time	Remarks
Emotion Data Retrieval	The system retrieves stored emotion data from the database for a selected session.	Emotion data successfully fetched and loaded for analysis.	1-2 secs	Passed
Emotion Graph Generation	The system displays a visual graph showing emotion trends during the session.	Emotion graph displayed accurately based on stored emotion data.	1-2 secs	Passed
Emotion Timeline Display	The system shows a timeline of emotions detected throughout the session.	Timeline displayed correctly, showing emotion transitions over time.	1-2 secs	Passed
Recommendation Generation	The system provides automated recommendations or insights based on the analyzed emotion patterns.	Recommendations generated and displayed based on detected emotional trends.	1 secs	Passed

Integration Testing Results

Table 20 shows the integration testing of VirtuSense; all the modules in the system are tested to see if each module is working as expected. Each module aligns with each expected output, resulting in a “Passed” remark for

all the modules. The Account Management Module tests whether the system can allow Google sign-in and accurately assign roles for the user, such as host and participants. The Conference Module is evaluated by testing if the system can allow users to create or join meetings and use all the authorised conference features. The third test is the Emotion Detection Module, which is focused on seeing if the system can detect emotion during the conference and log it in the database. Lastly, the Report Management Module is tested to see if the system can display and generate graphs and recommendations.

Table 20. Integration Testing Result of the System

Tested Case	Expected Output	Actual Output	Response Time	Remarks
Accounts Management Module	The system should allow users to sign in and correctly assign their roles as host or participant when creating or joining a session.	The system allows sign in for users and accurately assigns role for them when creating or joining a session	5 secs	Passed
Conference Module	The system should allow users to create or join a meeting, and access all integrated conference features without errors or conflicts.	The system allowed users to create or join a meeting, and use all conference features simultaneously without system crashes or feature conflicts.	5-7 secs	Passed
Emotion Detection Module	The system integrates emotion recognition during live sessions, accurately detecting participants' facial emotions while maintaining session performance.	The system successfully detects and processes participant emotions in real time without affecting the session's performance.	2-3 secs	Passed
Report Management Module	The system creates and successfully displays reports and graphs	The system generated and displayed emotion reports and visual graphs accurately after each session.	2-3 secs	Passed

Compatibility Testing Results

Table 21 shows the compatibility testing in different Windows versions. The system was tested in Windows 11, Windows 10, Windows 8, and Windows 7, all of which are capable of running the system smoothly and

without any problems. The remark “Passed” indicates that the system could run in that particular version.

Table 21. Different Windows Versions Compatibility Testing

Windows Version	Expected Output	Actual Output	Remarks
Windows 11	Compatible	Compatible	Passed
Windows 10	Compatible	Compatible	Passed
Windows 8	Compatible	Compatible	Passed
Windows 7	Compatible	Compatible	Passed

Table 22 displays the result of compatibility testing across various Android versions, specifically Android 16, Android 15, Android 14, and Android 13. The system works perfectly fine without any issues in all the Android versions, and all these versions achieved “Passed” remarks for the system’s compatibility. This shows that the system is versatile in terms of different devices and different Android Versions.

Table 22. Different Android Versions Compatibility Testing

Android Version	Expected Output	Actual Output	Remarks
Android 16	Compatible	Compatible	Passed
Android 15	Compatible	Compatible	Passed
Android 14	Compatible	Compatible	Passed
Android 13	Compatible	Compatible	Passed

Table 23 shows the result of browser testing, which indicates that the system is browser independent, which means that the system is functional, accessible, and can run smoothly in all the tested browsers, which are Google, Microsoft Edge, Mozilla Firefox, Opera, and Safari. All the expected output was met in all the browsers which resulted in the “Passed” remarks of the five different browsers.

Table 23. Different Web Browsers Compatibility Testing

Web Browser	Expected Output	Actual Output	Remarks
Google	Compatible	Compatible	Passed
Microsoft Edge	Compatible	Compatible	Passed
Mozilla Firefox	Compatible	Compatible	Passed
Opera	Compatible	Compatible	Passed
Safari	Compatible	Compatible	Passed

Performance Testing Results

Table 24 demonstrates the result of the stress testing using Apache JMeter. The team tested the capability of the system's server with different numbers of users. The system can handle requests smoothly to moderately with 100 - 400 users, with a maximum response time of 183ms for 100 users, with an excellent performance level, 2445ms for 200 users, with a good performance level, and 6459ms for 400 users, with a moderate performance level. However, when the system reaches 800-1600 user requests, the response time of the system is slower and can be extremely slow, but it can still handle the request successfully without error.

Table 24. Stress Testing Using JMeter

Users	Max Response Time (ms)	Error (%)	Performance Level	Remarks
100	183ms	0.00%	Excellent	System responded very quickly; stable performance
200	2445ms	0.00%	Good	Slight increase in response time; still stable
400	6459ms	0.00%	Moderate	Moderate response time; performance acceptable
800	11924ms	0.00%	Slow	Noticeable delay; system under high load but stable
1600	24004ms	0.00%	Very Slow	High response time

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

Summary

The problem between the declining academic performance of students, which can be caused by their emotions during online classes, has led to the development of VirtuSense. VirtuSense is an online conference platform with emotion detection using Face++ Web API, a pre-trained emotion detection model, which can detect happiness, sadness, anger, fear, surprise, disgust, and neutral emotion.

An Agile Scrum methodology was used in the development process of the system. This ensures that all the functions and features of the system have an allotted time to finish and evaluate. In addition, using this method ensures that the development of the system is organized in manageable sprints. A system flowchart was used to visualize the flow and blueprint of the whole system, including all its functions and features, which was made using React, Tailwind CSS, TypeScript, and Agora for the frontend side of the system. On the other hand, Firebase and NodeJS were used for the backend side of the project. A cloud database was utilized to ensure the availability and security of the data collected by the system. These materials enabled the creation of a robust and user-friendly online conference platform.

The team used a questionnaire survey as a data gathering tool that helped to gather insights from the target users. This survey was made in accordance with the Technology Acceptance Model (TAM), which was answered by instructors, and ISO 25010, which was answered by three (3) IT experts.

Conclusion

In conclusion, the web application achieved the following objectives:

1. Develop a responsive web application that serves as an online conference platform to support learners during online classes.

The system became an additional option for all the existing online conference platforms with interactive features and functions exceeding other platforms. This was made possible by using different kinds of programming tools such as React, TypeScript, TailwindCSS, Agora, and Firebase for both frontend and backend. In addition, the system can work across different browsers, Windows versions, and Android versions for mobile devices.

2. Integrate real-time emotion detection features through the integration of emotion detection that allows educators to monitor learners' emotional states as they participate in online classes.

The real-time emotion detection integrated in the system was achieved using Face++ Web API, which is a pre-trained emotion detection model that can detect seven specific emotions, which are happiness, sadness, disgust, surprise, fear, anger, and neutral. These gathered emotions were used in the data analytics of the system in the record section, which shows the emotion distribution graphs, recommendations and insights, statistical cards, emotion timeline, and student engagement table.

3. Evaluate the acceptability of the system using the Technology Acceptance Model (TAM) and ISO 25010.

The result of the questionnaire survey answered by instructors showed that the system has passed the Technology Acceptance Model and user acceptance. In addition, the questionnaire survey answered by IT experts who

focus on the functionality, reliability, and overall quality of the system has passed the ISO 25010 standards for a quality system.

Recommendations

These recommendations were based on the findings of the project and the recommendations of IT experts and some of the instructors:

1. This website is recommended to become a mobile application for convenience and accessibility, and to improve the overall performance of the system.
2. The system should have more features, such as reactions, raise hand, record of meeting, and the whiteboard feature of the system should be interactive and usable for both participants and hosts, and the screen sharing should also be available on mobile, not just on desktop.
3. The recommendation and suggestion system of the online conference platform in the record section could become an AI-generated content for more insights that the system could provide, which can make the system more informative and complex.
4. The system is recommended to have an archive section for sessions that are no longer the latest, or if the record section has so many sessions recorded.

REFERENCES

Acosta-Gonzaga, E., & Ruiz-Ledesma, E. F. (2022c). Students' Emotions and Engagement in the Emerging Hybrid Learning Environment during the COVID-19 Pandemic. *Sustainability*, *14*(16), 10236. https://doi.org/10.3390/su141610236

Agarwal, P. (2022, December 31). Emotional AI is no substitute for empathy. *WIRED*. https://www.wired.com/story/artificial-intelligence-empathy/

Aravindhar, S. M. D. J. (2024, March 26). *Deep learning based facial emotion recognition for analysing the effectiveness of online class*. https://ijisae.org/index.php/IJISAE/article/view/5602

Berking, M., & Wupperman, P. (2012). Emotion regulation and mental health. *Current Opinion in Psychiatry*, *25*(2), 128–134. https://doi.org/10.1097/yco.0b013e3283503669

Bhattacharjee, A., Pias, T., Ahmad, M., & Rahman, A. (2018b). *On the Performance Analysis of APIs Recognizing Emotions from Video Images of Facial Expressions* (pp. 223–230). https://doi.org/10.1109/icmla.2018.00040

Bryant, D., & Howard, A. (2019b). *A Comparative Analysis of Emotion-Detecting AI Systems with Respect to Algorithm Performance and Dataset Diversity* (pp. 377–382). https://doi.org/10.1145/3306618.3314284

Burić, I., Sorić, I., & Penezić, Z. (2016). Emotion regulation in academic domain: Development and validation of the academic emotion regulation questionnaire (AERQ). *Personality and Individual Differences*, *96*, 138–147. https://doi.org/10.1016/j.paid.2016.02.074

Camacho-Morles, J., Slemp, G. R., Pekrun, R., Loderer, K., Hou, H., & Oades, L. G. (2021). Activity Achievement Emotions and Academic Performance: A Meta-analysis. *Educational Psychology Review*, *33*(3), 1051–1095. https://doi.org/10.1007/s10648-020-09585-3

Clipa, O., & Mihai, I. D. (2019). The impact of evaluation on Learning Process – Teachers perceptions. *The European Proceedings of Social & Behavioural Sciences*, 1908–1917. https://doi.org/10.15405/epsbs.2019.08.03.235

Cohn, J., Schmidt, K., Gross, R., & Ekman, P. (2003). *Individual differences in facial expression: stability over time, relation to self-reported emotion, and ability to inform person identification*. https://doi.org/10.1109/icmi.2002.1167045

Day, S. E., Shore, D. M., & Krumhuber, E. G. (2024). *Contextual Influences on the Perception and Interpretation of Facial Expressions*. https://doi.org/10.1007/978-3-031-70064-4_4

De Freitas Pereira, T., Schmidli, D., Linghu, Y., Zhang, X., Marcel, S., & Günther, M. (2022). Eight years of face recognition research: Reproducibility, Achievements and open issues. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2208.04040

Deng, L., & Yu, D. (2014). Deep Learning: methods and applications. *Foundations and Trends® in Signal Processing*, *7*(3–4), 197–387. https://doi.org/10.1561/2000000039

Dennen, V. P., Darabi, A. A., & Smith, L. J. (2007). Instructor–Learner Interaction in Online Courses: The relative perceived importance of particular instructor actions on performance and satisfaction. *Distance Education*, *28*(1), 65–79. https://doi.org/10.1080/01587910701305319

Dupré, D., Krumhuber, E. G., Küster, D., & McKeown, G. J. (2020b). A performance comparison of eight commercially available automatic classifiers for facial affect recognition. *PLoS ONE*, *15*(4), e0231968. https://doi.org/10.1371/journal.pone.0231968

Felzmann, H., Villaronga, E. F., Lutz, C., & Tamò-Larrieux, A. (2019b). Transparency you can trust: Transparency requirements for artificial intelligence between legal norms and contextual concerns. *Big Data & Society*, *6*(1). https://doi.org/10.1177/2053951719860542

Fuad, M. T. H., Fime, A. A., Sikder, D., Iftee, M. a. R., Rabbi, J., Al-Rakhami, M. S., Gumaei, A., Sen, O., Fuad, M., & Islam, M. N. (2021). Recent advances in deep learning techniques for face recognition. *IEEE Access*, *9*, 99112–99142. https://doi.org/10.1109/access.2021.3096136

Guo, Y., Xia, Y., Wang, J., Yu, H., & Chen, R. (2020b). Real-Time facial affective computing on mobile devices. *Sensors*, *20*(3), 870. https://doi.org/10.3390/s20030870

Hassan, G., & Elgazzar, K. (2016b). *The case of face recognition on mobile devices*. https://doi.org/10.1109/wcnc.2016.7564975

Hassanat, A. B., Albustanji, A., Tarawneh, A. S., Alrashidi, M., Alharbi, H., Alanazi, M., Alghamdi, M., Alkhazi, I. S., & Prasath, V. B. S. (2021). Deep learning for identification and face, gender, expression recognition under constraints. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2111.01930

Hinduja, S., Canavan, S. J., & Aathreya, S. (2020). Impact of action unit occurrence patterns on detection. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2010.07982

Jan, A. (2017). Deep learning based facial expression recognition (Master's thesis, Brunel University). *Brunel University Research Archive.* https://bura.brunel.ac.uk/bitstream/2438/15944/1/FulltextThesis.pdf

Jia, S., Wang, S., Hu, C., Webster, P. J., & Li, X. (2021). Detection of genuine and posed facial expressions of emotion: Databases and Methods. *Frontiers in Psychology*, *11*, 580287. https://doi.org/10.3389/fpsyg.2020.580287

Karaivanova, K. (n.d.). *The effects of encouraging Student-Faculty interaction on academic success, identity development, and student retention in the first year of college*. University of New Hampshire Scholars Repository. https://scholars.unh.edu/dissertation/1355/

Kheddar, H., Hemis, M., & Himeur, Y. (2024). Automatic speech recognition using advanced deep learning approaches: A survey. *Information Fusion*, *109*, 102422. https://doi.org/10.1016/j.inffus.2024.102422

Küntzler, T., Höfling, T. T. A., & Alpers, G. W. (2021c). Automatic facial expression recognition in standardized and non-standardized emotional expressions. *Frontiers in Psychology*, *12*, 627561. https://doi.org/10.3389/fpsyg.2021.627561

LB, N. K., & Gg, N. L. P. (2016e). Student Emotion Recognition System (SERS) for e-learning Improvement Based on Learner Concentration Metric. *Procedia Computer Science*, *85*, 767–776. https://doi.org/10.1016/j.procs.2016.05.264

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436–444. https://doi.org/10.1038/nature14539

Lek, J. X., & Teo, J. (2023). Academic Emotion Classification Using FER: A Systematic Review. *Human Behavior and Emerging Technologies*, *2023*, 1–27. https://doi.org/10.1155/2023/9790005

MacIntyre, P. D., & Vincze, L. (2017). Positive and negative emotions underlie motivation for L2 learning. *Studies in Second Language Learning and Teaching*, *7*(1), 61–88. https://doi.org/10.14746/ssllt.2017.7.1.4

Maqableh, W., Alzyoud, F. Y., & Zraqou, J. (2022). The use of facial expressions in measuring students' interaction with distance learning environments during the COVID-19 crisis. *Visual Informatics*, *7*(1), 1–17. https://doi.org/10.1016/j.visinf.2022.10.001

Marengo, D., Settanni, M., & Montag, C. (2022b). Dataset on individual differences in self-reported personality and inferred emotional expression in profile pictures of Italian Facebook users. *Data in Brief*, *41*, 107899. https://doi.org/10.1016/j.dib.2022.107899

Mestari, S. Z. E., Lenzini, G., & Demirci, H. (2023b). Preserving data privacy in machine learning systems. *Computers & Security*, *137*, 103605. https://doi.org/10.1016/j.cose.2023.103605

Mishra, P., Verma, A. S., Chaudhary, P., & Dutta, A. (2024). *Emotion Recognition from Facial Expression Using Deep Learning Techniques* (pp. 1–6). https://doi.org/10.1109/i2ct61223.2024.10543313

Mollahosseini, A., Chan, D., & Mahoor, M. H. (2016). *Going deeper in facial expression recognition using deep neural networks* (pp. 1–10). https://doi.org/10.1109/wacv.2016.7477450

Molina-Moreira, A. A., Velásquez-Orellana, O. J., Zambrano-Murillo, D. J., & Zambrano-Villamil, M. E. (2023). Importance of feedback in the student evaluation process. *International Journal of Social Sciences*, *6*(3), 168–172. https://doi.org/10.21744/ijss.v6n3.2176

Munna, A. S., & Kalam, M. A. (2021, April 16). *Impact of active Learning strategy on the student engagement*. http://gnosijournal.com/index.php/gnosi/article/view/96

Nadeem, A., Umer, F., & Anwar, M. J. (2023). Emotion regulation as predictor of academic performance in university students. *Journal of Professional & Applied Psychology*, *4*(1), 20–33. https://doi.org/10.52053/jpap.v4i1.157

Nonis, F., Dagnes, N., Marcolin, F., & Vezzetti, E. (2019). 3D Approaches and Challenges in Facial Expression Recognition Algorithms—A Literature review. *Applied Sciences*, *9*(18), 3904. https://doi.org/10.3390/app9183904

Oducado, R. M. F., & Estoque, H. (2021). Online learning in nursing education during the COVID-19 pandemic: stress, satisfaction, and academic performance. *Journal of Nursing Practice*, *4*(2), 143–153. https://doi.org/10.30994/jnp.v4i2.128

Ong, S. G. T., & Quek, G. C. L. (2023). Enhancing teacher–student interactions and student online engagement in an online learning environment. *Learning Environments Research*, *26*(3), 681–707. https://doi.org/10.1007/s10984-022-09447-5

Panichkriangkrai, C., Silapasuphakornwong, P., & Saenphon, T. (2021). Emotion recognition of students during e-learning through online conference meeting. *Science, Engineering and Health Studies*, 15.

Paul Ekman Group, LLC. (2024, October 11). *Universal Emotions | What are Emotions? | Paul Ekman Group*. Paul Ekman Group. https://www.paulekman.com/universal-emotions/

Pekrun, R. (2006c). The Control-Value Theory of Achievement Emotions: Assumptions, corollaries, and implications for educational research and practice. *Educational Psychology Review*, *18*(4), 315–341. https://doi.org/10.1007/s10648-006-9029-9

Pekrun, R., & Linnenbrink-Garcia, L. (2012b). *Academic Emotions and Student Engagement*. https://doi.org/10.1007/978-1-4614-2018-7_12

Richards, J. C. (2020). Exploring emotions in language teaching. *RELC Journal*, *53*(1), 225–239. https://doi.org/10.1177/0033688220927531

Rothkrantz, L. (2017). Affective didactic models in higher education. *International Journal of Human Capital and Information Technology Professionals*, *8*(4), 50–66. https://doi.org/10.4018/ijhcitp.2017100105

Sirianansopa, K. (2024). Evaluating students' learning achievements using the formative assessment technique: a retrospective study. *BMC Medical Education*, *24*(1), 1373. https://doi.org/10.1186/s12909-024-06347-5

Tan, J., Mao, J., Jiang, Y., & Gao, M. (2021). The Influence of Academic Emotions on Learning Effects: A Systematic review. *International Journal of Environmental Research and Public Health*, *18*(18), 9678. https://doi.org/10.3390/ijerph18189678

Tcherkassof, A., & Dupré, D. (2020). The emotion–facial expression link: evidence from human and automatic expression recognition. *Psychological Research*, *85*(8), 2954–2969. https://doi.org/10.1007/s00426-020-01448-4

Vaijayanthi, S., & Arunnehru, J. (2024). Deep neural network-based emotion recognition using facial landmark features and particle swarm optimization. *Automatika*, *65*(3), 1088–1099. https://doi.org/10.1080/00051144.2024.2343964

Valladolid, V.C., & Valladolid, R.S. Students' Academic Self-Efficacy, Engagement, Satisfaction, and Perceived Academic Performance in Online Learning.

Valstar, M. F., Jiang, B., Mehu, M., Pantic, M., & Scherer, K. (2011). *The first facial expression recognition and analysis challenge*. https://doi.org/10.1109/fg.2011.5771374

Wang, K., Peng, X., Yang, J., Meng, D., & Qiao, Y. (2020). Region attention networks for pose and occlusion robust facial expression recognition. *IEEE Transactions on Image Processing*, *29*, 4057–4069. https://doi.org/10.1109/tip.2019.2956143

Wenkel, S., Alhazmi, K., Liiv, T., Alrshoud, S., & Simon, M. (2021). Confidence Score: The Forgotten Dimension of Object Detection Performance Evaluation. *Sensors*, *21*(13), 4350. https://doi.org/10.3390/s21134350

White, H. (n.d.). *Computer Vision Basics: confidence & accuracy*. Leverege. https://www.leverege.com/blogpost/computer-vision-basics-how-confidence-accuracy-and-thresholds-impact-performance

Whitehill, J., Serpell, Z., Lin, Y., Foster, A., & Movellan, J. R. (2014b). The Faces of Engagement: Automatic recognition of student engagement from facial expressions. *IEEE Transactions on Affective Computing*, *5*(1), 86–98. https://doi.org/10.1109/taffc.2014.2316163

Wibawanto, S., & Kirana, K. C. (2017). *Recognition of student emotion based on matrix-1 median fisher's face and backpropagation algorithm* (pp. 103–108). https://doi.org/10.1109/iceltics.2017.8253255

Wyman, A., & Zhang, Z. (2023b). API face value. *Journal of Behavioral Data Science*, *3*(1), 1–11. https://doi.org/10.35566/jbds/v3n1/wyman

Zeidner, M. (2007). Test anxiety in educational contexts. In *Elsevier eBooks* (pp. 165–184). https://doi.org/10.1016/b978-012372545-5/50011-3

APPENDICES

APPENDIX A

PLANNING & REQUIREMENTS ANALYSIS PHASE

Republic of the Philippines
Laguna State Polytechnic University
College of Computer Studies
Province of Laguna

Technology Acceptance Model (TAM) Survey Questionnaire

Dear Respondents,

We are students from Laguna State Polytechnic University-Los Baños Campus under the College of Computer Studies conducting a capstone project **DEVELOPMENT OF AN ONLINE CLASS CONFERENCE PLATFORM: INTEGRATING FACIAL EXPRESSION RECOGNITION UTILIZING FACE++ WEB API FOR LEARNERS EMOTION DETECTION**. We would like to know your opinion on the acceptability of the project.

Please evaluate the statements below by checking the appropriate number where, **four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree**. Be reassured that any of your information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for academic purposes only.
Thank you for your cooperation.

Regards:
Geñoso, Kenneth Rasheed A.
Mangabat, Ma. Cristina M.
Maquiddang, Philip Louise C.

Perceived Usefulness (PU)	**4**	**3**	**2**	**1**
1. The platform improves how I monitor students' engagement in online classes.				
2. The emotion detection feature helps me understand the overall emotional atmosphere of the class.				
3. The system can contribute to a more effective online learning experience.				
4. The platform's emotion detection provides useful insights into the class's general emotional state during online sessions.				

Perceived Ease of Use (PEOU)	**4**	**3**	**2**	**1**
1. The platform is easy to navigate and use.				
2. I learned to use the system without much difficulty.				
3. Users can quickly access the features needed during an online class.				
4. Interacting with the platform, such as starting sessions and viewing feedback, is straightforward.				

Social Influence (SI)	4	3	2	1
1. I recommend the platform to my peers.				
2. My institution supports using emotion-monitoring technologies.				
3. I am more likely to use the system because others recommend it.				
4. Using this system is positively viewed by my peers.				

Information Awareness (IA)	4	3	2	1
1. The platform provides real-time emotional feedback from students.				
2. I am more aware of student engagement during lessons.				
3. The system helps me become aware of the overall emotional trends among students during class.				
4. The platform helps me make better teaching strategies.				

Perceived Cost (PC)	4	3	2	1
1. The platform does not require costly equipment.				
2. The platform is affordable to implement.				
3. The value of the system justifies any cost involved.				
4. The system saves time and resources in the long run.				

Willingness to Adopt (WTA)	4	3	2	1
1. I am willing to use this platform in future classes.				
2. I would encourage my institution to adopt this platform for online learning&				
3. I am open to using this system regularly.				
4. I am interested in exploring similar systems. .				

Republic of the Philippines
Laguna State Polytechnic University
College of Computer Studies
Province of Laguna

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Rubric for 4-Point Likert Scale to Evaluate Product/ System Quality Using ISO25010

	4	3	2	1
I. FUNCTIONAL SUITABILITY				
Functional Completeness The system provides all necessary features for conducting an online class conference with emotion detection.				
Functional Accuracy The platform accurately detects and displays learners' emotions using the Face++ API.				
Functional Appropriateness The system's features are appropriate and useful for online class engagement and monitoring.				
II. PERFORMANCE EFFICIENCY				
Time Behaviour The system responds quickly and processes user actions without noticeable delay.				
Resource Utilization				

The system operates efficiently without consuming excessive device or network resources.				
III. COMPATIBILITY				
Co-Existence The system performs properly when multiple browser tabs or applications are open simultaneously.				
Interoperability The platform functions consistently and displays correctly across major browsers such as Chrome, Firefox, Edge, Safari, and Opera.				
IV. USABILITY				
Learnability The system is easy to understand and use, even for first-time users.				
Operability Users can easily navigate and control the system's features during online conferences.				
User Error Protection The system minimizes user errors and provides clear guidance or feedback.				
User Interface Aesthetics The system's interface is visually appealing and appropriately designed for online learning.				
V. RELIABILITY				
Maturity The system performs consistently and stably during operation.				
Fault Tolerance The system continues to function correctly even when minor errors occur.				
Recoverability The system can quickly recover from crashes or interruptions without data loss.				
VI. SECURITY				
Confidentiality The system ensures that users' personal and facial data are kept secure and private.				

Integrity The system protects data from unauthorized access or alteration.				
Non-Repudiation The platform records user actions to verify accountability when needed.				
Accountability User roles and permissions are clearly identified and managed in the system.				
Authenticity The login process ensures that only legitimate users can access the platform.				
VII. MAINTAINABILITY				
Analysability The system allows easy identification and analysis of errors or problems.				
Modifiability The system can be modified or updated without affecting other functions.				
Modularity The system's components are organized so each can be maintained independently.				
Testability The system can be tested easily after modifications or updates.				
VIII. PORTABILITY				
Adaptability The system runs well on different devices and operating systems.				
Installability The system is easily accessible through any web browser without requiring installation.				
Replaceability The system can be updated or replaced without major disruption or data loss.				

APPENDIX B

USER MANUAL

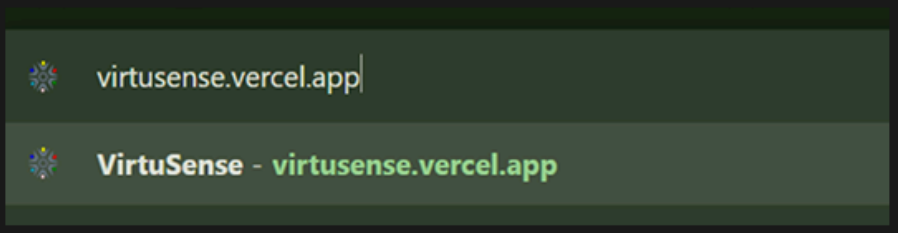
VirtuSense

User Manual

VirtuSense
1 Accessing the Website
2 User Sign in
3 Conference Module
4 Report Management Module

VirtuSense

Accessing the Website

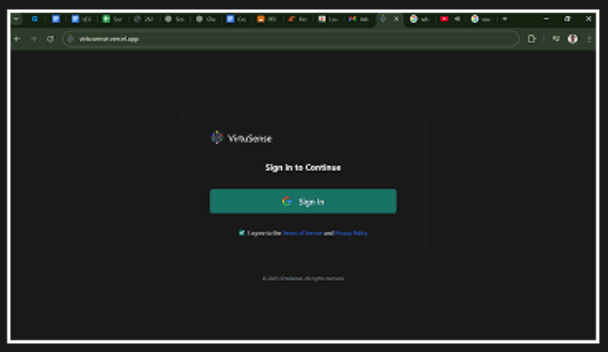


In your mobile phone, laptop, or computer' browser search or type "virtusense.vercel.app". Then press enter and it will redirect you to the sign in page of VirtuSense

Note:
It is important that you open it in an external browser and not in app browsers like messenger
because the system is only compatible with external browsers such as Chrome, Safari, Mozilla Firefox, Edge etc

VirtuSense

User Sign In

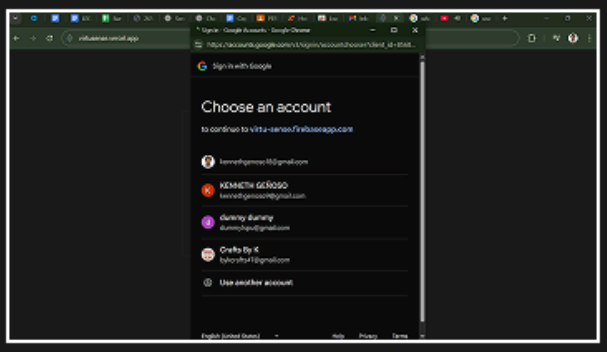


1

Check the checkbox to be able to sign in, then click the Sign in Button

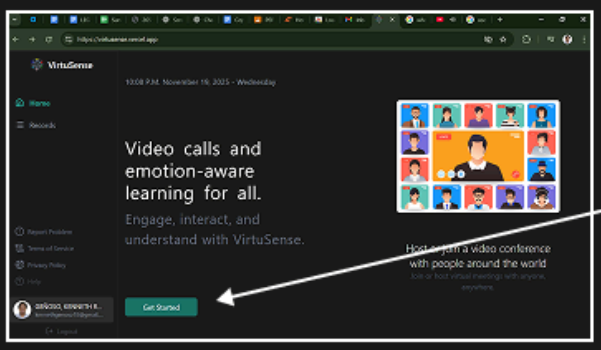
2

This pop window will show and click the google account you want to register and then it will redirect you to the homepage of VirtuSense if the sign in process is successful



VirtuSense

Conference Module
(Online Room Creation)

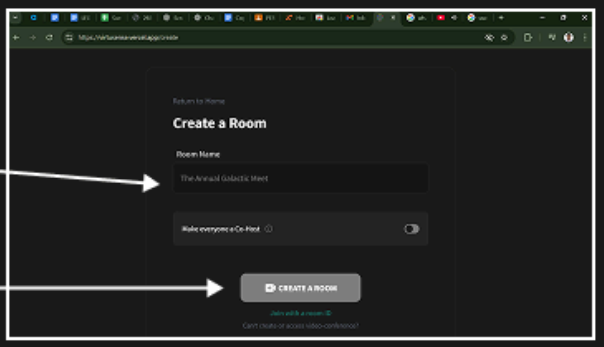


1

Click the Get Started Button to be able to create an online room

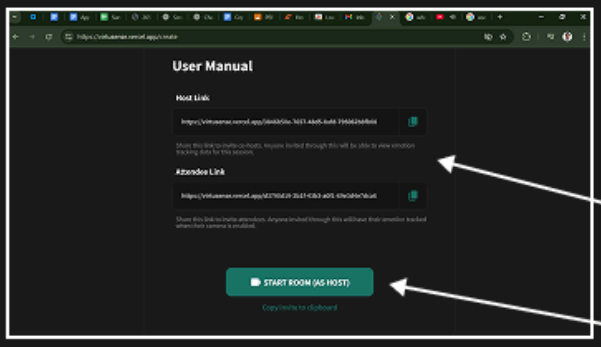
2

Enter a name for your online room and then click “Create a Room” button



VirtuSense

Conference Module
(Online Room Creation)



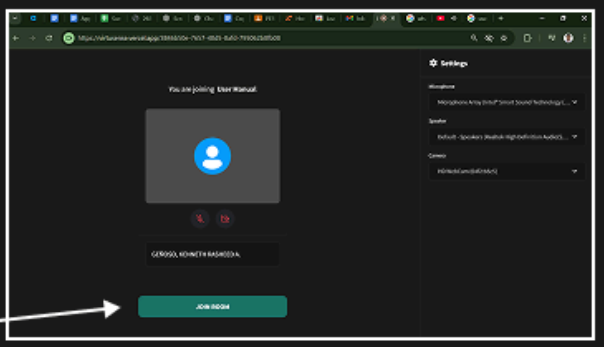
3

Copy the joining links to invite participants in your room, then click “Start Room”

Note: Host link is for the people you want to invite as host and attendee link is for the people you want to invite as a participant

4

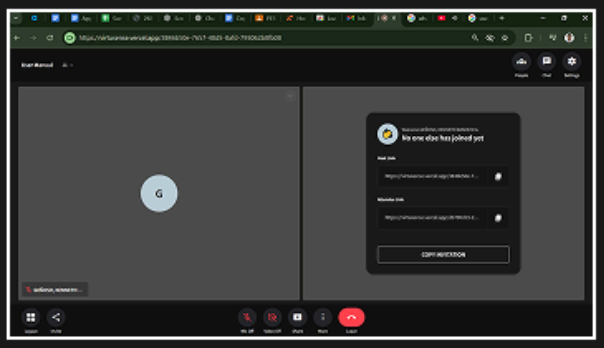
Manage your camera and microphone settings and then click “Join Room”



VirtuSense

Conference Module (Online Room Creation)

5



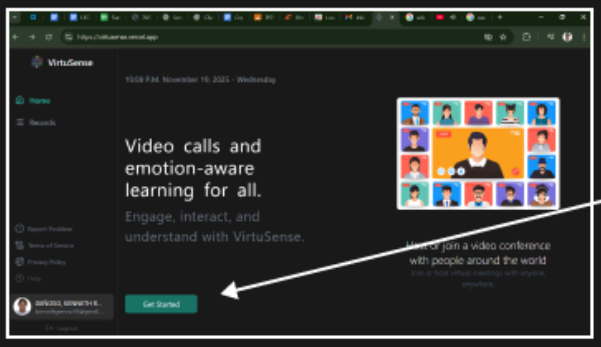
This is the main interface of the conference room, you can use all the features offered by VirtuSense such as Camera and Microphone Setting, Virtual Whiteboard, Screen Sharing.

In this interface, emotion detection is running once there are participant in your room, all the emotion gathered will be saved and display in the record section of the web application

Note: Virtual Whiteboard will only be triggered by host and interacted by host, the participants will just be able to see it

VirtuSense

Conference Module
(Joining Online Room w/ ID)

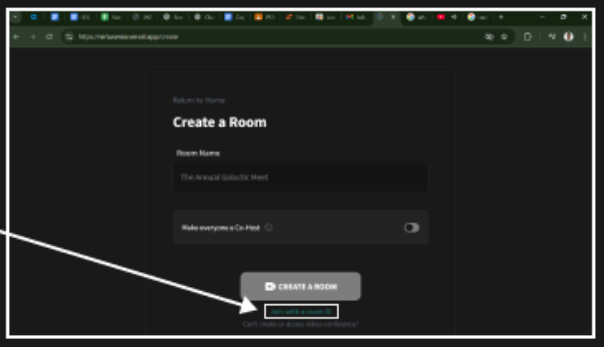


1

Click the Get Started Button to be able to create an online room

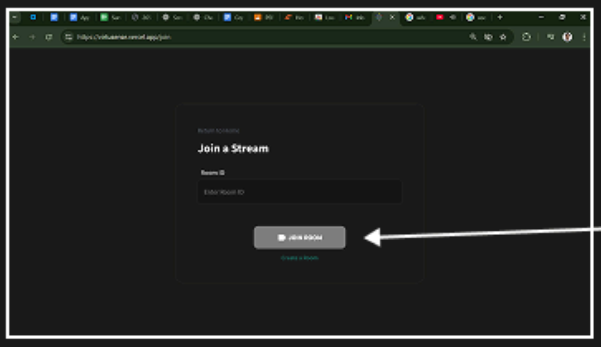
2

Enter a name for your online room and then click "Create a Room" button



VirtuSense

Conference Module
(Joining Online Room w/ ID)

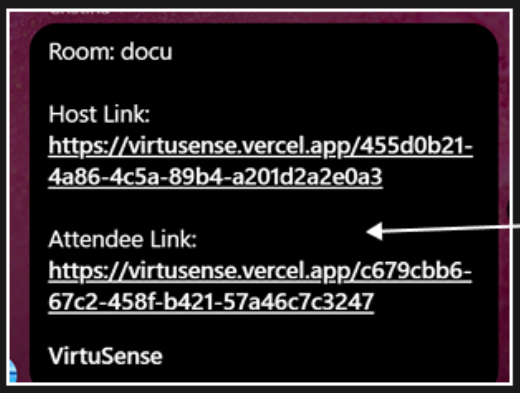


3

Enter the Room ID provided by host then click Join Room

VirtuSense

Conference Module
(Joining Online Room w/ Link)

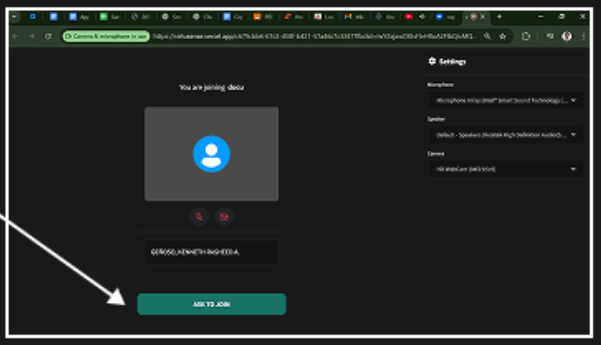


1

Click the link provided by the host

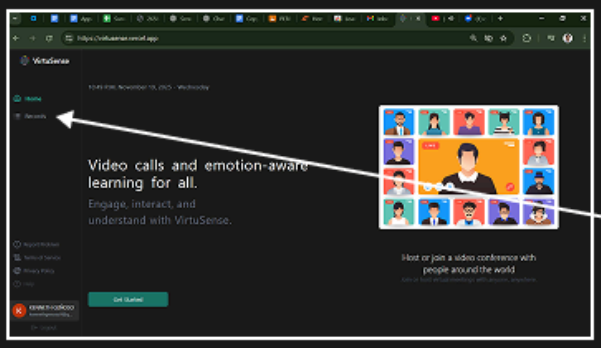
2

Manage your camera and microphone settings and then click “Ask to Join” then wait for the host to accept you



VirtuSense

Report Management Module (General Data)

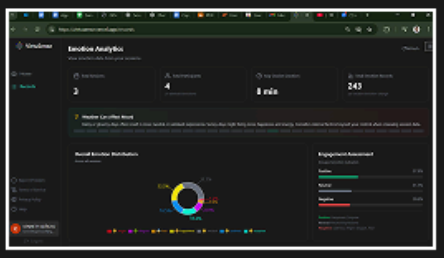


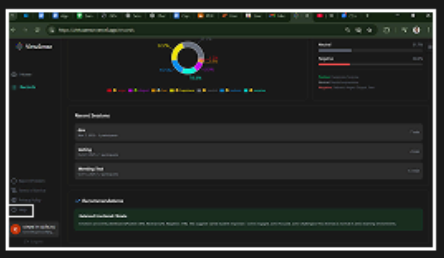
1

Click the “Record” button to access the data analytics

2

This is the summary report and records of all the sessions you hosted. You can see stats cards, graphs, facts, and reccommendation

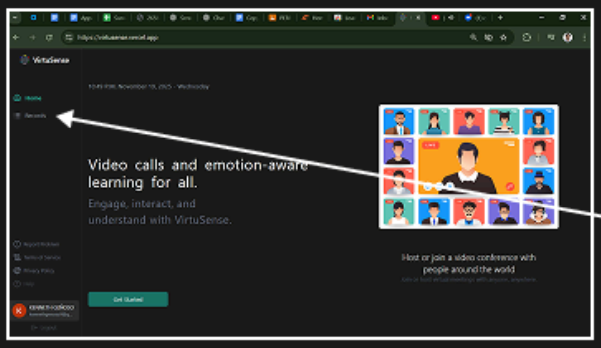




Note: You can click the “Help” button to view all the definitions and explanation of all the graphs and figures displayed in the record section

VirtuSense

Report Management Module (General Data)



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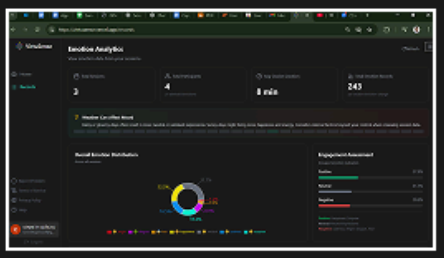
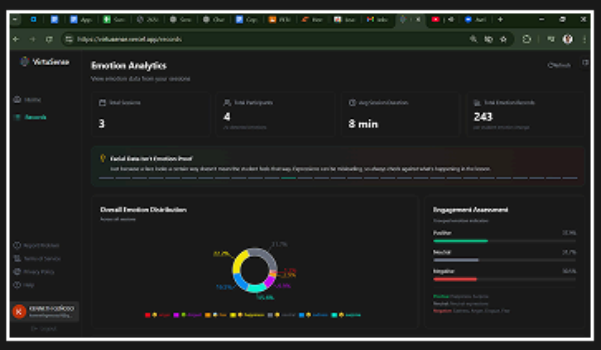


Note: You can click the “Help” button to view all the definitions and explanation of all the graphs and figures displayed in the record section

VirtuSense

Report Management Module (Per Session)

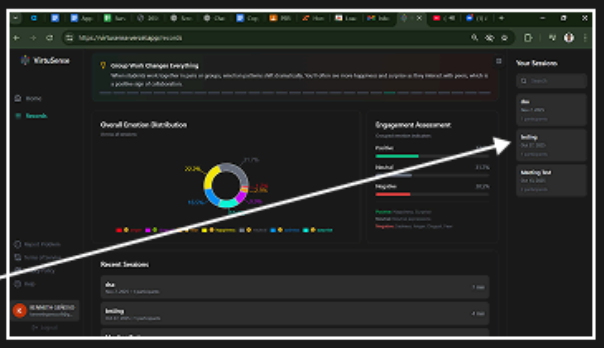


1

Click the more icon in the upper right of the interface beside the refresh button

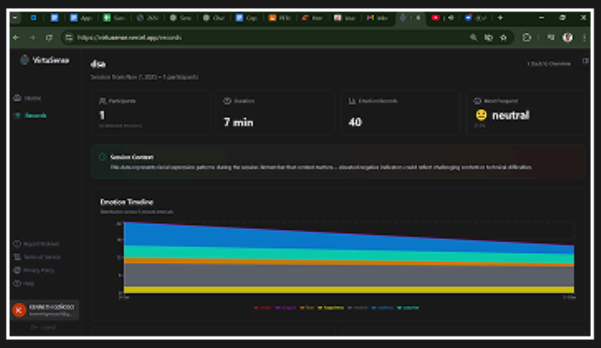
2

Select the session you want to view the data



VirtuSense

Report Management Module (Per Session)

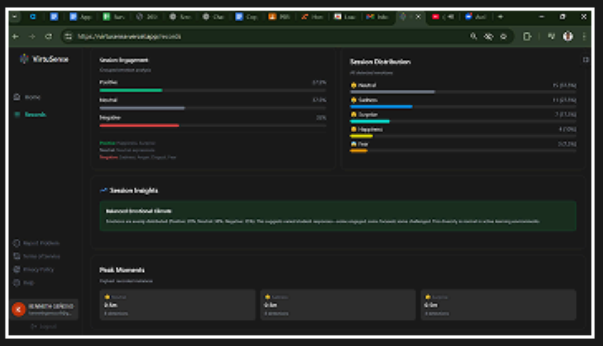


1

Here you can see the stats card, and emotion timeline graph

2

Here you can see the emotion distribution graph, session engagement and recommendation



Note: You can click the “Help” button to view all the definitions and explanation of all the graphs and figures displayed in the record section

APPENDIX C

SOURCE CODE

```
private ferCheckTimeout: NodeJS.Timeout | null = null;
private currentHostCount: number = 0;
private meetingTitle = '';
private isHost = false;
private username = '';
private userUID = '';
private sessionId = '';

private emotionBatch: Array<{
  emotion: string;
  confidence: number;
  timestamp: Timestamp;
}> = [];
private lastSavedEmotion: string | null = null;
private batchInterval: NodeJS.Timeout | null = null;
private readonly FER_CAPTURE_INTERVAL = 600; // 600ms
private readonly BATCH_WRITE_INTERVAL = 10000; // 10 seconds
private readonly MIN_CONFIDENCE_THRESHOLD = 75; // 75%

private hostAbsentTimer: NodeJS.Timeout | null = null;
private readonly HOST_ABSENT_GRACE_PERIOD = 30000; // 30 seconds

private sessionStartTime: Timestamp | null = null;
private sessionCheckInterval: NodeJS.Timeout | null = null;
private readonly MAX_SESSION_DURATION = 3600000; //
```

```
private handleHostCountChange = (event: CustomEvent) => {
  const newHostCount = event.detail.totalHosts;

  console.log('RTCENGINE: Received host count event:', newHostCount);

  if (this.currentHostCount !== newHostCount) {
    console.log(`RTCENGINE: Host count changed: ${this.currentHostCount} → ${newHostCount}`);
    this.currentHostCount = newHostCount;
    this.checkFERState();
  }
};

setMeetingMetadata(metadata: {
  meetingTitle?: string;
  isHost?: boolean;
  username?: string;
  userUID?: string;
  sessionId?: string;
}) {
  if (metadata.meetingTitle !== undefined) {
    this.meetingTitle = metadata.meetingTitle;
  }
  if (metadata.isHost !== undefined) {
    this.isHost = metadata.isHost;
  }
  if (metadata.username !== undefined) {
    this.username = metadata.username;
  }
```


```
private handleHostCountChange = (event: CustomEvent) => {
  const newHostCount = event.detail.totalHosts;

  console.log('RTCENGINE: Received host count event:', newHostCount);

  if (this.currentHostCount !== newHostCount) {
    console.log(`RTCENGINE: Host count changed: ${this.currentHostCount} → ${newHostCount}`);
    this.currentHostCount = newHostCount;
    this.checkFERState();
  }
};

setMeetingMetadata(metadata: {
  meetingTitle?: string;
  isHost?: boolean;
  username?: string;
  userUID?: string;
  sessionId?: string;
}) {
  if (metadata.meetingTitle !== undefined) {
    this.meetingTitle = metadata.meetingTitle;
  }
  if (metadata.isHost !== undefined) {
    this.isHost = metadata.isHost;
  }
  if (metadata.username !== undefined) {
    this.username = metadata.username;
  }
```

```
  console.log('▶️ RTCENGINE: Starting FER');

  //Detect platform and choose best capture method
  const userAgent = navigator.userAgent.toLowerCase();
  const isIOS = /iphone|ipad|ipod/.test(userAgent);
  const isSafari = /safari/.test(userAgent) && !/chrome|crios|crmo/.test(userAgent);
  const isFirefox = /firefox/.test(userAgent);

  //Use ImageCapture for Android Chrome, Video Element for iOS/Safari/Firefox/fallback
  const useImageCapture = !isIOS && !isSafari && !isFirefox && typeof ImageCapture !== 'undefined';

  console.log(`Platform: iOS=${isIOS}, Safari=${isSafari}, Firefox=${isFirefox}, Method=${useImageCapture ? 'ImageCapture' : 'VideoElem

  // Start batch writing interval
  this.startEmotionBatchWriter();

  if (useImageCapture) {
    this.startFERWithImageCapture(track);
  } else {
    this.startFERWithVideoElement(track);
  }
}
```

```
// ImageCapture method
private startFERWithImageCapture(track: MediaStreamTrack) {
  console.log('📸 Starting FER with ImageCapture (Android Chrome)');

  const imageCapture = new ImageCapture(track);
  const canvas = document.createElement('canvas');
  const ctx = canvas.getContext('2d');

  this.ferInterval = setInterval(async () => {
    try {
      if (!this.shouldContinueFER()) {
        this.stopFER();
        return;
      }

      // Capture frame using ImageCapture API
      const bitmap = await imageCapture.grabFrame();
      canvas.width = bitmap.width;
      canvas.height = bitmap.height;
      ctx?.drawImage(bitmap, 0, 0);

      const imageData = canvas.toDataURL('image/jpeg', 0.6);
      const base64Image = imageData.replace(/^data:image\/\w+; base64,/, '');

      await this.processFERImage(base64Image);
    } catch (err) {
```

```
    } catch (err) {
      if (err.name !== 'InvalidStateError') {
        console.error('FER Error (ImageCapture):', err);
        // If ImageCapture fails, fall back to video element
        console.warn('⚠️ ImageCapture failed, switching to Video Element');
        this.stopFER();
        this.startFERWithVideoElement(track);
      }
    }
  }, this.FER_CAPTURE_INTERVAL);
}

// Video Element method (for iOS/Safari/Firefox/fallback)
private startFERWithVideoElement(track: MediaStreamTrack) {
  console.log('📹 Starting FER with Video Element (iOS/Safari/Firefox)');

  const video = document.createElement('video');
  video.srcObject = new MediaStream([track]);
  video.autoplay = true;
  video.muted = true;
  video.playsInline = true;
  video.setAttribute('playsinline', '');

  const canvas = document.createElement('canvas');
```

```
video.onloadedmetadata = () => {
  canvas.width = video.videoWidth;
  canvas.height = video.videoHeight;

  console.log(` FER Video ready: ${canvas.width}x${canvas.height}`);

  this.ferInterval = setInterval(async () => {
    try {
      if (!this.shouldContinueFER()) {
        this.stopFER();
        video.srcObject = null;
        return;
      }

      if (video.readyState < 2) {
        return;
      }

      ctx?.drawImage(video, 0, 0, canvas.width, canvas.height);
      const imageData = canvas.toDataURL('image/jpeg', 0.6);
      const base64Image = imageData.replace(/^data:image\/\w+; base64,/, '');

      await this.processFERImage(base64Image);
    } catch (err) {
```

```
//FER helper
private shouldContinueFER(): boolean {
  if (this.currentHostCount === 0) {
    console.warn('⏸️ FER: No hosts detected, stopping');
    return false;
  }

  if (!this.isVideoEnabled) {
    console.warn('⏸️ FER: Video disabled, stopping');
    return false;
  }

  const currentTrack = this.localStream.video?.getMediaStreamTrack();
  if (!currentTrack || currentTrack.readyState !== 'live') {
    console.warn('FER: Track no longer live, stopping');
    return false;
  }

  return true;
}
```

```
//FER processor (POST/GET)
private async processFERImage(base64Image: string) {
  try {
    const response = await fetch('/api/fer', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ image_base64: base64Image }),
    });

    // Handle API errors
    if (!response.ok) {
      console.warn(`⚠️ FER API returned ${response.status}, skipping frame`);
      return;
    }

    const result = await response.json();
    const emotion = result.faces?.[0]?.attributes?.emotion;

    if (emotion) {
      const highestEmotion = Object.entries(emotion).reduce((max, [emotionName, confidence]) => {
        return (confidence as number) > max.confidence
          ? { emotion: emotionName, confidence: confidence as number }
          : max;
      }, { emotion: '', confidence: 0 });

      if (highestEmotion.confidence >= this.MIN_CONFIDENCE_THRESHOLD) {
        if (highestEmotion.emotion !== this.lastSavedEmotion) {
```

```
private stopFER() {
  if (this.ferInterval) {
    console.log('🔲 RTCENGINE: Stopping FER');
    clearInterval(this.ferInterval);
    this.ferInterval = null;
  }

  // Stop batch writer and flush remaining emotions
  if (this.batchInterval) {
    clearInterval(this.batchInterval);
    this.batchInterval = null;
  }

  // Write any remaining emotions before stopping
  if (this.emotionBatch.length > 0) {
    console.log(`💾 Flushing ${this.emotionBatch.length} remaining emotions...`);
    this.writeEmotionBatch();
  }

  // Reset state
  this.lastSavedEmotion = null;
  this.emotionBatch = [];
}
```

```
// Clean up FER monitoring
private cleanupFERMonitoring() {
  console.log('🔴 RTCENGINE: Cleaning up FER monitoring');

  // Remove event listener
  window.removeEventListener('fer:totalHostsChanged', this.handleHostCountChange);

  // Stop FER if running
  this.stopFER();
}

private startEmotionBatchWriter() {
  // Clear any existing interval
  if (this.batchInterval) {
    clearInterval(this.batchInterval);
  }

  this.batchInterval = setInterval(async () => {
    if (this.emotionBatch.length > 0) {
      await this.writeEmotionBatch();
    }
  }, this.BATCH_WRITE_INTERVAL);
}
```

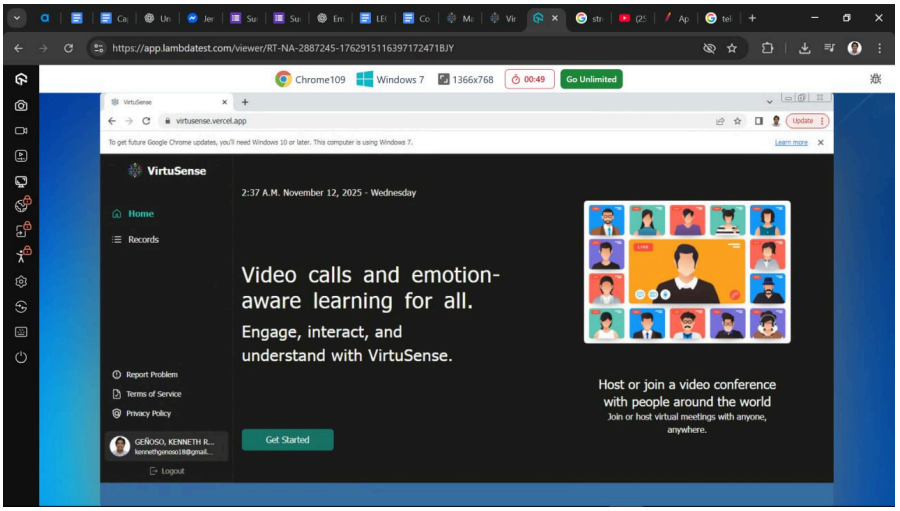
```
//write emotions per batch
private async writeEmotionBatch() {
  if (this.emotionBatch.length === 0) return;

  try {
    const attendeeRef = doc(
      db,
      'sessions',
      this.sessionId,
      'attendees',
      this.userUID
    );

    const attendeeSnap = await getDoc(attendeeRef);

    if (!attendeeSnap.exists()) {
      // attendee doesn't exist in records
      await setDoc(attendeeRef, {
        userUID: this.userUID,
        username: this.username || 'Unknown Attendee',
        createdAt: serverTimestamp(),
        emotions: this.emotionBatch,
      });
    } else {
      // append emotions to existing array
      const currentData = attendeeSnap.data();
```

APPENDIX D

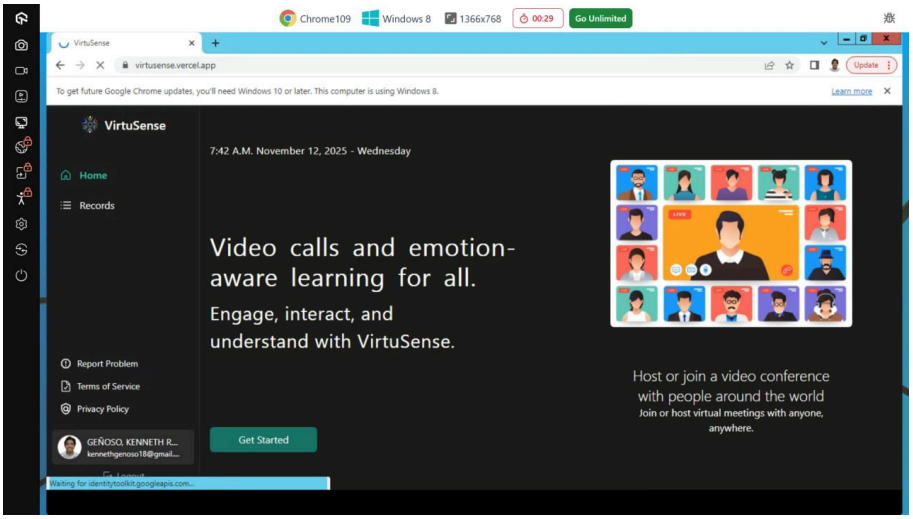
TESTING RESULTS

COMPATIBILITY TESTING USING LAMBDA TEST ON DIFFERENT WINDOWS VERSIONS

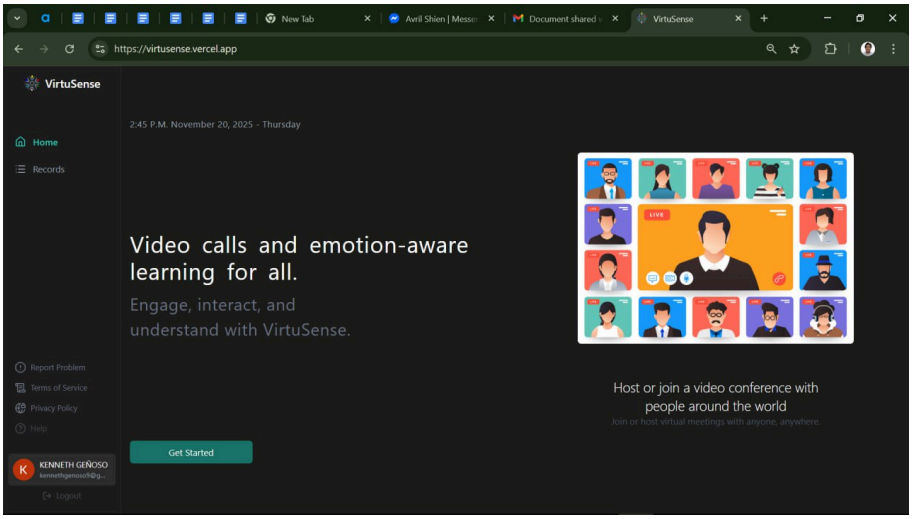
WINDOWS 7



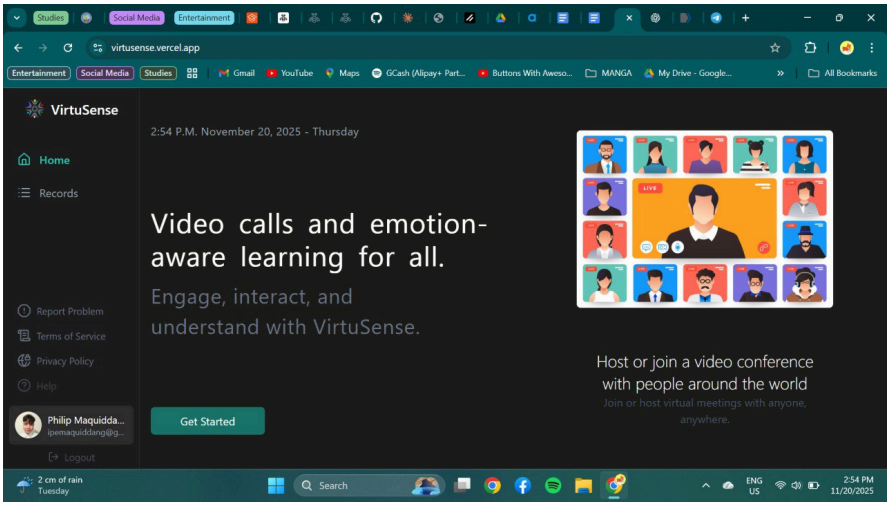
WINDOWS 8



WINDOWS 10

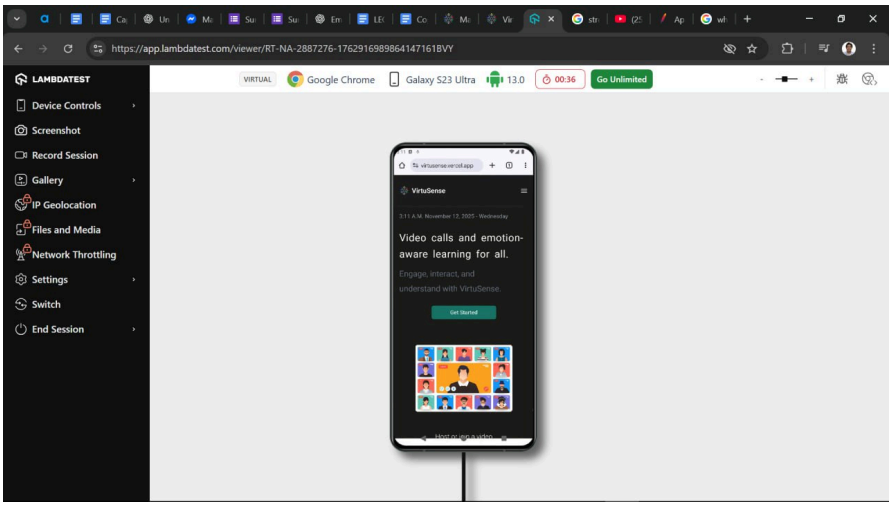


WINDOWS 11

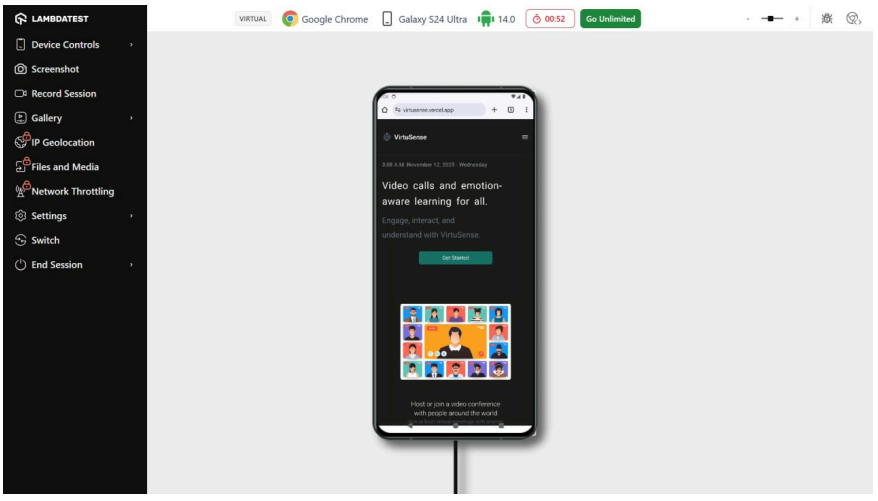


COMPATIBILITY TESTING USING LAMBDA TEST ON DIFFERENT WINDOWS VERSIONS

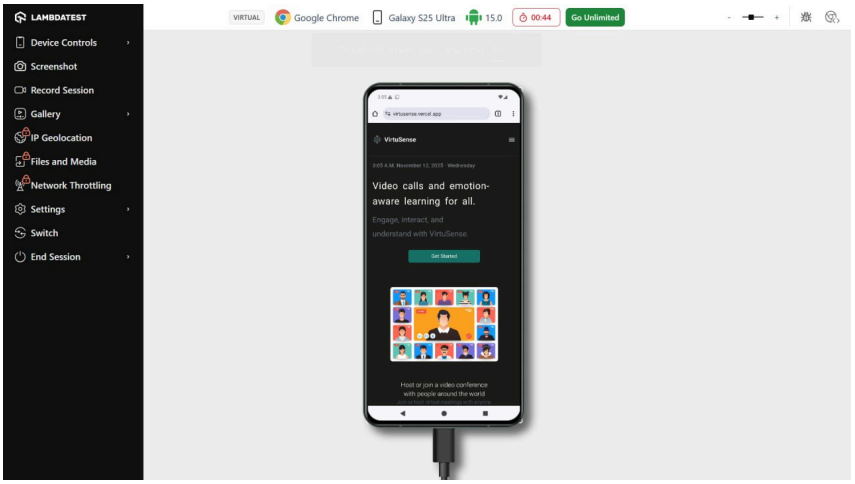
ANDROID 13



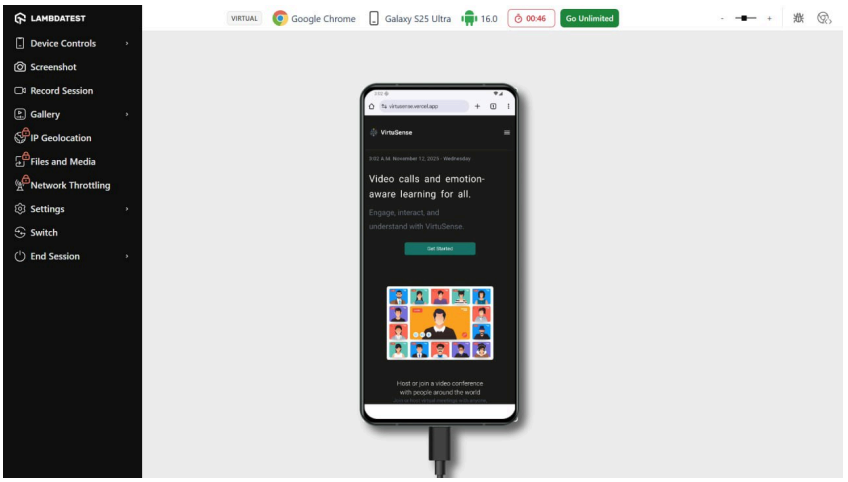
ANDROID 14



ANDROID 15

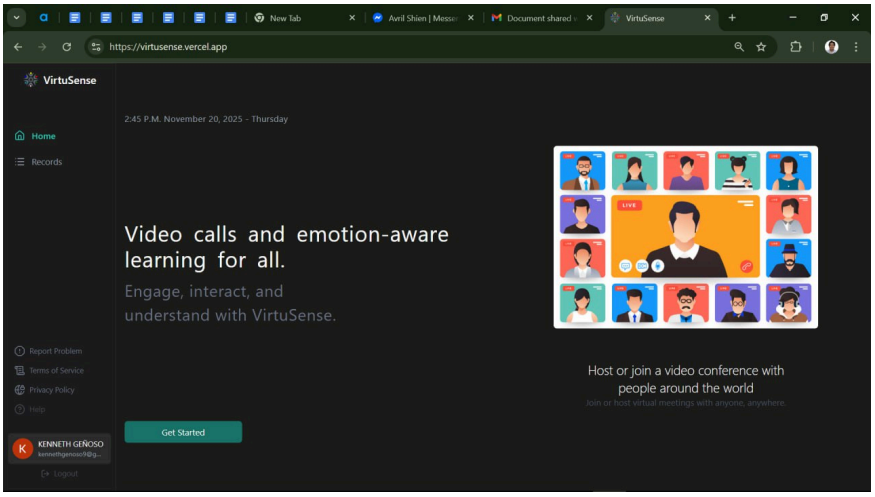


ANDROID 16

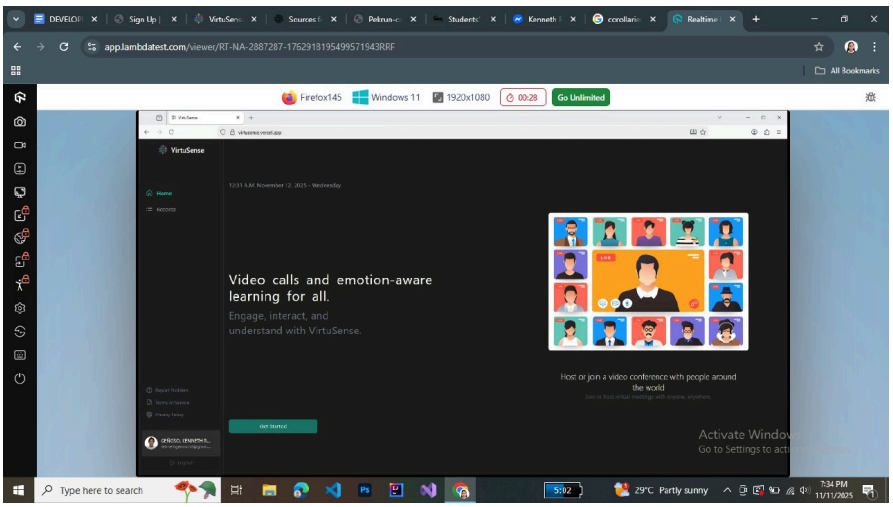


COMPATIBILITY TESTING USING LAMBDA TEST ON DIFFERENT WEB BROWSERS

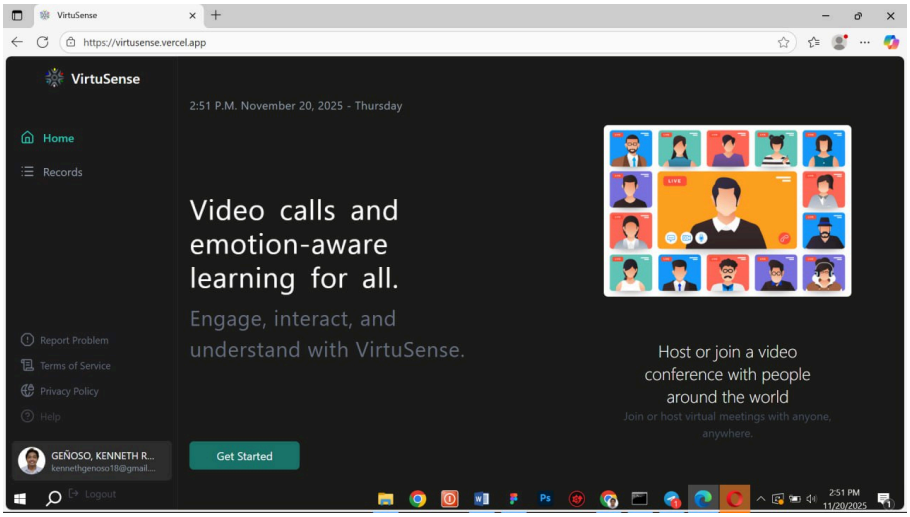
CHROME



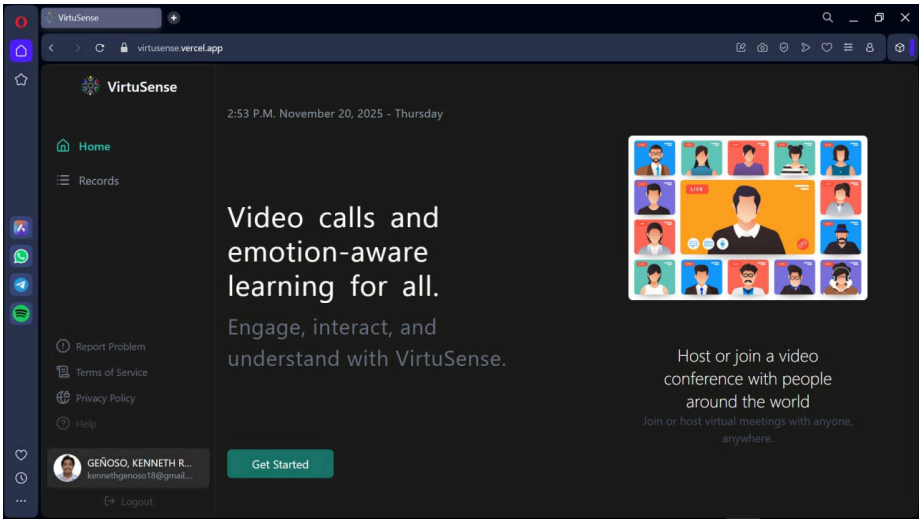
MOZILLA FIREFOX



MICROSOFT EDGE

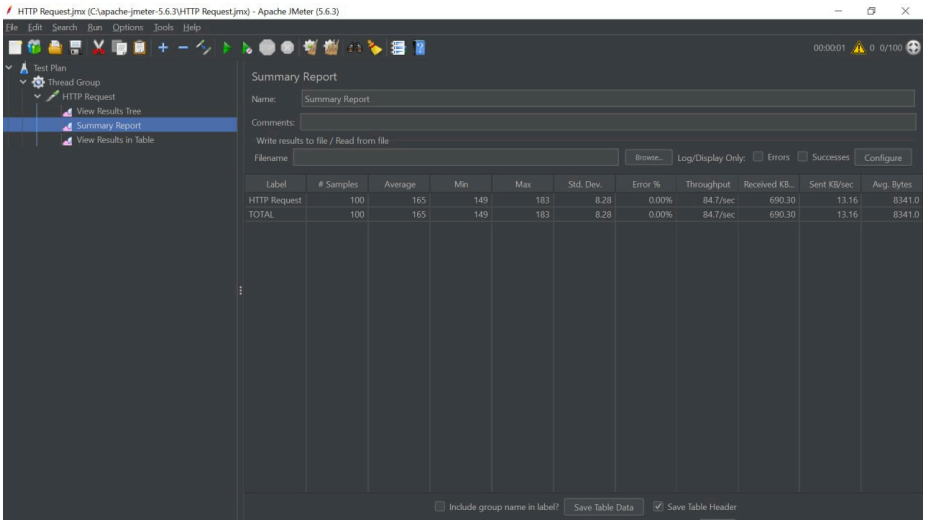


OPERA

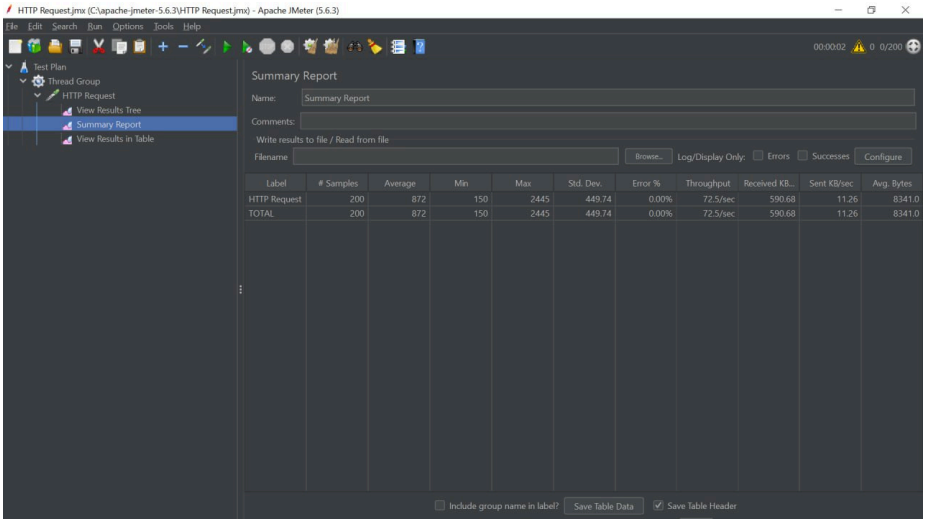


STRESS TESTING USING JMETER

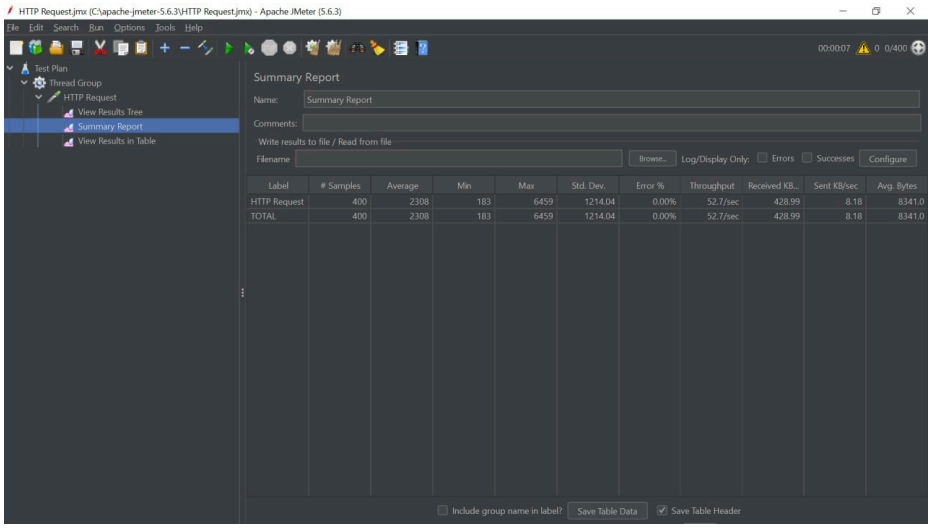
100 USERS



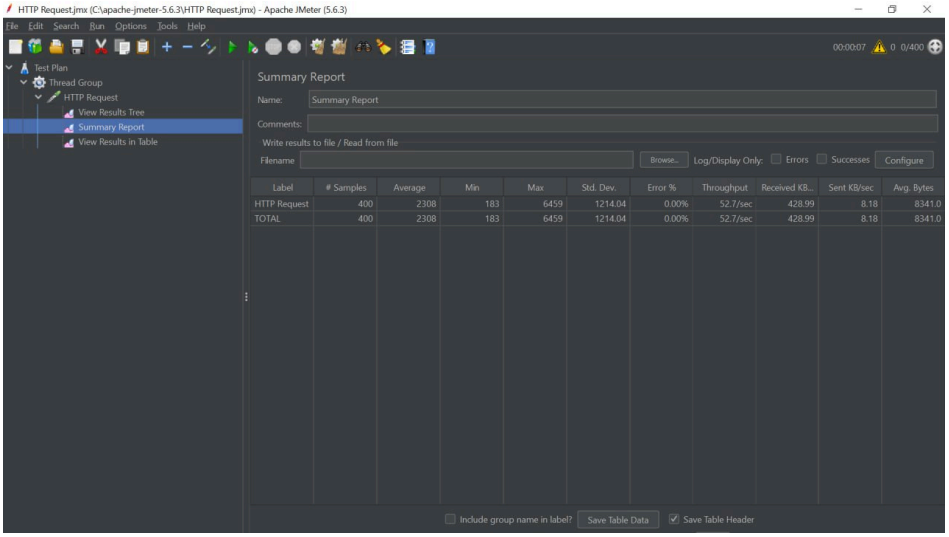
200 USERS



400 USERS



800 USERS


HTTP Request.jmx (C:\apache-jmeter-5.6.3\HTTP Request.jmx) - Apache JMeter (5.6.3)
File Edit Search Run Options Tools Help
00:00:07 0 0/400
Test Plan
Thread Group
HTTP Request
View Results Tree
Summary Report
View Results in Table
Summary Report
Name: Summary Report
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only: Errors Successes
Configure
Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB... Sent KB/sec Avg. Bytes
HTTP Request 400 2308 183 6459 1214.04 0.00% 52.7/sec 428.99 8.18 8341.0
TOTAL 400 2308 183 6459 1214.04 0.00% 52.7/sec 428.99 8.18 8341.0
Include group name in label?
Save Table Data
Save Table Header

APPENDIX E

PUBLISHABLE PAPER

DEVELOPMENT OF AN ONLINE CLASS CONFERENCE PLATFORM: INTEGRATING FACIAL EXPRESSION RECOGNITION UTILIZING FACE++ WEB API FOR LEARNERS EMOTION DETECTION

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College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

***Abstract* —** VirtuSense: An online conference platform with emotion detection through facial expression recognition was developed to address the problem in the academic performance of students during online classes. This project is developed by utilizing Agora for video conferencing, TypeScript, React for the frontend side, Firebase for the backend side of the system, and Face++ Web API for emotion detection. The project used agile scrum methodology. Software testing, stress testing, compatibility testing, and user acceptance testing was used to test the project's reliability and usability and achieved "Passed" and "Compatible" remarks in all the testing. The ISO25010 answered by three (3) IT experts achieved a 3.53 overall mean which indicates that the system has strongly passed all the overall criteria of the model. This means that the system is a reliable and usable tool for an online conference platform with emotion detection.

Keywords: Online Conference Platform, Emotion Detection, Face++, Agile Scrum, ISO25010, Software Testing, Stress Testing, Compatibility Testing, User Acceptance Testing.

Introduction

Emotion serves as a crucial indicator of a student's engagement or interest in online learning or even in physical class, it also plays a significant role for creating strategies for teaching and learning process.[1] Also, academic emotion has a strong impact on the learners' academic achievement or performance.[2] Individuals' positive emotions such as happiness during classes can help to boost one's interest in studying. However, negative emotions like sadness, fear, and anger can negatively affect the students' behavior during exams or the learning process.[3] Having these problems leads to the emergence of emotion detection through facial expression recognition. A study shows that the level of students' engagement can be identified through emotion detection particularly with their facial expression.[4]

Recent advancements in emotion detection have been made. Last 2021, one of the universities in Thailand created an online conference platform with emotion detection and achieved 88.97% satisfactory results after applying it to real online classes. The system detects boredom, neutral, happy, and confused and attains higher accuracy results than any other convolutional neural network-based emotion recognition system.[5] India developed a system that also detects facial expression during online classes using artificial intelligence which allows educators to gain valuable feedback on the effectiveness of their online teaching methods and make data-driven decisions to improve the learning experience for their students.[6]

Having a deep understanding of students' emotion during online classes can give numerous benefits for the educators or the institutions such as improving their teaching style, creating new and engaging strategies that can improve learners' interest in online classes. Through this advancement in artificial intelligence technology, the development of an online conference platform with the integration of emotion detection using Application Programming Interface (API) such as Face++ will be possible and the creation of this type of software is important for refining the online learning environment not only for the students' gain but also for the educators.

The aim of this capstone project is to develop a smart conference platform capable of detecting the emotions of learners in online classes through facial expression recognition. Specifically, the project seeks to develop a responsive web application that functions as an online conference platform to support learners during virtual classes, integrate real-time emotion detection features that allow educators to monitor learners' emotional states as they participate, and evaluate the system's acceptability using the Technology Acceptance Model (TAM) and ISO 25010.Related Literature.

The purpose of this project is to track emotional identification of the students during online classes through Facial Expression Recognition (FER) technology. The development of this system will greatly help the education sector that still implements online classes as one of their class modalities, specifically the teaching strategies during distance learning. Furthermore, this project aligns with the Sustainable Development Goal (SDG) 4: Quality Education by improving the learning environment of the students even in online classes in which their engagement and emotion are being constantly monitored and addressed. Aside from education, this project also targets the SDG 9: Industry, Innovation, and Infrastructure by innovating the online learning platforms as the project integrates automatic emotion recognition through FER technology which can result in more personalized and data-driven teaching strategies. Lastly, it also contributes to SDG 3: Good Health and Well Being goal by assessing students' emotional and mental health during online classes which can lead to a healthier educational experience.

This project will benefit educators through identifying and understanding their students' emotions during online class sessions by automatically collecting their emotions through facial expressions. This study will also serve as a pillar for future researchers that are interested in innovating the current technology in the educational sector by integrating Facial Expression Recognition (FER) and those researchers who want to study the behavioural identity of students during online classes.

Conceptual Framework

Figure 1 shows the conceptual framework which includes all the components, architecture, and methods used to create the online conference platform with emotion detection. This framework visually illustrates the interaction between various parts of the system, such as the frontend interface, the backend services and the emotional recognition API. It describes the flow of information through the system and user interaction, and provides insight into how API processes and be used to recognize, analyze, and interpret emotions over a range of real-time or recorded online sessions.

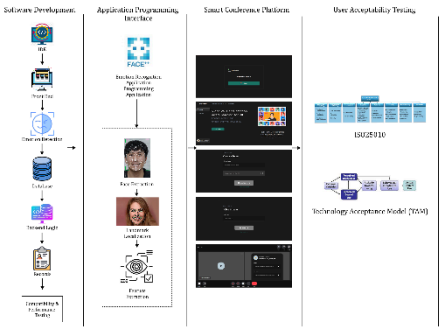

Figure 1. Conceptual Framework

METHODOLOGY

Project Design

Figure 2 presents the system architecture of the smart conference platform, showing how its four main modules work together. The Account Module manages secure user login and registration, while the Conference Module provides core meeting functions such as joining sessions, controlling audio and video, chatting, using the virtual whiteboard, and screen sharing. The Emotion Detection Module captures and processes facial images using an emotion recognition API to classify emotions and generate confidence scores. The Record Module visualizes emotional data, tracks trends, and produces automatic suggestions based on detected emotions. All modules are connected to a central database that securely stores, manages, and retrieves data, enabling smooth interaction and providing users with meeting logs and emotion profiles.

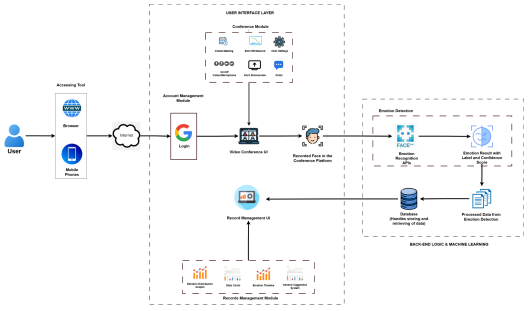

Figure 2. System Architecture

Project Development

The development of the online conference platform follows the Agile Scrum methodology. The project will start with the listing of product backlogs, followed by sprint planning then sprint backlogs. Daily scrum meetings will be done at the end of every sprint.

Figure 3. Scrum Sprint Cycle

Evaluation Procedure

Figure 10 shows that the software product quality was based on the ISO25010 standards. It was used as a standard for the systems evaluation procedure. It focuses on the functionality, reliability, usability, efficiency, maintainability, and portability of the Smart Conference Platform. Every module and feature of the system will be tested based on these standards.

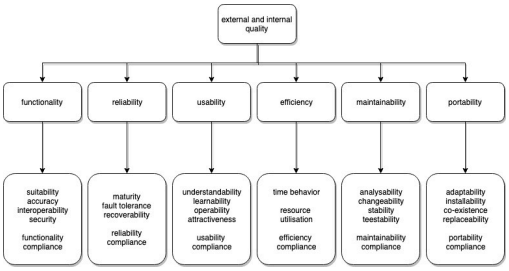

Figure 4. Quality Characteristics of Software Products

RESULTS AND DISCUSSION

Account Management Module

Figure 5 and 6 shows the sign in page and homepage of the online conference platform. The sign in page offers a Google Account sign in for safer and easy registration of users. In the homepage, it displays the "Get Started" button which will lead to the creation of the online meeting session.

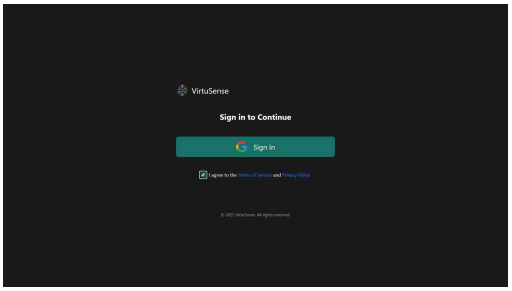

Figure 5. VirtuSense Sign in page

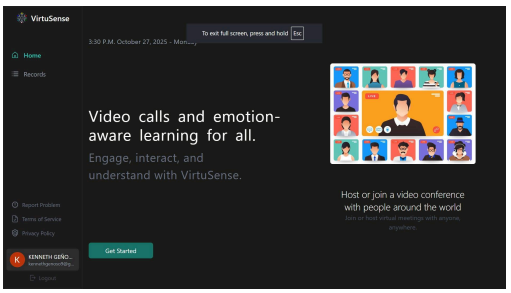

Figure 6. VirtuSense Home Page

Conference Module

Figure 7 and 8 illustrates the creation page of online meeting sessions in which the users will be required to put a name or title in the session they are making, what is all about or anything they want to put in. In addition, the users will be given an option to make every participant who will join become a co-host. Also, inside the creation page and after the user fills the required name of the session, the button "Create a Room" will direct them to another process page in which the joining links of the online room are displayed whether the joining link for participants or a co-host. Lastly, the "Start Room" button will redirect them to the main conference platform interface.

However, for the option of joining a meeting, if the user will just join an existing room, they will just need to click the "Join with a Room ID" text and it will direct them to the participant joining process page which will require them to put the room ID provided by the host. Then they will be able to join the conference if the host allows their admission after clicking the "Join Room" button which is shown in Figure 9.

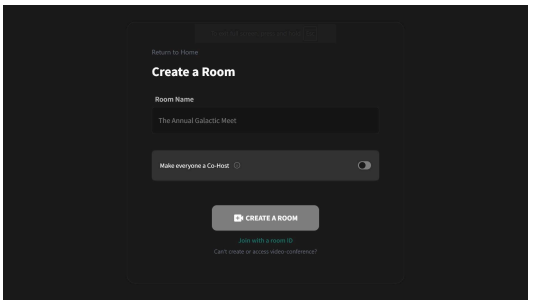

Figure 7. VirtuSense Online Room Creation

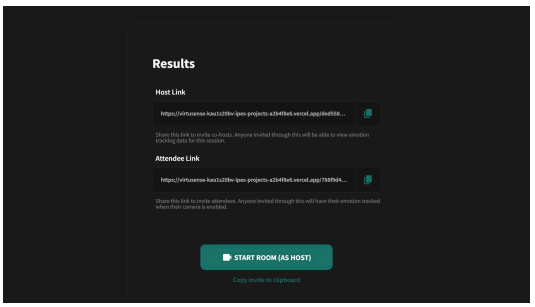

Figure 8. VirtuSense Online Room Creation Links

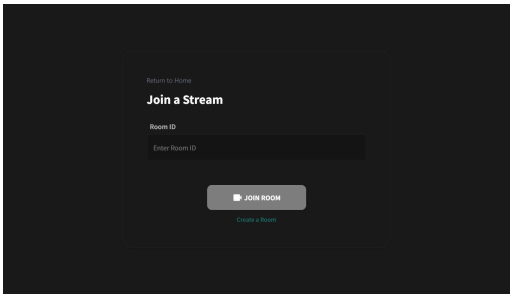

Figure 9. VirtuSense Joining a Room Page

Figure 10 shows the main interface of the online conference room. This page caters the most important feature of the project which is the emotion detection which will work if the participants have their cameras on. Also, in this page the users can see all the features available and they can use it to connect with their participants such as the on/off camera and microphone, sharescreen, public and private chats, host settings, virtual whiteboard, page layout, and the invitation link.

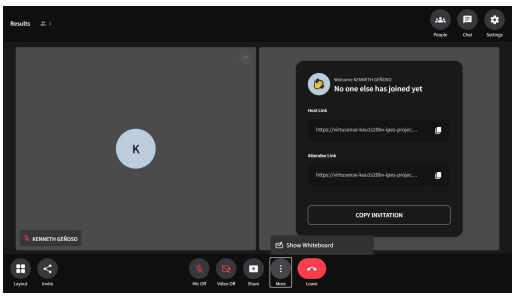

Figure 10. VirtuSense Online Conference Room

Report Management Module

Figure 11 shows the record section of the online conference platform in which the host can see the accumulated detection among his/her participants during the online session. The system will provide data analytics in this section such as the general statistics card which displays the total session, total participants, average session duration, and total emotion recorded. The general record section also shows general reminders and suggestions, emotion distribution graph, engagement assessment graph, and general recommendation.

The record section has another part where the host can view the data inside a particular session, and inside that page the host can see the number of participants in that session, duration, emotion recorded, and the most frequent emotion displayed during the session. In addition, there is also an emotion timeline graph, session engagement and emotion distribution graph, and session insights based on the gathered data.

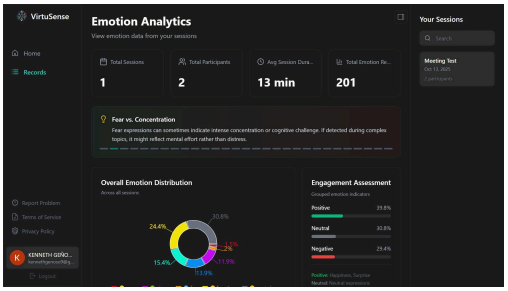

Figure 11. VirtuSense Record Section

Figure 11.1. VirtuSense Records Section

Emotion Detection Module

Figure 12 illustrates the emotion detection during online conference sessions, although the system's emotion detection doesn't really have emotion logs during the conference to make the participants stay anonymous for data privacy, the team made a temporary console emotion logs to demonstrate that the emotion detection is working. These logs will be saved or transported to the database of the system and will be analyzed and displayed in the record section of the web application. The emotions will just be saved if the system detects a change in facial expression of the participants and it passes the required confidence level which is 75%.

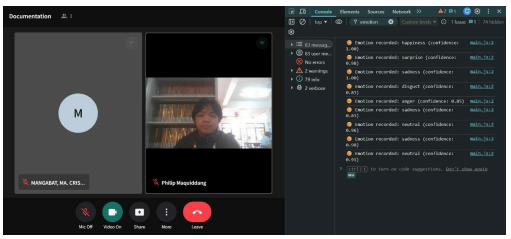

Figure 12. Emotion Detection Temporary Logs

Figure 13 shows the emotion logs being absorbed in the database, the data being transported in the backend includes the emotion detected, timestamp, and confidence level. These three data is necessary for the report management module since the data analytics will all be based on the collected data of the system during each session.

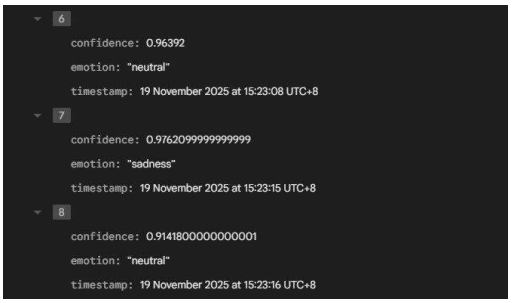

Figure 13. Emotion Detection Logs in Database

Implementation of the System in Real Online Class Session

Figure 23 shows the interface of VirtuSense in a real online class session. This session is titled "CMSC 307" with 32 participants at its peak. The students are required to open their camera to trigger the emotion detection of the system.

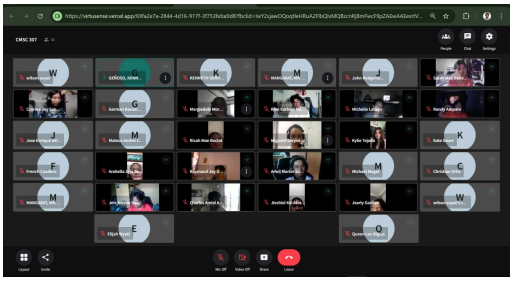

Figure 23. Real-Life Online Class Session

Figure 26 illustrates the statistics card and emotion timeline graph which is based on the gathered data in the class "CMSC 307". The statistical cards show that there are 25 detected participant's emotions and its duration is thirty-two (32) minutes. It also displayed the emotion records which indicates that there are 222 emotions detected during the session. Also, the cards show the most frequent emotion detected in the class which is "neutral". Lastly, the emotion timeline graph which shows emotion patterns during the class. In 0-10 minutes, the system detected 0 emotion since the participants had their camera off. However, in 10-15 minutes, the system started to detect some emotions such as happiness, surprise, and sadness.

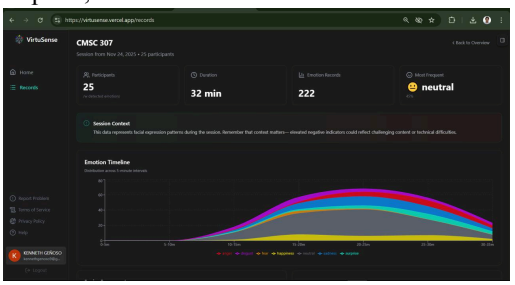

Figure 26. Statistics Card and Emotion Timeline Graph

Figure 27 displayed the session engagement, session emotion distribution, session insights graphs, and the peak moments. Session engagement categorized each emotion into positive, neutral, and negative. Happiness and surprise emotion is categorized as positive, neutral emotion is categorized as neutral and all the remaining emotions are categorized as negative. The session engagement in the class "CMSC 307" shows neutral emotion with 10% margin with negative emotion. This result will be used as the basis for the session insights which displayed low positive engagement since the session engagement displayed that there are only 19% positive emotion detected during that class. Another graph is the session emotion distribution which displays neutral emotion as dominant in all other emotions detected. Lastly, the peak moments information in which it displayed the timeline where most emotions are detected and what emotion is the most detected in that timeline.

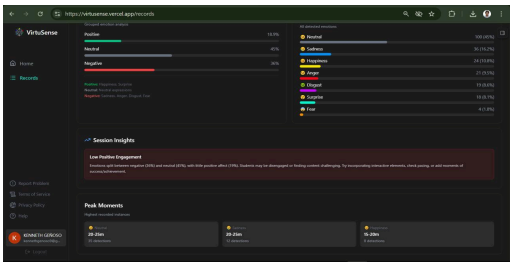

Figure 27. Session Engagement, Session Emotion Distribution, Session Insights, and Peak Moments

Test and Evaluate the Software using ISO 25010

Table 1 illustrates the ISO25010 answered by three (3) IT experts, it indicates that VirtuSense achieved an acceptable evaluation in each criteria. However, the system still has a lot to improve in terms of functional accuracy and appropriateness, user

interface aesthetics, maturity, recoverability, non-repudiation, accountability, analysability, modifiability, modularity, testability since it only got a score ranging from 3.00 - 3.33. Overall, the system's weighted mean is 3.53 which indicates that the system is a reliable tool for online class learning which needs constant improvement to achieve the outstanding performance of the platform.

Table 1. Results of BetaTesting answered by the IT Specialist

Item	Mean	Interpretation
Functional Completeness	3.67	Strongly Agree
Functional Accuracy	3.33	Agree
Functional Appropriateness	3.33	Agree
Time Behaviour	4.00	Strongly Agree
Resource Utilization	3.67	Strongly Agree
Co-Existence	3.67	Strongly Agree
Interoperability	3.67	Strongly Agree
Learnability	4.00	Strongly Agree
Operability	3.67	Strongly Agree
User Interface Aesthetics	3.33	Agree
Maturity	3.00	Agree
Fault Tolerance	3.67	Strongly Agree
Recoverability	3.33	Agree
Confidentiality	3.67	Strongly Agree
Integrity	4.00	Strongly Agree
Non-Repudiation	3.00	Agree
Accountability	3.33	Agree
Authenticity	4.00	Strongly Agree
Analysability	3.33	Agree
Modifiability	3.00	Agree
Modularity	3.00	Agree
Testability	3.00	Agree
Adaptability	4.00	Strongly Agree
Installability	3.67	Strongly Agree
Replaceability	4.00	Strongly Agree
Overall Weighted Mean	**3.53**	**Strongly Agree**

n=3, Legend: four (4) = Strongly Agree; three (3) = Agree; two (2) = Disagree; one (1) = Strongly Disagree

Compatibility Testing

Table 3 shows the compatibility testing in different Windows versions. The system was tested in Windows 11, Windows 10, Windows 8, Windows 7 which are all capable of catering the system smoothly and without any problems. The remarks "Passed" indicates that the system could run in that particular version.

Table 3. Compatibility Testing of VirtuSense on Different Windows Versions

Windows Version	Expected Output	Actual Output	Remarks
Windows 11	Compatible	Compatible	Passed
Windows 10	Compatible	Compatible	Passed
Windows 8	Compatible	Compatible	Passed
Windows 7	Compatible	Compatible	Passed

Table 4 displays the result of compatibility testing across various Android versions, specifically Android 16, Android 15, Android 14, and Android 13. The system works perfectly fine without any issues in all the Android versions and all these versions achieved "Passed" remarks for the system's compatibility. This shows that the system is versatile in terms of different devices and different Android Versions.

Table 4. Compatibility Testing of VirtuSense on Different Android Versions

Android Version	Expected Output	Actual Output	Remarks
Android 16	Compatible	Compatible	Passed
Android 15	Compatible	Compatible	Passed
Android 14	Compatible	Compatible	Passed
Android 13	Compatible	Compatible	Passed

Table 5 shows the result of browser testing which indicates that the system is browser independent which means that the system is functional, accessible and can run smoothly in all the tested browsers which are Google, Microsoft Edge, Mozilla Firefox, Opera, and Safari. All the expected output was met in all the browsers which resulted in the "Passed" remarks of the five different browsers.

Table 5. Compatibility Testing of VirtuSense on Different Browsers

Web Browser	Expected Output	Actual Output	Remarks
Google	Compatible	Compatible	Passed
Microsoft Edge	Compatible	Compatible	Passed
Mozilla Firefox	Compatible	Compatible	Passed
Opera	Compatible	Compatible	Passed
Safari	Compatible	Compatible	Passed

Result of Performance Testing

Table 6 demonstrated the result of the stress testing using Apache Jmeter. The system can handle requests smoothly to moderately with 100 - 400 users with a maximum response time of 219ms for 100 users, 598ms for 200 users and 400ms for 400 users. However, when the system reaches 800-1600 user requests the response time of the system is slower and can be extremely slow but still can handle the request successfully without error.

Table 6. Stress Testing Using JMeter

Users	Max Response Time (ms)	Error (%)	Performance Level	Remarks
100	219ms	0.00%	Excellent	System responded very quickly; stable performance
200	598ms	0.00%	Good	Slight increase in response time; still stable
400	1158ms	0.00%	Moderate	Moderate response time; performance acceptable
800	5412ms	0.00%	Slow	Noticeable delay; system under high load but stable
1600	9285ms	0.00%	Very Slow	High response time; system approaching stress limit but still functional

Conclusion

1. Develop a progressive web application that serves as an online conference platform to support learners during online classes.

The system became an additional option for all the existing online conference platforms with interactive features and functions exceeding other platforms. This was made possible by using different kinds of programming tools such as React, TypeScript, TailwindCSS, Agora and Firebase for both frontend and backend. In addition, the system can work across different browsers, Windows version, and Android version for mobile devices.

2. Integrate real-time emotion detection features through the integration of emotion detection that allow educators to monitor learners' emotional states as they participate in online classes.

The real-time emotion detection integrated in the system was achieved using Face++ Web API which is a pre-trained emotion detection model that can detect seven specific emotions which are happiness, sadness, disgust, surprise, fear, anger, and neutral. These gathered emotions were used in the data analytics of the system in the record section which shows the emotion distribution graphs, recommendations and insights, statistical cards, emotion timeline and student engagement table.

3. Evaluate the acceptability of the system using Technology Acceptance Model (TAM) and ISO 25010.

The result of the questionnaire survey answered by instructors showed that the system has passed the Technology Acceptance Model and user acceptance. In addition, the questionnaire survey answered by IT experts that focus on the functionality, reliability, and overall quality of the system has passed the ISO 25010 standards for a quality system.

Recommendations

These recommendations were based on the findings of the project and the recommendations of IT experts and some of the instructors:

1. This website is recommended to become a mobile application for convenience and accessibility and to improve the overall performance of the system.

2. The system should have more features such as reactions, raise hand etc. and also the whiteboard feature of the system should be interactive and usable for both participant and host.

3. The recommendation and suggestion system of the online conference platform in the record section could become an AI generated content for more insights that the system could provide that can make the system become more informative and complex.

4. The system is recommended to have an archive section for sessions that are no longer latest or if the record section has so many sessions recorded.

References

[1] LB, N. K., & Gg, N. L. P. (2016d). Student Emotion Recognition System (SERS) for e-learning Improvement Based on Learner Concentration Metric. Procedia Computer Science, 85, 767–776. https://doi.org/10.1016/j.procs.2016.05.264

[2] Camacho-Morles, J., Slemp, G. R., Pekrun, R., Loderer, K., Hou, H., & Oades, L. G. (2021). Activity Achievement Emotions and Academic Performance: A Meta-analysis. Educational Psychology Review, 33(3), 1051–1095. https://doi.org/10.1007/s10648-020-09585-3

[3] Rothkrantz, L.J.M. (2017). "Automatic Emotion Recognition: An Overview." In Proceedings of the 9th International Conference on Agents and Artificial Intelligence (ICAART 2017), pages 65–73.

[4] Maqableh, W., Alzyoud, F. Y., & Zraqou, J. (2022). The use of facial expressions in measuring students' interaction with distance learning environments during the COVID-19 crisis. Computers and Education: Artificial Intelligence, 3, Article 100073. https://doi.org/10.1016/j.caeai.2022.100073

[5] Panichkriangkrai, C., Silapasuphakornwong, P., & Saenphon, T. (2021). Emotion recognition of students during e-learning through online conference meetings. li01.tci-thaijo.org. https://doi.org/10.14456/sehs.2021.37

[6] Aravindhar, S. M. D. J. (2024, March 26). Deep learning based facial emotion recognition for analysing the effectiveness of online class. https://ijisae.org/index.php/IJISAE/article/view/5602

[7] Bhattacharjee, A., Pias, T., Ahmad, M., & Rahman, A. (2018, December 1). On the performance analysis of APIs recognizing emotions from video images of facial expressions. https://doi.org/10.1109/icmla.2018.00040

[8] Küntzler, T., Höfling, T. T. A., & Alpers, G. W. (2021). Automatic facial expression recognition in standardized and non-standardized emotional expressions. Frontiers in Psychology, 12, 627561. https://doi.org/10.3389/fpsyg.2021.627561

[9] White, H. (2025, March 25). Computer Vision Basics: Confidence & Accuracy https://www.leverege.com/blogpost/computer-vision-basics-how-confidence-accuracy-and-thresholds-impact-performance

[10] Wenkel, S., Alhazmi, K., Liiv, T., Alrshoud, S., & Simon, M. (n.d.). Confidence Score: The Forgotten Dimension of Object Detection Performance Evaluation. Sensors, 21(13), 4350. https://doi.org/10.3390/s21134350

[11] Dupré, D., Krumhuber, E. G., Küster, D., & McKeown, G. J. (2020). A performance comparison of eight commercially available automatic classifiers for facial affect recognition. PLoS ONE, 15(4), e0231968. https://doi.org/10.1371/journal.pone.0231968

[12] Hassanat, A. B., Albustanji, A., Tarawneh, A. S., Alrashidi, M., Alharbi, H., Alanazi, M., . . . Prasath, V. B. S. (2021b). Deep learning for identification and face, gender, expression recognition under constraints. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2111.0193

APPENDIX F

GRAMMARLY RESULT

DEVELOPMENT OF AN ONLINE CLASS CONFERENCE PLATFORM: INTEGRATING FACIAL EXPRESSION RECOGNITION UTILIZING FACE++ WEB API FOR LEARNS EMOTION RECOGNITION

by Ronah joy Ramirez

General metrics

73,213	11,027	543	44 min 6 sec	1 hr 24 min
characters	words	sentences	reading time	speaking time

Score

82

This text scores better than 82% of all texts checked by Grammarly

Writing Issues

322	✓	322
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APPENDIX G

PLAGIARISM RESULT

DEVELOPMENT-OF-AN-ONLINE-CLASS-CONFERENCE-PLATFORM_-INTEGRATING-FACIAL-EXPRESSION-RECOGNITION-UTILIZING-FACE-WEB-API-FOR-LEARNERS-EMOTION-DETECTION-1.pdf

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APPENDIX H

CURRICULUM VITAE

Kenneth Rasheed A. Geñoso

Summary

Motivated Information Technology student with strong academic performance and a track record of awards and honors. Skilled in programming, web development, database management, and UI/UX design, with practical experience in academic projects involving system testing, debugging, and responsive application development. Possesses strong analytical, problem-solving, and teamwork skills, with a commitment to continuous learning and contributing effectively to technical projects.

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Education

2011 - 2016
Elementary
Tayabas West Central School I

2016 - 2020
Junior Highschool
Luis Palad Integrated Highschool

2020 - 2022
Senior Highschool
Luis Palad Integrated Highschool

2020 - 2022
College
Laguna State Polytechnic University - Los Baños Campus

Language

English

Filipino

Experience

Academic Projects – Information Technology Program

- Conducted system testing, debugging, and documentation for group and individual projects.
- Developed user-friendly interfaces and implemented basic UI/UX principles
- Created interactive web applications with responsive design for desktop and mobile platforms.

Skills

Soft Skills

- Leadership
- Critical Thinking
- Teamwork and Collaboration
- Work Ethic and Professionalism
- Analytical and Problem-Solving Skills
- Communication Skills

Technical Skills

- Programming: Java, Python, C++, C#, JavaScript, PHP
- Web Development: HTML, CSS, JavaScript, Bootstrap, React
- Database Management: MySQL, SQL Server, MongoDB
- UI/UX Design (Figma, Canva)
- System Documentation & Technical Writing

AWARDS/ACHIEVEMENTS

ACADEMIC

- With honors 2017-2020
- With honors 2021-2022
- Division School Press Conference Sports writing 1st Place
- Division School Press Conference Online Publishing 1st Place
- Dean's Lister Award 2022-2025

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2016 - 2020
Junior Highschool
St. Thomas Montessori Learning Center

2020 - 2022
Senior Highschool
First College

2022 - Present
College
Laguna State Polytechnic University - Los Baños Campus

Language

English

Filipino

Ma. Cristina M. Mangabat

Summary

Information Technology student with practical experience in web development, technical support, and basic design tools. Demonstrates strong analytical skills, adaptability, and the ability to work well with others. Experienced in completing academic IT projects, using version control, and presenting technical work clearly. Consistently recognized for academic excellence and a commitment to learning.

Experience

Academic Projects – Information Technology Program

- Applied problem-solving and analytical skills in developing technical solutions based on project specifications.
- Presented final projects to faculty and peers, incorporating feedback to improve system performance.
- Applied version control using Git and GitHub to manage project development collaboratively.

Skills

Soft Skills

- Work Ethic and Professionalism
- Critical Thinking
- Teamwork and Collaboration
- Communication Skills
- Adaptability and Willingness to Learn
- Time Management

Technical Skills

- Web Development: HTML, CSS, JavaScript, Bootstrap, React
- IT Support & Troubleshooting
- UI/UX Design (Figma, Canva)
- Data Analysis (Excel, Python Pandas)
- System Documentation & Technical Writing

AWARDS/ACHIEVEMENTS

ACADEMIC

- With honors 2016-2020
- With honors 2020-2022
- Dean's Lister Award 2022-2025

Philip Louise C. Maquiddang

Summary

Information Technology student with solid experience in programming, database development, and web technologies. Skilled in building functional academic systems, integrating APIs, and managing both front-end and back-end components. Demonstrates strong analytical thinking, attention to detail, and the ability to collaborate effectively in team-based projects. Recognized for consistent academic achievement and a strong commitment to quality and continuous learning.

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Obando National Highschool
Brgy. Paliwas, Obando, Bulacan

2019 - 2021
Senior Highschool
Obando Senior Highschool
Brgy. Paliwas, Obando, Bulacan

2022 - Present
College
Laguna State Polytechnic University - Los Baños Campus

Language

English

Filipino

Experience

Academic Projects – Information Technology Program

- Designed and implemented database systems for academic requirements using SQL-based tools.
- Conducted system testing, debugging, and documentation for group and individual projects.
- Implemented front-end and back-end functionality, integrating databases and APIs

Skills

Soft Skills

- Critical Thinking
- Creativity and Innovation
- Work Ethic and Professionalism
- Attention to Detail
- Teamwork and Collaboration
- Analytical and Problem-Solving Skills

Technical Skills

- Programming: Java, Python, C++, C#, JavaScript, PHP
- Web Development: HTML, CSS, JavaScript, Bootstrap, React
- Database Management: MySQL, SQL Server, MongoDB
- API Integration (RESTful APIs)
- Version Control: Git, GitHub

AWARDS/ACHIEVEMENTS

ACADEMIC

- MTAP District Level 4th Place
- With Honors 2015
- With Honors 2019-2021
- Dean's Lister Award 2022